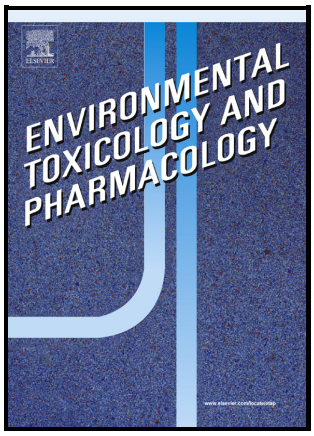
Journal Pre-proof



Comprehensive Analysis of Prohibited Substances and Methods in Sports: Unveiling Trends, Pharmacokinetics, and WADA Evolution

Patrik Oleksak, Eugenie Nepovimova, Marian Valko, Saleh Alwasel, Suliman Alomar, Kamil Kuca

PII: S1382-6689(24)00087-5

DOI: https://doi.org/10.1016/j.etap.2024.104447

Reference: ENVTOX104447

To appear in: *Environmental Toxicology and Pharmacology*

Received date: 28 November 2023
Revised date: 24 March 2024
Accepted date: 11 April 2024

Please cite this article as: Patrik Oleksak, Eugenie Nepovimova, Marian Valko, Saleh Alwasel, Suliman Alomar and Kamil Kuca, Comprehensive Analysis of Prohibited Substances and Methods in Sports: Unveiling Trends, Pharmacokinetics, and WADA Evolution, *Environmental Toxicology and Pharmacology,* (2024) doi:https://doi.org/10.1016/j.etap.2024.104447

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2024 Published by Elsevier.

Journal Pre-proof

Comprehensive Analysis of Prohibited Substances and Methods in Sports: Unveiling Trends, Pharmacokinetics, and WADA Evolution

Patrik Oleksak[1], Eugenie Nepovimova[1], Marian Valko[2,3], Saleh Alwasel[3], Suliman Alomar[3]*, and Kamil Kuca[1,4,5]*

1 Department of Chemistry, Faculty of Science, University of Hradec Kralove, 500 03 Hradec Kralove, Czech Republic.
2 Faculty of Chemical and Food Technology, Slovak University of Technology, 812 37, Bratislava, Slovakia.
3 Zoology Department, College of Science, King Saud University, Riyadh, 11451, Saudi Arabia.
4 Biomedical Research Center, University Hospital Hradec Kralove, Hradec Kralove, Czech Republic.
5 Andalusian Research Institute in Data Science and Computational Intelligence (DaSCI), University of Granada, 18071, Granada, Spain.

Abstract

This review systematically compiles sports-related drugs, substances, and methodologies based on the 2023 Prohibited List. Aligned with the list's structure, it covers all proscribed items and details the pharmacokinetics and pharmacodynamics of five representatives from each section. Notably, it explores significant metabolites and metabolic pathways associated with these substances. Adverse analytical findings are summarized in tables for clarity, and the prevalence is visually represented through charts. The review includes a concise historical overview of doping and World Anti-Doping Agency's role, examining modifications in the prohibited list for an understanding of evolving anti-doping measures.

Key words: Prohibited substances; Prohibited methods; Anti-doping drugs; World Anti-Doping Agency; Sport

Journal Pre-proof

1 Introduction

Sport may be in general term defined as an activity involving physical exertion in which individuals or squads compete against each other for entertainment, health and/or money according to fixed rules. In addition to physical exertion, sport is commonly associated with measures of performance. However, it is important to note that also other forms of competition, not involved in this definition, could be considered as a sport (Varmus et al. 2021). Thus, an activity which is considered as the sport in one instance may not be considered in another one. There are different forms of sport worldwide and based on the global trends, societal norms, and new directions the sport is constantly changing and modifying (Gould and Carson 2008).

In order to achieve success in sport, some athletes are using an unacceptable cheating practice to enhance performance which is in sports commonly termed as doping. This involves a use of variable

prohibited substances or methods. Historically, doping substances were used to improve performance and vitality as well as to cure diseases. In ancient Egypt and Mesopotamia, poppy and opiate derivatives were used for their stimulant effects. Several cultures; such as Aztecs, Indians, and Egyptians; practiced organotherapy which is based on the consumption of human or animal organs. Many other substances with stimulant effect such as hallucinogenic mushrooms, substances extracted from natural sources, wine, brandy, and others were used to overcome fatigue and thus improve performance. In 19th century, advances in modern medicine provided a plenty of pharmacologically attractive compounds, isolated and purified from natural sources or via their synthetic optimalization. Thus, along with modern medicine, the era of modern doping has begun. However, the use of pharmaceutically active compounds as a doping may represent a potential risk for athletes, even has fatal consequences. Therefore, in order to protect health of athletes as well as to promote fair competition in sports, the anti-doping analysis has been introduced in almost all sports (Gómez-Guerrero et al. 2022).

This review summarizes all adverse analytical/atypical findings reported within last 20 years by the World Anti-Doping Agency in order to identify commonly detected prohibited substances and methods in sports, that could be potentially used as a doping. The main aim of this review is to provide summarized information about different categories of substances and methods in sports; pervious and current trends of doping in sports; the most often detected and reported members; pharmacokinetics of commonly misused substances; their mode of action as doping agents; and potential methods of detection. The review also incorporates notable changes and reclassifications within the categories; and elucidates important policies, rules, and regulations of anti-doping program. In addition, description of a concise anti-doping history as well as discussion regarding infrequent biological matrices and notable anti-doping cases are present in the review.

2 A brief history of anti-doping tests

In 1960, the Summer Olympic Games in Rome were the first Summer Games streamed live to an international audience. This significant advancement in broadcasting technology allowed to watch real-time sport events to millions of spectators all around the globe. However, the 1960 Summer Olympics are associated with a tragedy of a Danish cyclist, who collapsed and died during a team road race. At the time, athletes often used stimulating cocktails of different drugs to increase performance. In following years, these circumstances turned out to be the main trigger for the organized fight against drugs in sport. In 1961, the International Olympic Committee (IOC) Medicinal Commission was founded as a working group to design a strategy against the drug use in Olympic sports (Ljungqvist 2017). In 1967, IOC Medicinal Commission has been officially established with an aim to develop a list of prohibited substances and methods. The first introductions of drug tests were implemented during the 1968 Winter Olympics in Grenoble, France and the 1968 Summer Olympics in Mexico City (Fraser 2004). Later, during the 1972 Summer Olympics in Munich, a comprehensive anti-doping program was introduced, in which about 2000 samples were collected and analyzed for different stimulants. In the late 1970s, the procedural guidelines and specific requirements for laboratories evaluating doping analysis were drafted by the International Association of Athletics Federations (IAAF). In 1979, has begun the accreditation of laboratories for doping control. On the basis of the IAAF Council decision, only laboratories that adhered to drafted guidelines and requirements were qualified for doping control. In 1981, the IAAF accreditation system was officially overtaken by IOC. In following years doping laboratories were jointly accredited by both, IAAF and IOC. In 1986, the full responsibility of accreditation was taken over by the IOC (Ljungqvist 2017).

Journal Pre-proof

3 World Anti-Doping Agency

The World Anti-Doping Agency (WADA) is an international independent agency established on 10th November 1999 under initiative of the IOC with the support and participation of intergovernmental organizations, governments, public authorities, and other public and private bodies fighting doping in sport. The main aim of WADA is to protect athletes, promote the values of clean sport, and preserve the spirit of sport internationally. As the key role, WADA is developing, harmonizing, and coordinating the anti-doping rules and policies across all sports and countries. Basic activities of the agency involve scientific and social science research, education, investigation, development of anti-doping capacity, and monitoring of compliance with the World Anti-Doping Program. WADA is composed and funded by the sport movement and governments of the world (WADA 2023a). WADA developed the Anti-Doping Administration & Management System (ADAMS) as a secure and cost-free web-based system that centralizes doping control-related information. The aim of ADAMS is to allow a safe storage and sharing of the anti-doping data among relevant organizations. The system promotes efficiency, transparency, and effectiveness in all anti-doping activities (WADA 2023b). The document harmonizing anti-doping policies, rules, and regulations within sport organization and among public authorities around the world is termed as the World Anti-Doping Code (Code). The aim of the Code is to harmonize anti-doping policies and ensure the same standards for all athletes. The code works in conjunction with eight international standards to promote consistency among anti-doping organizations in various areas (WADA 2023c). The Athlete Biological Passport (ABP) has been introduced to monitor selected biological variables over time that indirectly reveal the effects of doping, rather than attempting to detect the doping substance or method itself. Thus, ABP primarily reflects the changes in biological markers of an athlete instead of relying upon the detection of particular prohibited drug (WADA 2023d). The ABP program utilizes a combination of statistical analysis of laboratory results and data of individual passports to identify athletes or specific samples that require further attention. Currently, ABP program is composed of two Modules. The first, known as Hematological Module, has been introduced in December 2009 to identify oxygen transport enhancement such as erythropoiesis-stimulating agents, any form of blood transfusion or manipulation. The Hematological Module is based on the evaluation of biomarkers in an athlete's blood sample. The second, known as Steroidal Module, has been introduced in January 2014 to identify exogenously administered anabolic androgenic steroids or other anabolic agents. The Steroidal Module is based on the evaluation of biomarkers in an athlete's urine sample. In addition, the third, known as Endocrine Module, is now under development via WADA's 2020–2024 Strategic Plan with aim to enhance detection of misused hormones such as growth hormone (WADA 2023e). Implementation of two existing Modules and development of the novel Module proves that ABP is a dynamic anti-doping tool. Generally, the main purpose of ABP is to provide anti-doping program more effective. Even with advanced techniques the evaluation of misused compounds that are naturally produced by human body is difficult. The long-term monitoring of athlete's biomarkers in blood and urine represents a useful tracking system to reveal abnormalities associated with potential doping. However, the ABP data have to be interpreted with caution because there are still several factors which may provide false results (Mahendru et al. 2020).

The prohibited list reported by WADA is a mandatory document, which is used as an international standard to identify prohibited substances and methods in sports. The list is annually updated, following an extensive consultation process, and represents one of the most important parts of harmonization globally across the anti-doping movement. Prohibited list categorize prohibited substance and methods into several sections and classes. It also identifies which substances and

Journal Pre-proof

methods are prohibited at all times (in- and out-of-competition), in-competition, or substances that are prohibited in particular sports (USADA 2019). The summary of all doping control samples analyzed and reported by WADA-accredited laboratories is provided annually in the Anti-Doping Testing Figures report (WADA 2023f). The report includes adverse analytical findings (AAFs); reports from WADA-accredited laboratories or other WADA-approved laboratories that, consistent with the International Standard for Laboratories (ISL) and related Technical Documents, identify in a sample the presence of prohibited substances, its metabolites, markers, or evidences of the use of prohibited methods; as well as atypical findings (ATFs); reports from WADA-accredited laboratories or other WADA-approved laboratories that require further investigation, as provided by the ISL or related Technical Documents, prior to the determination of AAFs (WADA 2021a). Before 2014, the summary of AAFs and ATFs has been reported combined in the Anti-Doping Testing Figures report however, since 2014, the annular reports by WADA involve both, AAFs and ATFs, summarized separately. The annual reports (2003–2021) are available on the official website of WADA (WADA 2022a).

As mentioned, the prohibited list reported by WADA is annually updated. In addition to the list and other documents, WADA also report the Summary of Major Modifications and Explanatory Notes. Implementation of these modifications and changes is necessary and allows WADA to explore and identify the latest doping trends. This ensures that any new or existing substance or method that may meet the criteria for addition to prohibited list is considered, so as to protect athlete health and the integrity of clean sport worldwide (WADA 2022b). Modifications in WADA involve reclassification within the section, for example, in 2022 tibolone was transferred from the group of other anabolic agents (S1.2) to the group of anabolic androgenic steroids (S1.1), because it reveals clinical effects on the androgen receptor, mainly due to its biotransformation to δ^4-tibolone, the metabolite which exhibits a potent androgenic effect (WADA 2022c). As well as reclassification among the sections, for example, in 2006 clenbuterol has been transferred from the section of β_2 agonists (S3) into the section of anabolic agents (S1) to the group of other anabolic agents (S1.2) (WADA 2006). Selected notable modifications are described below in following sections however, it is important to note that these modifications are out of scope of this review and their mention is only informative.

Presence of analytical targets involved in the prohibited list is evaluated by the WADA accredited laboratories. Anti-doping laboratories process collected biological fluids, usually the urine (50–60 mL) or in some cases also the blood (3–5 mL) or its components such as serum (1.5–2.5 mL). Biological samples intended for analysis are delivered to the laboratories anonymized with only limited information about the athlete, the discipline, and the type of test. Initially, the analytical anti-doping procedures were based on the detection/quantification of “markers of exposure”, represented by small-molecule xenobiotics. However, expanding of the prohibited list in following years increased demand for development of novel detection methods capable to detect novel doping strategies, previously considered as “invisible for detection”. Subsequently, novel developed methods also provided an improvement of detectability of targets that were previously considered as “visible for detection”. To ensure detection of different doping substances and methods incorporated in the prohibited list, the optimal combination of direct and indirect analytical methods is required. Direct methods targeting markers of exposure are effective to detect molecular structures significantly different from any endogenous substance and/or those that are present in biological fluids in a sufficient high concentration. Indirect methods targeting markers of effect are utilized to detect those substances that are undistinguishable from their endogenous analogs and/or those that are present in a low concentration (Botrè et al. 2020).

Journal Pre-proof

4 Drugs, substances and methods prohibited by WADA

The 2023 prohibited list includes following sections of substances and methods prohibited at all times (in- and out-of-competition): non-approved substances (S0); anabolic agents (S1); peptide hormones, growth factors, related substances, and mimetics (S2); β_2 agonists (S3); hormone and metabolic modulators (S4); diuretics and masking agents (S5); prohibited methods (M1-3). Substances and methods prohibited in-competition are involved in following sections: stimulants (S6); narcotics (S7); cannabinoids (S8); glucocorticoids (S9). Substances prohibited in particular sports are represented by the section of β-blockers (P1). The most commonly detected representants of these and previously involved sections are disclosed in following sub-chapters. In addition, the summary of findings in the sections, as reported in the Anti-Doping Testing Figures reports (2003–2021), are summarized in Table 1 and graphically depicted in Chart 1.

Table 1. Summary of sections involving ATFs and AAFs of drugs, substances and methods prohibited by WADA in all sports during 2003–2021. The table involves sections based on the 2023 prohibited list reported by WADA. Sections are ordered from left to right according to decreasing number of total ATFs and AAFs in ADAMS. The findings are referred according to original WADA reports in each year. The table involves annual summary and percentage of total findings as well as complete summary and percentage of each section from 2003 to 2021. It is important to note, that the authors found small discrepancies in the original data reports. These discrepancies are represented by 8 particular findings, this means that they not affect the data, however for the sake of completeness, the authors disclosing them in following sentences to provide the reader a comprehensive overview. Following discrepancies were found: S1 section in 2016 - table summarizing identified substances reported 117 ATFs (specified findings), however table of final summary reported 116 ATFs, the authors use data from table summarizing identified substances (117 specified findings); S1 section in 2018 - table summarizing identified substances reported 167 ATFs (specified findings), however table of final summary reported 168 ATFs, the authors use data from table summarizing identified substances (167 specified findings); S2 section in 2016 - table summarizing identified substances reported 463 ATFs (specified findings), however table of final summary reported 460 ATFs, the authors use data from table summarizing identified substances (463 specified findings); S2 section in 2021 - table summarizing identified substances reported 80 ATFs (specified findings), however table of final summary reported 79 ATFs, the authors use data from table summarizing identified substances (80 specified findings); S9 section in 2019 - table summarizing identified substances reported 231 AAFs (specified findings), however table of final summary reported 230 AAFs, the authors use data from table summarizing identified substances (231 specified findings); S9 section in 2020 - table summarizing identified substances for ATFs (specified findings) was not included in the report, however table of final summary reported 1 ATF, the authors cannot determine specificity of the finding thus this one particular finding was not incorporated in the summary.

	Anabolic agents (S1)	Stimulants (S6)	Diuretics and masking agents (S5)	Cannabinoids (S8)	Glucocorticoids (S9)	β_2 agonists (S3)	Peptide hormones, growth factors, related substances, and mimetics (S2)	Hormones and metabolic modulators (S4)	Narcotics (S7)	β-Blockers (P1)	Previously involved sections and others	Prohibited methods (M1-3)	Non-approved substances (S0)	Total (annually)	Percentage of total (annually)
2003	872	516	142	378	286	297	79	6	26	30	82	2		**2,716**	**3.3%**
2004	1,191	382	157	518	548	381	78	8	15	25		2		**3,305**	**4.0%**
2005	1,864	509	246	503	325	609	162	21	17	42				**4,298**	**5.2%**
2006	1,966	490	290	553	282	631	42	30	16	28		4		**4,332**	**5.3%**
2007	2,322	793	359	576	288	399	41	18	21	27		6		**4,850**	**5.9%**
2008	3,259	472	436	496	316	350	106	29	28	31				**5,523**	**6.7%**
2009	3,297	325	273	399	265	303	100	50	24	38	5	5		**5,084**	**6.2%**
2010	3,374	574	396	533	234	209	86	75	20	30	9	6		**5,546**	**6.8%**
2011	3,325	718	368	445	274	225	125	70	20	21	5	4		**5,600**	**6.8%**
2012	2,279	697	322	406	365	131	181	74	26	13	5	1		**4,500**	**5.5%**
2013	3,320	530	393	188	330	138	202	93	43	25	8	1		**5,271**	**6.4%**
2014	1,812	474	389	73	252	122	192	145	34	25	49	3		**3,570**	**4.3%**

Journal Pre-proof

2015	1,845	531	428	127	215	115	1,876	155	27	19	3	1		**5,342**	**6.5%**
2016	1,921	568	499	110	184	172	572	721	54	14		4		**4,819**	**5.9%**
2017	1,929	577	614	154	225	157	146	321	73	12		1		**4,209**	**5.1%**
2018	1,990	605	590	141	285	164	197	351	24	18	1	12		**4,378**	**5.3%**
2019	2,147	611	677	130	232	153	194	362	30	20	7	9		**4,572**	**5.6%**
2020	878	225	213	42	69	67	88	127	4	14		2		**1,729**	**2.1%**
2021	1,097	348	320	92	125	68	178	248	23	9		5		**2,513**	**3.1%**
Total (2003-2021)	**40,688**	**9,945**	**7,112**	**5,864**	**5,100**	**4,691**	**4,645**	**2,904**	**525**	**441**	**174**	**68**	**0**	**82,157**	
Percentage of total	**49.5 %**	**12.1 %**	**8.7 %**	**7.1 %**	**6.2 %**	**5.7 %**	**5.7%**	**3.5%**	**0.6 %**	**0.5 %**	**0.2%**	**0.1 %**	**0.0 %**		

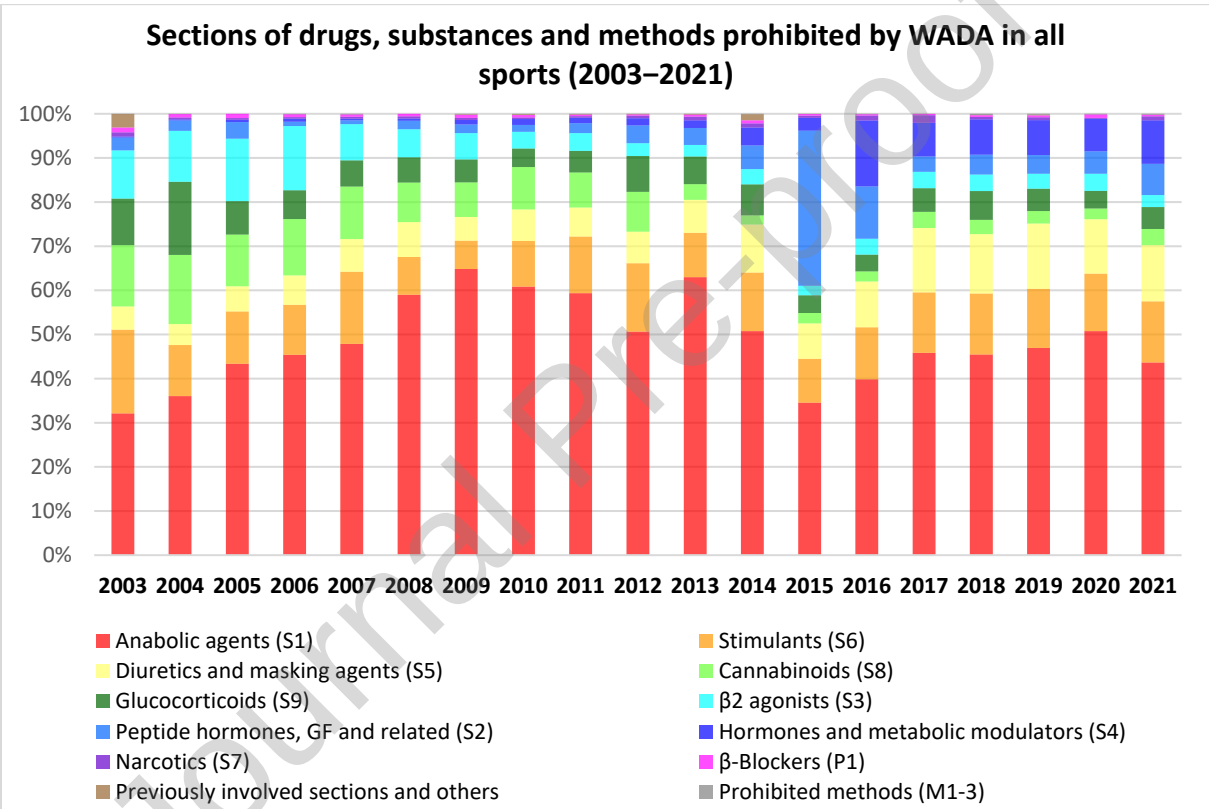


Chart 1. Sections of drugs, substances and methods prohibited by WADA in all sports (2003–2021), represented by ATFs and AAFs in ADAMS. Sections refer to the 2023 prohibited list reported by WADA. In the chart, the S2 section name is abbreviated, its name is originally reported as "Peptide hormones, growth factors, related substances, and mimetics" in the 2023 prohibited list.

4.1 Non-approved substances (S0)

This section of the 2023 prohibited list includes any pharmacological substance which is not incorporated in any subsequent sections of the list and is not currently approved by any governmental regulatory health authority for human therapeutic use. The section encompasses for example drugs under pre-clinical or clinical development, discontinued drugs, designer drugs, or substances approved only for veterinary use. All of these compounds are prohibited at all times (WADA 2022d). The non-approved substances section was included by WADA to the prohibited list in 2011 (WADA 2010). This section covers diverse structures, one particular example is body protection compound-157 (BPC-157), a pentadecapeptide consisting of Gly-Glu-Pro-Pro-Pro-Gly-Lys-Pro-Ala-Asp-Asp-Ala-Gly-Leu-Val with

anti-inflammatory and wound healing effects on multiple tissues and organs (Jung et al. 2022). Based on the summarization of the ATFs and AAFs reported in ADAMS between 2003 and 2021, none representative of this section has been reported by WADA within the period (Table 1).

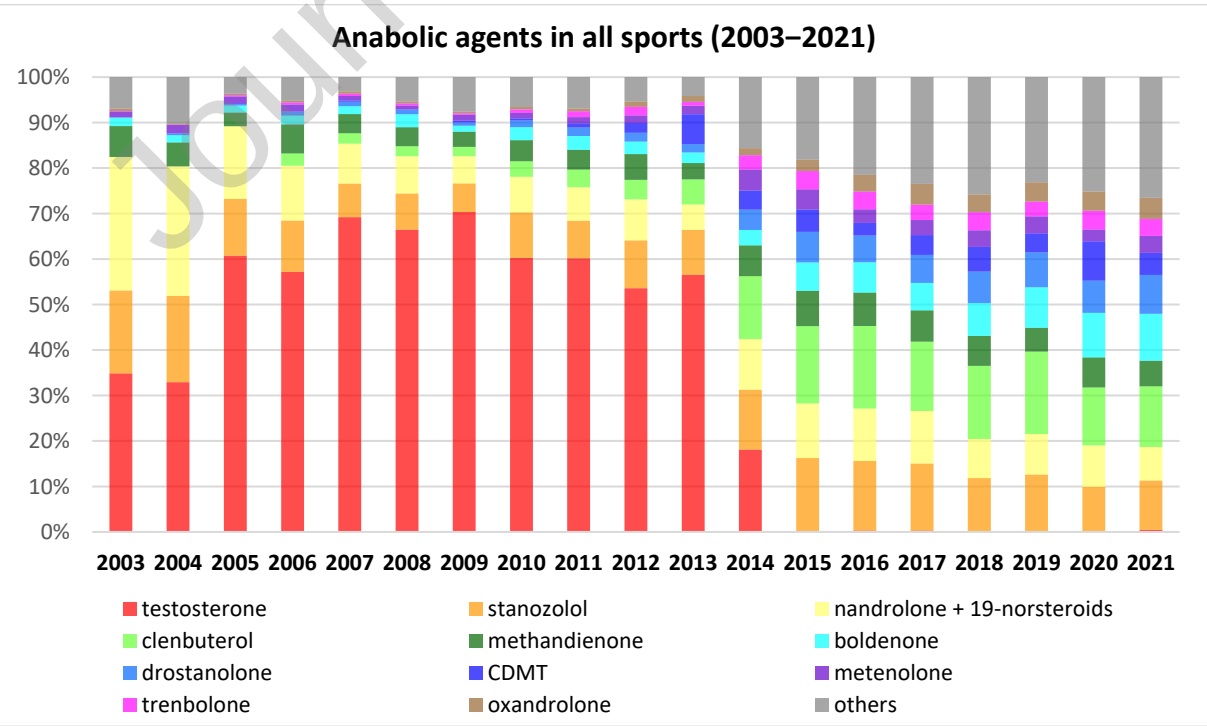
4.2 Anabolic agents (S1)

Anabolism (anabolic metabolism) is defined as the coordinated metabolic activity that allows cells to construct larger molecules from smaller units. It is an endergonic process, the process requiring energy, which is associated with the growth of muscles, bones and other body structures. The opposite exergonic process, in which large molecules are broken down into smaller structures, is known as catabolism (DeBerardinis and Thompson 2008). Regarding drug misuse in sport, the anabolic effects are associated with protein synthesis, muscle growth and erythropoiesis (Mottram and George 2000). Anabolic agents belong to substances prohibited at all times (in- and out-of-competition). Some of the substances incorporated to the S1 section may be found, without limitation, in medications used for treatment of various conditions such as male hypogonadism (WADA 2022d). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003-2021, the section of anabolic agents (S1) is the most common section of compounds misused as a potential doping in all sports, representing 49.5% of overall findings. In the section of anabolic agents (S1), the most frequently individual compounds are represented by: testosterone (40.6%), stanozolol (11.0%), nandrolone and 19-norsteroids (10.0%), clenbuterol (7.2%), methandienone (5.2%), boldenone (3.8%), drostanolone (3.0%), 4-chlorodehydromethyltestosterone (CDMT) (2.4%), metenolone (2.2%), trenbolone (1.7%), oxandrolone (1.6%), and others (11.2%). However, it is important to note that the implementation of Steroidal Module of the ABP in ADAMS since 2014 significantly affected annual laboratory reports provided by WADA-accredited laboratories. This phenomenon can be markedly observable on extensive decrease of testosterone findings since 2014. The Steroidal Module of the ABP is based on the measurement of steroidal markers in athlete's urine for the detection of doping with endogenous anabolic androgenic steroids, in particular testosterone, its precursors (for example, 4-androstenediol, androstenedione, and prasterone), its active metabolite (dihydrotestosterone), or its epimer (epitestosterone) (WADA 2021b). Furthermore, clenbuterol findings are incorporated in the section since 2006, the drug was previously (2003–2005) classified as β_2 agonist in S3 section. Thus, the complete summarization of clenbuterol findings within the whole time period is 3,076 findings (the sum of findings from S1.2 and S3 sections). However, to reflect reports from WADA-accredited laboratories the findings are interpreted according to original WADA reports. Consequently, the data have to be interpreted with caution. The annual laboratory reports of the most commonly detected ATFs and AAFs of anabolic agents reported during 2003–2021 by WADA are summarized in Table 2 and graphically depicted in Chart 2. According to the 2023 prohibited list published by WADA, anabolic agents involve two main subgroups of compounds: anabolic androgenic steroids (S1.1) and other anabolic agents (S1.2), both subgroups are disclosed below (WADA 2022d).

Table 2. Anabolic agents in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: significant decrease of testosterone ATFs and AAFs is caused by the implementation of Steroidal Module of the ABP in ADAMS since 2014. Clenbuterol has been incorporated into the section since 2006, previously classified as β_2 agonist (S3) section. Notes to findings: boldenone – includes findings reported as “boldenone IRMS” and “the GC/C/IRMS result for boldenone and/or boldenone metabolite(s) is consistent with an exogenous origin”; CDMT - reported by WADA as “dehydrochloromethyltestosterone”, herein reported under common name as 4-chlorodehydromethyltestosterone (CDMT); nandrolone and 19-norsteroids – findings include “nandrolone”, “19-norandrosterone”, and “19-norandrostenedione” as well as findings reported as “ 19-norandrosterone IRMS”, “the GC/C/IRMS result for 19-norandrosterone is consistent with an exogenous origin”, and “nandrolone metabolite(s)”; testosterone - including findings reported as "elevated T/E ratio" and

"elevated TE ratio (IRMS exogenous)" between 2008-2011 ,and "T/E ratio > 4" reported between 2012-2015. Abbreviations: CDMT, 4-chlorodehydromethyltestosterone; T/E, testosterone/epitestosterone.

	testosterone	stanozolol	nandrolone and 19-norsteroids	clenbuterol	methandienone	boldenone	drostanolone	CDMT	metenolone	trenbolone	oxandrolone	others	Total (annually)	Percentage of total (annually)
2003	304	159	256		59	16	1		11	1	5	60	**872**	**2.1%**
2004	392	226	339		63	19	5		22	1	3	121	**1,191**	**2.9%**
2005	1,132	233	298		56	28	6	4	28	5	6	68	**1,864**	**4.6%**
2006	1,124	222	237	53	126	37	17	2	30	9	7	102	**1,966**	**4.8%**
2007	1,607	172	203	53	98	40	23	7	23	11	12	73	**2,322**	**5.7%**
2008	2,166	259	266	73	136	93	32	3	28	15	14	174	**3,259**	**8.0%**
2009	2,319	208	197	67	110	43	24	13	44	10	11	251	**3,297**	**8.1%**
2010	2,034	337	263	116	156	97	49	16	40	26	18	222	**3,374**	**8.3%**
2011	2,001	275	244	129	145	100	63	30	46	43	20	229	**3,325**	**8.2%**
2012	1,221	240	205	98	129	63	44	49	38	43	27	122	**2,279**	**5.6%**
2013	1,877	329	184	183	120	77	62	217	62	28	43	138	**3,320**	**8.2%**
2014	328	239	201	251	123	61	81	76	84	57	28	283	**1,812**	**4.5%**
2015	4	296	221	314	143	115	124	91	82	74	47	334	**1,845**	**4.5%**
2016	6	294	221	349	141	128	113	53	58	75	70	413	**1,921**	**4.7%**
2017	6	284	223	294	133	116	119	84	64	65	88	453	**1,929**	**4.7%**
2018	1	235	171	320	131	143	138	108	73	79	78	513	**1,990**	**4.9%**
2019	4	267	191	389	113	191	165	90	80	69	92	496	**2,147**	**5.3%**
2020	1	86	80	112	58	86	62	76	23	36	37	221	**878**	**2.2%**
2021	5	119	81	146	62	113	94	53	41	41	51	291	**1,097**	**2.7%**
Total (2003-2021)	**16,532**	**4,480**	**4,081**	**2,947**	**2,102**	**1,566**	**1,222**	**972**	**877**	**688**	**657**	**4,564**	**40,688**	
Percentage of total	**40.6%**	**11.0%**	**10.0%**	**7.2%**	**5.2%**	**3.8%**	**3.0%**	**2.4%**	**2.2%**	**1.7%**	**1.6%**	**11.2%**		



Journal Pre-proof

Chart 2. Anabolic agents in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: significant decrease of testosterone ATFs and AAFs is caused by the implementation of Steroidal Module of the ABP in ADAMS since 2014. Clenbuterol has been incorporated into the section since 2006, previously classified as β_2 agonist (S3) section.

4.2.1 Anabolic androgenic steroids (AAS) (S1.1)

An androgen, also termed as androgenic hormone, is natural or synthetic steroid compound that binds to androgen receptors, expressed in myosatellite cells that are the precursors of skeletal muscle cells, to stimulate or control the development or maintenance of male characteristic in vertebrates. Although circulating concentrations of androgens are 20-to 25-fold lower in women in comparison with men, androgens are the precursors of all estrogens, the female sex hormones. The primary and most well-known androgen is the primary sex hormone in males, testosterone (Horstman et al. 2012; Rao 2014). Testosterone upon antagonizing the androgen receptor induces gene expression that causes growth and development of male reproductive tissues and organs such as testes and prostate. In addition, testosterone promotes increased growth of secondary sexual characteristics such as muscle and bone mass, and the growth of body hairs (Patel et al. 2022).

Anabolic androgenic steroids (AAS) mimic the physiological effect of testosterone, acting primarily via androgen receptors (Sessa et al. 2018). The history of AAS is associated with the primary sex hormone in males, testosterone. In 1935, the male steroid hormone was isolated and characterized by Ernst Laqueur, the compound has been named testosterone (David et al. 1935). In the same year, the chemical synthesis of testosterone was simultaneously reported by Adolf Butenandt (Butenandt and Hanisch 1935) and Leopold Ruzicka (Ruzicka and Wettstein 1935). In 1940s, testosterone found an application in medicine, for example in treatment of depression (Altschule and Tillotson 1948). In addition, numerous testosterone derivatives having unique properties were synthetized in these years (Ding et al. 2021). In 1950s, athletes and bodybuilders popularized these medications with the aim to enhance athletic performance or improve aesthetics. In 1960s, the prevalent utilization of AAS in athletics resulted in commencing screening programs for AAS and related compounds, by IOC. Since 1974, the use of AAS has been banned (Wu and Kovac 2016). In 1970s, effective methods of AAS detection became available. However, earlier methods of detection were not able to distinguish between synthetic AAS and the endogenous steroid, testosterone. In 1984, this problem was partially overcome using gas chromatography and mass spectrometry for the scanning of AAS. The evaluation of testosterone/epitestosterone (T/E) ratios were introduced to indicate testosterone misuse. However, in spite of widespread screening the misuse of AAS continued further (Mottram and George 2000). Indeed, testosterone and its precursors represent the most misused anabolic agents of nowadays. Their detection is articulated in two different levels involving the longitudinal evaluation of the urinary androgenic steroid profile using gas chromatography mass spectrometry (GC-MS), and the evaluation of steroid origin based on the differences in the isotopic abundance using gas-chromatography/combustion/isotope ratio mass spectrometry (GC/C/IRMS). This method evaluates the different isotopic abundance of ^{13}C in steroids of endogenous and synthetic origin. Endogenous steroids, in comparison to synthetic steroids, usually contain higher levels of ^{13}C isotope, as the result of metabolic processes (de la Torre et al. 2021). Novel methods for the detection of AAS are still under development (Albertsdóttir et al. 2021). In medicine, AAS are used to treat specific human diseases and disorders, such as hypogonadism or sexual impotence in men or breast cancer in women (Tauchen et al. 2021).

Use and abuse of AAS is frequently linked with various health damage effects on different organs or organ systems (Sessa et al. 2018). In cardiovascular system, AAS-related adverse effects

Journal Pre-proof

include coronary heart disease, cardiomyopathy, hypertension, and decrease of high-density lipoprotein (HDL) cholesterol levels in plasma. In central nervous system (CNS), these effects are associated with emotional lability, aggressive behavior, violence, nervousness, depression, insomnia, headache, anosmia, body pain and others. In gastrointestinal system, side effects of AAS include gingivitis, mouth irritation, decreased appetite, dysgeusia, and more. In genitourinary system, adverse effects are associated with increased levels of prostate-specific antigen, benign prostatic hyperplasia, testicular atrophy, and other side effects. In musculoskeletal system, undesired condition can involve myalgia or limb pain. In integumentary system, the adverse effects mediated by AAS can be associated with skin blister, skin rash, contact dermatitis, and acne (Ganesan et al. 2022). Illicit use of AAS as performance enhancers in sport is also associated with sudden cardiac death (Torrisi et al. 2020). AAS are typically used in patterns known as “cycling”, in which multiple doses of AAS are taken over a specific period of time, followed by time period without use of AAS. Afterwards, the usage of AAS starting again. People misusing AAS often “stacking” the drugs, using two or more different AAS, combining oral and/or injectable types of the steroids, even using steroids designed for veterinary purposes. Another pattern associated with misuse of AAS is commonly referred to as “pyramiding”, when doses of the steroid gradually increasing, then decreasing over the period of the cycle. This cycle can be followed by a second cycle in which the training continues without using of the steroids. Interestingly, another technique termed as “plateauing” consists of the integrated or exclusive use of other steroids, with the aim to avoid the onset of tolerance. However, the efficiency of these methods is still debatable (Favretto et al. 2022).

Structurally, AAS involve more than 1000 compounds related to testosterone. Many of the AAS members have been designed to improve bioavailability of testosterone. This resulted in three main classes of AAS, commonly labeled as A, B, and C. However, due to a large number of prepared AAS derivatives, various AAS members belong to more than one class or involve additional modifications. Class A represents 17β-esters of testosterone having a higher lipid solubility if compared with testosterone. The AAS members of class A are administered intramuscularly and have increased androgenic properties. Class B consists of 17α-alkylated testosterone derivatives that are administrated orally. The members of class B exhibit weaker potency in comparison to class A compounds. Moreover, hepatotoxic properties have been observed in class B members. Class C comprises of testosterone derivatives alkylated on the A, B, or C ring of the steroid scaffold, that can be administered orally. The bioactivity of class C members is similar to those of class B (Hall and Hall 2005; van Amsterdam et al. 2010).

Testosterone represents the most misused AAS in all sports, based on the summarization of the ATFs and AAFs in S1 section, reported by WADA-accredited laboratories during 2003–2021 (Table 2). Testosterone, the primary sex hormone and anabolic steroid in males, antagonizes the androgen receptor which results in the induction of gene expression (Ashique et al. 2022). The hormone regulates distinct functions at different stages of life, such as formation of the male phenotype during embryonic development or manifestation of second sexual characteristics during puberty (Saudan et al. 2006). Testosterone can be used to treat symptoms of sexual dysfunction in men. Further potential benefits involve improvement of libido, increase of bone mass as well as sense of being well (Ashique et al. 2022). It is an endogenous steroid, synthetized from cholesterol, however only a small portion of the hormone is free and active. Most of the testosterone in blood circulation binds to sex hormone binding globulin (SHBG) providing an inactive form of the hormone which is protected from metabolic degradation. Biotransformation of testosterone involves phase I and II metabolism. For example, the metabolism of testosterone involves aromatase-promoted conversion to estradiol or 5α-reductase-promoted conversion to dihydrotestosterone which is a more potent derivative of testosterone (Durdiakova et al. 2011). The measurement of concentrations and/or ratios of specific steroid markers

Journal Pre-proof

in athlete's urine is a valuable tool to determine the urinary steroid profile. This provides an information about possible testosterone misuse (WADA 2021b). Testosterone can be administered by injection, pill, topical gel, patch, or implant. The side effects of testosterone include polycythemia and acne in men, and hepatotoxicity, virilization and acne in women (Margo and Winn 2006).

Stanozolol, the second commonly found AAS in all sports in S1 section (Table 2), is a synthetic 17α-alkylated derivative of testosterone. The drug is commonly used to treat hereditary angioedema (Tse et al. 2017). Stanozolol is also used in veterinary, especially in dogs (Skoupá et al. 2022). The alkylation at C17 position allows the drug to be taken orally, because this modification markedly reduces its metabolism by the liver. Incorporation of a pyrazole ring enables to favor anabolic over androgenic effects. As a result, stanozolol reveals less physiologic side effects caused by androgens such as virilization, acne, frontal baldness, and psychological and behavioral chances including depression and insomnia (Carson et al. 2015). In addition, a pyrazole core attached to A-ring of the steroid scaffold provides selected specific metabolic products, such as 3'-hydroxy-stanozolol, 4β-hydroxy-stanozolol, 16β-hydroxy-stanozolol, epistanozolol, and others. The unique metabolic profile of stanozolol facilitate to detect its misuse in sport (Balcells et al. 2017). The adverse effects associated with stanozolol involve menstrual irregularities, hirsutism, weight gain in women as well as acne and mood changes (Maini et al. 2014). In addition, stanozolol exhibits hepatotoxic effects regardless of its application route (oral or intramuscular) (Büttner and Thieme 2010).

Nandrolone and 19-norsteroids, the third most commonly found compounds among AAS in all sports during 2003–2021 in S1 section (Table 2), are mainly represented by nandrolone and its metabolite 19-norandrosterone. Nandrolone, also known as 19-nortestosterone, is a derivative of testosterone devoid of C19 methyl group. Due to its improved anabolic properties, nandrolone found an utilization in treatment of chronic diseases associated with chronical state such as cornea healing, burns, and osteoporosis (Agriesti et al. 2020). However, it is no longer recommended for the treatment of osteoporosis. Nandrolone is commonly utilized in the form of different esters, such as phenpropionate or decanoate. Type of ester has a marked influence on the pharmacokinetics. The drug is administered by intramuscular injection (Furman 2017). Metabolites of nandrolone that can be identified in human urine are represented by 19-norandrosterone, 19-noretiocholanolone and 19-norepiandrosterone (Ayotte 2006). The adverse effects are expected to be similar to other AAS (Furman 2017).

Methandienone, also known as methandrostenolone, is a 17α-alkylated derivative of testosterone with incorporated double bond between C1 and C2 positions of steroidal A-ring. It was synthetized to improve oral bioavailability and attenuate side effects of testosterone. The drug is administered orally. Biotransformation of methandienone involve phase I and II metabolism, such as 17-epimerization providing epimethandienone; hydroxylations providing 6β-hydroxy-methandienone, 16-dihydroxy-methandienone and 6β,16-dihydroxy-methandienone; hydrogenation of A-ring resulting in 17β-hydroxy-17α-methyl-5β-androsta-1-en-3-one; dehydrogenation of B-ring affording 6-ene-methandienone; and other reactions. The metabolites are excreted mainly via urine (Mazzarino et al. 2018). Adverse effects of methandienone are associated with cholestasis of the intrahepatic bile ducts which can result in elevated aminotransferases, clinical jaundice, and hyperbilirubinemia (Rosenfeld et al. 2011).

Boldenone is an AAS which differs from testosterone only in additional double bond between C1 and C2 positions of steroidal A-ring. Boldenone undecylenate, an ester of boldenone, is a veterinary drug used to improve growth and feed conversions in veal calves as well as to increase muscle size. Hence, the drug can be misused indirectly via consumption of meat from animals treated with boldenone esters, or directly via voluntary intramuscular administration of an athlete (Neamat-Allah

Journal Pre-proof

2014). The use of boldenone is forbidden in humans, food-producing animals, and for purposes of doping in racehorses (Skoupá et al. 2022). However, boldenone and its metabolites can be naturally produced in a variety of species that were not treated with the drug. For example, boldenone can be formed from phytosterols present in vegetable fat (De Brabander et al. 2004). Biotransformation of boldenone in human body provides glucuronide and sulfate conjugates as well as 5β-androst-1-en-17β-ol-3-one, as a product of hydrogenation, and other derivatives (Wu et al. 2015). The data reporting health risks associated with boldenone use are still limited (Behairy et al. 2021).

Further AAS reported between 2003 and 2021 by WADA-accredited laboratories in all sports in S1 section are represented by: drostanolone, 4-chlorodehydromethyltestosterone (CDMT), metenolone, trenbolone, oxandrolone, mesterolone (469 findings in total, but not shown in Table 2, incorporated among "others"), and others. Structures of the most commonly found AAS by WADA in S1 section as a potential doping are shown on Figure 1.

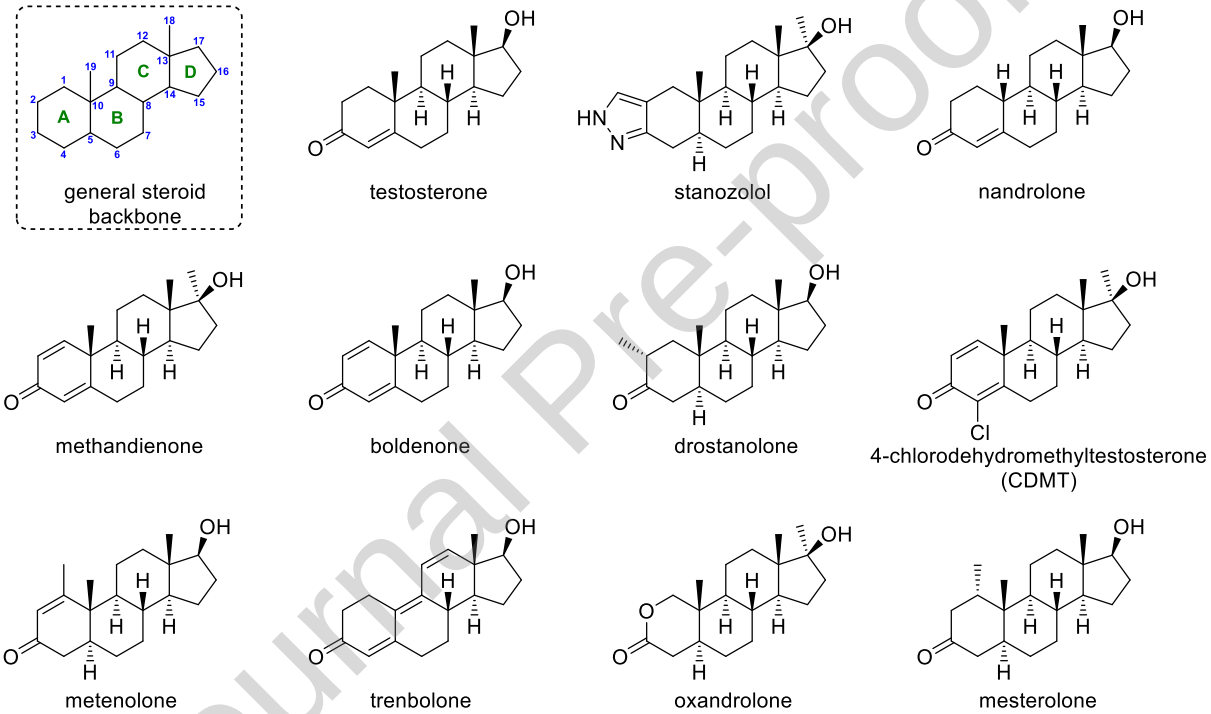


Figure 1. Structures of the most misused anabolic androgenic steroids (AAS) in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS. General steroid backbone (top-left) illustrates scaffold composting of A-D rings (green letters) with atom numbering (blue numbers).

4.2.2 Other anabolic agents (S1.2)

This class of compounds comprise of additional drugs with anabolic properties that are not incorporated in AAS. Several drugs of this class are known as selective androgen receptor modulators (SARMs), that exhibit ideal anabolic properties such as anabolic steroids with reduced androgenic effects (Kintz 2022). SARMs have been designed to target the androgen receptor more specifically with minimizing off-target effects. In addition, SARMs are investigated as a promising treatment for a wide range of medicinal challenges (Giovanelli and Quinton 2022). Potential SARMs-mediated medicinal applications involve treatment of osteoporosis, stress-induced urinary incontinence, muscular dystrophy, cachexia, Alzheimer's disease, prostate cancer, breast cancer and others (Christiansen et al. 2020). Many of the SARMs are available for purchase online, however the verifiability of their source can be questionable. Despite no SARMs have been clinically approved to date, a potential for their

Journal Pre-proof

abuse is significant among competitive athletes. A potential misuse of the substances has led to prohibition of SARMs in all sports, by WADA since 2008 (Solomon et al. 2019). Interestingly, more than half of SARMs users experienced side effects such as mood swings, decreased testis size, and acne. These effects are in accordance with those reported for anabolic steroid usage (Efimenko et al. 2022). Besides SARMs, class of other anabolic steroids involves further compounds with different mode of action.

Clenbuterol is the most detected drug belonging to the class of other anabolic agents (S1.2), as reported by WADA-accredited laboratories during 2003–2021 in all sports (Table 2). Clenbuterol is an agonist of β_2-adrenergic receptor. Based on the mode of action, clenbuterol was previously (2003–2005) classified in the section of β_2 agonists (S3) of the prohibited list, however since 2006 the drug has been reclassified and inserted into the section of anabolic agents (S1) to the group of other anabolic agents (S1.2) (WADA 2006). For a sake of completeness, the sum of 2,947 clenbuterol findings reported in S1 section since 2006 and 129 findings of the drug reported during 2003–2005 in S3 section results in a total of 3,076 clenbuterol findings within the whole time period. The summarization only slightly increases the total number of clenbuterol findings. In 1977, clenbuterol was marketed as an asthma reliever but due its adverse cardiac effects it is no longer applied for human use in most countries (Jessen et al. 2020). In addition, clenbuterol promotes muscle growth and reduces fat deposition in animals. The use of clenbuterol to improve growth in livestock is banned in many countries. The drug can be deposited in muscle or liver which can lead to adverse effects in consumers (Guo et al. 2022). In the United States (US), the drug is not approved for usage in humans (Bonney et al. 2019). It was approved by US Food and Drug Administration (FDA) as a prescription-only drug for the treatment of inflammatory airway disease and airway obstruction in horses (Rankins et al. 2022). However, due to its potent anabolic effects clenbuterol is misused by athletes as a doping agent (Slíž et al. 2022). Adverse effects associated with extensive intake of clenbuterol include severe acute poisoning, hyperthyroidism symptoms, and arrythmia (Cong et al. 2019). Besides clenbuterol voluntary misuse, the drug also can be detected after consumption of food contaminated with clenbuterol. This still remains a real risk to athletes leaving in the countries where clenbuterol is not banned to improve growth of livestock (Mottram and Chester 2018). Despite its mode of action, clenbuterol is classified by WADA as other anabolic agent (S1.2) (Slíž et al. 2022).

Except of clenbuterol, total findings (2003–2021) of other individual compounds belonging to S1.2 section are significantly lower and represented by: enobosarm (also known as ostarine), ligandrol, non-specified types of SARMs, zilpaterol, testolone, zeranol, andarine, and others. The structures of other anabolic agent representants are depicted on Figure 2.

Journal Pre-proof

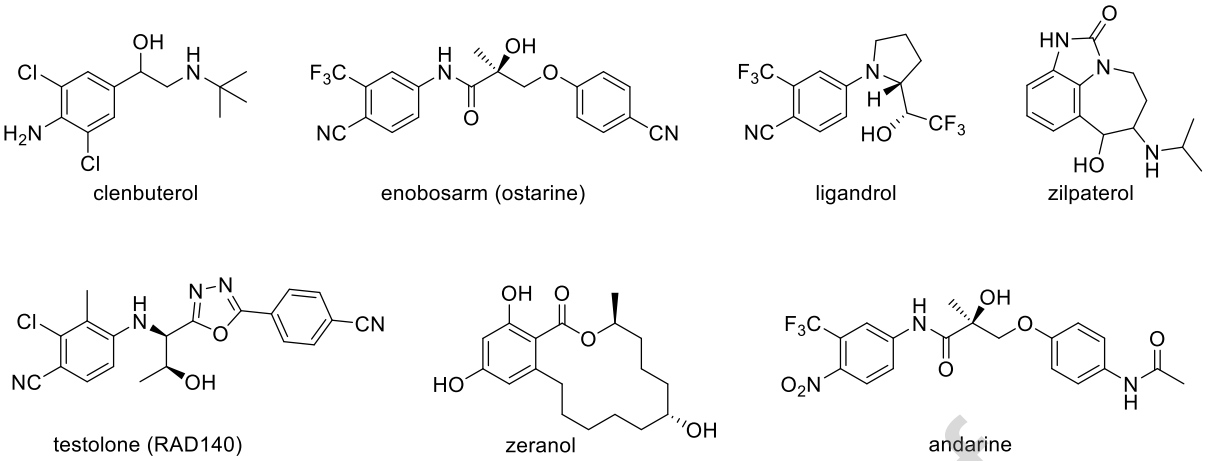


Figure 2. Structures of the most misused other anabolic agents in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.3 Peptide hormones, growth factors, related substances, and mimetics (S2)

This section of the 2023 prohibited list includes substances prohibited at all times (in- and out-of-competition). Substances of the section can be classified into three main groups as follows: 1.) erythropoietins and agents affecting erythropoiesis; 2.) peptide hormones and their releasing factors; 3.) growth factors and growth factors modulators. Findings of each group are reported together within the section S2. Some of the substance involved in the S2 section may be found, without limitation, in medications used for the treatment of various diseases or conditions such as anemia, male hypogonadism, growth hormone deficiency and others (WADA 2022d). According to ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, S2 section represents the seventh commonly detected section of substances misused as a potential doping in all sports, representing 5.7% of overall findings. The summarization of ATFs and AAFs reported by WADA in S2 section in all sports between 2003 and 2021 provides following order of the most common occurred findings: luteinizing hormone (LH) (53.8%), erythropoietin (EPO) (20.1%), human chorionic gonadotrophin (hCG) (17.7%), darbepoetin (dEPO) (1.9%), human growth hormone (hGH) (1.9%), methoxy polyethylene glycol-epoetin beta (MPG-EPO) that is also known as continuous erythropoietin receptor activator (CERA) (1.5%), ibutamoren (1.0%), growth hormone releasing peptide (GHRP)-2 (0.7%), GHRP-6 (0.6%), ipamorelin (0.3%), and others (0.4%). However, as mentioned above the data have to be interpreted with caution. For example, findings of EPO have been reported during the whole period in a range of 15–90 per year. Similarly, findings of LH were usually represented in ones or tens of occurrences. However, this trend has been disrupted in two years, particular in 2015 and 2016 when 1,747 and 453 findings were reported, respectively. Noteworthy, majority of this findings belong to ATFs. Further it is important to mention that in 2015 WADA reported the guidelines for reporting & management of hCG and LH findings which declares that detection of total LH below 1 international unit/L (IU/L) may serve as an evidence of LH suppression due to use of exogenous androgens or indirect androgen doping. Therefore, detection of LH values below 1 IU/L should be reported as ATFs for LH and interpreted in parallel with the values obtained for the markers of the steroid profile and any previous LH measurements (WADA 2015). Later, in 2017, the technical document TD2017CG/LH reported by WADA describes, among other things, that in cases of analytical testing for LH when the LH is not detectable, the laboratory shall report the finding as “the concentration of LH was less than the limit of detection (LOD)” and specify the applicable LOD (WADA 2017a). The complete annual laboratory reports of ATFs

Journal Pre-proof

and AAFs for peptide hormones, growth factors, related substances, and mimetics (S2) in all sports during 2003–2021 are summarized in Table 3, and graphically expressed in Chart 3.

Table 3. Peptide hormones, growth factors, related substances, and mimetics in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: significant increase of ATFs for LH in 2015 and 2016 is associated with changes incorporated by WADA. Notes to findings: erythropoietin (EPO) – includes findings reported as "EPO" and "EPO (active)"; methoxy polyethylene glycol-epoetin beta (CERA) – includes findings reported as "Micera (CERA)"; human chorionic gonadotropin (hCG) – includes findings reported as "hCG" and "gonadotrophins (hCG)"; luteinizing hormone (LH) – includes findings reported as "LH" and "gonadotrophins (LH)". Abbreviations: dEPO, darbepoetin; EPO, erythropoietin; GHRP, growth hormone releasing peptide; hCG, human chorionic gonadotrophin; LH, luteinizing hormone; MPG-EPO, methoxy polyethylene glycol-epoetin beta.

	luteinizing hormone (LH)	erythropoietin (EPO)	human chorionic gonadotrophin (hCG)	darbepoetin (dEPO)	human growth hormone (hGH)	MPG-EPO	ibutamoren (MK-677)	GHRP-2 (pralmorelin)	GHRP-6	ipamorelin	others	Total (annually)	Percentage of total (annually)
2003	7	51	14	7								**79**	**1.7%**
2004	16	38	24									**78**	**1.7%**
2005	3	15	143	1								**162**	**3.5%**
2006	2	17	22	1								**42**	**0.9%**
2007	2	22	15	2								**41**	**0.9%**
2008	1	51	46	2		5					1	**106**	**2.3%**
2009	6	56	25	4	1	8						**100**	**2.2%**
2010	5	36	33	8	3	1						**86**	**1.9%**
2011	23	44	47	5	6							**125**	**2.7%**
2012	29	45	93	4	8						2	**181**	**3.9%**
2013	15	56	124	5		2						**202**	**4.3%**
2014	6	59	109	7	2	3					6	**192**	**4.1%**
2015	1,747	42	56	5	7		2	7	3	3	4	**1,876**	**40.4%**
2016	453	48	17	4	11	20	4	6	4	1	4	**572**	**12.3%**
2017	12	69	15	3	3	22	11	5	2	2	2	**146**	**3.1%**
2018	61	73	12	13	9	6	3	11	8	1		**197**	**4.2%**
2019	41	90	14	7	14	2	13		10	3		**194**	**4.2%**
2020	30	33	2	4	5		7	2	2	2	1	**88**	**1.9%**
2021	42	90	11	6	18		8	2		1		**178**	**3.8%**
Total (2003-2021)	**2,501**	**935**	**822**	**88**	**87**	**69**	**48**	**33**	**29**	**13**	**20**	**4,645**	
Percentage of total	**53.8%**	**20.1%**	**17.7%**	**1.9%**	**1.9%**	**1.5%**	**1.0%**	**0.7%**	**0.6%**	**0.3%**	**0.4%**		

Journal Pre-proof

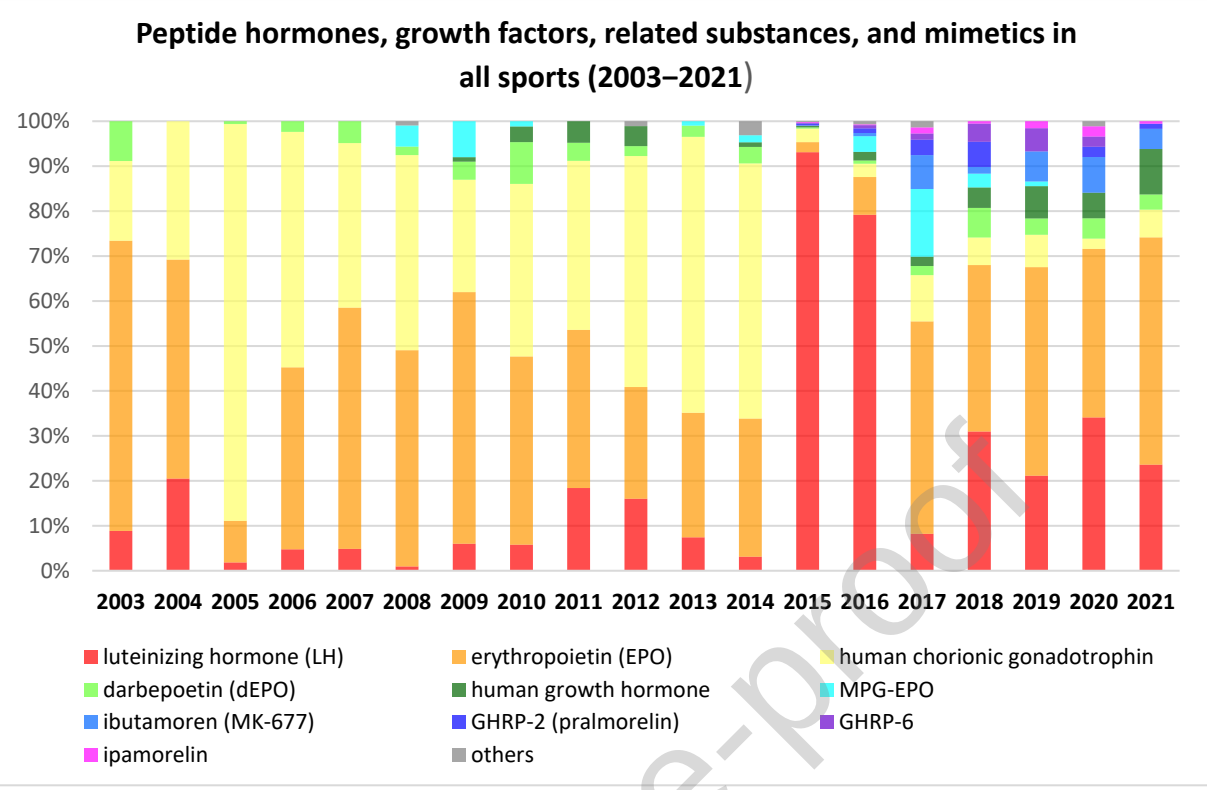


Chart 3. Peptide hormones, growth factors, related substances, and mimetics in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: significant increase of ATFs for LH in 2015 and 2016 is associated with changes incorporated by WADA. Abbreviations: GHRP, growth hormone releasing peptide; MPG-EPO, methoxy polyethylene glycol-epoetin beta.

4.3.1 Erythropoietins and agents affecting erythropoiesis (S2.1)

Erythropoiesis is a complex physiologic process associated with the proliferation and differentiation of hematopoietic stem cells into red blood cells (RBCs) (Klei et al. 2017). In adult mammals, erythropoiesis almost exclusively occurs within the bone marrow. RBCs represent the most abundant cell type in the blood with a concentration of about 5 million RBCs per µL of blood. In order to maintain concentration of RBCs in the blood, bone marrow produces 2 million RBCs per second. RBCs play a critical role in oxygen transfer to tissues. Dysregulation of erythropoiesis is associated with the development of anemia or polycythemia. A key regulator of erythropoiesis is an endogenous glycoprotein hormone erythropoietin (EPO) (Eggold and Rankin 2019). Among other agents affecting erythropoiesis belong recombinant versions of EPO, commonly termed as erythropoiesis-stimulating agents (ESAs), involving for example darbepoetin (dEPO) and methoxy polyethylene glycol-epoetin beta (MPG-EPO) (Schoener and Borger 2022). In addition, last generation of ESAs, such as EPO fusion protein (EPO-Fc) have a prolonged half-life, therefore require less frequent administration (Heiland et al. 2022). The EPO-Fc have been designed to treat anemia in humans by stimulating erythrocyte production. However, it can also be used by athletes in endurance sports as a performance-enhancing agent (Mesonzhnik et al. 2018). The EPO-Fc consists of monomeric or dimeric EPO and the dimeric Fc region of human immunoglobulin G molecules. The EPO-Fc have been prohibited by WADA since 2012 (Postnikov et al. 2015).

Erythropoietin (EPO), the second commonly detected hormone in S2 section (Table 3), is a 30.4-kDa glycoprotein composed of 165 amino acids (Jelkmann 2016). Human EPO was first isolated in 1977 from the urine of patients with aplastic anemia (Miyake et al. 1977). Its mode of action is based

on the interaction with EPO receptor (Elliott et al. 2008). Natural EPO, as the hormone inducing erythropoiesis and circulating concentration of RBCs, maintains delivery of oxygen to tissues. Recombinant human EPO, also known epoetin-α, has been developed to improve levels of circulating RBCs in patients with late-stage renal failure (Trinh et al. 2020). Indeed, recombinant EPO is clinically used to treat anemia in chronic kidney disease and other disorders (Lappin et al. 2021). In sport, the potential of EPO to increase oxygen uptake and thus increase endurance of an athlete resulted in abuse of EPO. In the late 1980s, recombinant human EPO became available in Europe, and in the early 1990s was banned its use in sport. The effect of EPO on athletic performance has been recently reviewed by Trinh *et al.*, and the authors concluded that there is low-to-moderate evidence suggesting that human recombinant EPO may be more beneficial than placebo. The authors noticed that more trials are needed to further elucidation of these effects in athletes (Trinh et al. 2020). Chronical administration of EPO, particularly at high doses, results in a high hematocrit which is associated with adverse effects such as hypertension and thromboembolism. In addition, this could further counteract neuroprotective and cardioprotective effects of EPO (Suresh et al. 2020).

Darbepoetin (dEPO), also known as novel erythropoiesis stimulating protein (NESP), the fourth commonly found substance in S2 section (Table 3), is also known as darbepoetin alpha is 37.1-kDa glycoprotein composed of 165 amino acids. The protein differs from human recombinant EPO only in positions of 5 amino acids, which provide two additional *N*-linked oligosaccharide chains. As the result of the structural changes, dEPO exhibits longer half-life, increased biological activity, and decreased receptor affinity than human recombinant EPO (Powell and Gurk-Turner 2002). In 2001, FDA approved dEPO for the treatment of anemia in patients with chronic renal faulire. In 2002, dEPO was approved by FDA to treat chemotherapy-related anemia in patients with cancer (Kanavos and Saka 2005). Its mode of action is mediated upon binding to EPO receptor, which results in increased production of RBCs (Powell and Gurk-Turner 2002). Similar as other ESAs, dEPO improves delivery of oxygen to the muscle tissues, thereby it can be misused by athletes as a doping (Yu et al. 2010). The most frequent adverse effects associated with clinical use of dEPO are represented by fatigue, diarrhoea, and edema. Less frequent adverse effects, but clinically significant, are represented by hypertension, seizures, trombotic events and death (Siddiqui and Keating 2006).

4.3.2 Peptide hormones and their releasing factors (S2.2)

In general, peptide hormones play an important role in regulation of energy homeostasis and metabolism. They control functions of gastrointestinal and cardiovascular systems, as well as energy expenditure and reproduction (Kołodziejski et al. 2021). Peptide hormones can be misused as doping agents by athletes. Their detection can be difficult, for example due to a short half-life in the blood which causes relative short time window for their detection in body fluids. Moreover, hormones such as human growth hormone (hGH) and human chorionic gonadotropin (hCG) can exist as a mixture of several isomers (Gómez-Guerrero et al. 2022). Based on the WADA-accredited laboratory reports, luteinizing hormone (LH), hCG, and hGH are the most commonly detected agents in this group (Table 3).

Luteinizing hormone (LH), the most commonly found hormone in S2 section (Table 3), is a heterodimeric glycoprotein comprised of α- and β-subunit. The α-subunit consists of 92 amino acids, while β-subunit contains 121 amino acids. The mode of action of LH is based on the binding to the luteinizing hormone/choriogonadotropin receptor (LHCGR), a member of the G protein-coupled receptors, which results in activation of adenylyl cyclase, then production of cyclic adenosine monophosphate (cAMP), followed by activation of protein kinase A (PKA) (Choi and Smitz 2014). In males, LH stimulates release of testosterone by the Leydig cells of the testes. In females, LH stimulates

Journal Pre-proof

steroid release from the ovaries, ovulation, and release of the progesterone after ovulation (Nedresky and Singh 2022). Recombinant human LH, also known as lutropin alpha, is used in a combination of fixed-ratio with recombinant human follicle stimulating hormone (FSH), also known as follitropin alpha, for the stimulation of follicular development (Abramova et al. 2019). However, due to ability to increase the production of testosterone, LH can be misused by male athletes as a doping. Therefore, LH is banned for male athletes in the 2023 prohibited list reported by WADA (WADA 2022d).

Human chorionic gonadotropin (hCG), third commonly detected substance in S2 section, is a heterodimeric glycoprotein comprised of α- and β-subunit. The α-subunit contains 92 amino acids and is identical with α-subunit of LH. The β-subunit of hCG involves 145 amino acids. As a result of differences in the structure and post-translational modifications, hCG is more stable with a longer circulating half-life in comparison to LH. The longer half-life of hCG as well as greater binding affinity of hCG to LHCGR results in higher biological activity of hCG in comparison with LH (Choi and Smitz 2014). Due to its structural similarity with LH, hCG acts as an LH agonist and stimulates production of testosterone from Leydig cells, moreover, hCG can also stimulate spermatogenesis (Hawksworth and Burnett 2019). Therefore, hCG is used in the treatment of male infertility (Fink et al. 2021). However, similar as LH, hCG can also be misused by athletes as a doping, to enhance muscle strength. Both, LH and hCG, are less effective than testosterone and anabolic steroids (Stenman et al. 2008). The hormone is banned for male athletes in the 2023 prohibited list (WADA 2022d). The side effects associated with clinical use of hCG involves headache, irritability, depression, and fatigue (Hawksworth and Burnett 2019).

Human growth hormone (hGH), fifth commonly found substance in S2 section, is a single-chain polypeptide composed of 191 amino acids produced by somatotropic cells with anterior pituitary gland. Originally, as implies its name, the hormone was found to be responsible for the regulation of growth during childhood. However, hGH regulates many other basal metabolic functions in the body (Brinkman et al. 2022). For example, hGH regulates production of insulin-like growth factor 1 (IGF-1), and both together, hGH and IGF-1, stimulate skeletal growth. In addition to muscle mass increase and bone formation, hGH also reduces fat mass and increases total body water (Höybye and Christiansen 2015). On the basis of above-mentioned properties, recombinant hGH found an utilization in the treatment of various growth disorders (Gou et al. 2022). Misuse of hGH in sport as a doping likely began in the early 1980s and became more frequent with the implementation of recombinant technology. Initially, expectation that hGH improves physical functions was based on its anabolic and lipolytic properties before any scientific evidence of benefit. Nowadays, the misuse of hGH in sport is widespread due to difficulty of detection, although many of its benefits are still unproven (Holt and Ho 2019). Recently, WADA established two procedures to detect abuse of recombinant hGH in sport. The hGH isoforms test utilizes the separate measurement of growth hormone isoforms using immunoassay. The hGH biomarkers test involves analysis of two growth hormone-responsive proteins, IGF-1 and procollagen III amino-terminal propeptide (P-III-NP), using either immunoassays and/or mass spectrometry-based techniques. However, both tests have their limitations. In particular, hGH isoforms test has the short window of opportunity whereas the hGH biomarkers test dependences on age and sex parameters and the more rapid elimination of IGF-1 compared to P-III-NP reducing retrospectivity. Despite these limitations both tests are applied in routine analysis in anti-doping laboratories (WADA 2017b).

4.3.3 Growth factors and growth factor modulators (S2.3)

Growth factors were initially defined as secreted biologically active substances that can affect cellular growth. However, more expanded definition of growth factors involves secreted substrates

Journal Pre-proof

that promote or inhibit mitosis or affect cellular differentiation. These substances commonly act as signaling molecules. Growth factors act on specific receptors located on the surface of cells which in turn transmit signals to other intracellular components and can result in altered gene expression (Stone et al. 2022). The growth factors and growth factor modulators banned by the 2023 prohibited list include for example: IGF-1, fibroblast growth factors (FGFs), hepatocyte growth factor (HGF), vascular endothelial growth factor (VEGF), and other growth factors or growth factor modulators affecting muscle, tendon or ligament protein synthesis/degradation, vascularization, energy utilization, regenerative capacity or fiber type switching (WADA 2022d). The ATFs and AAFs of IGF-1, a representant of this group, include only one finding reported in 2020.

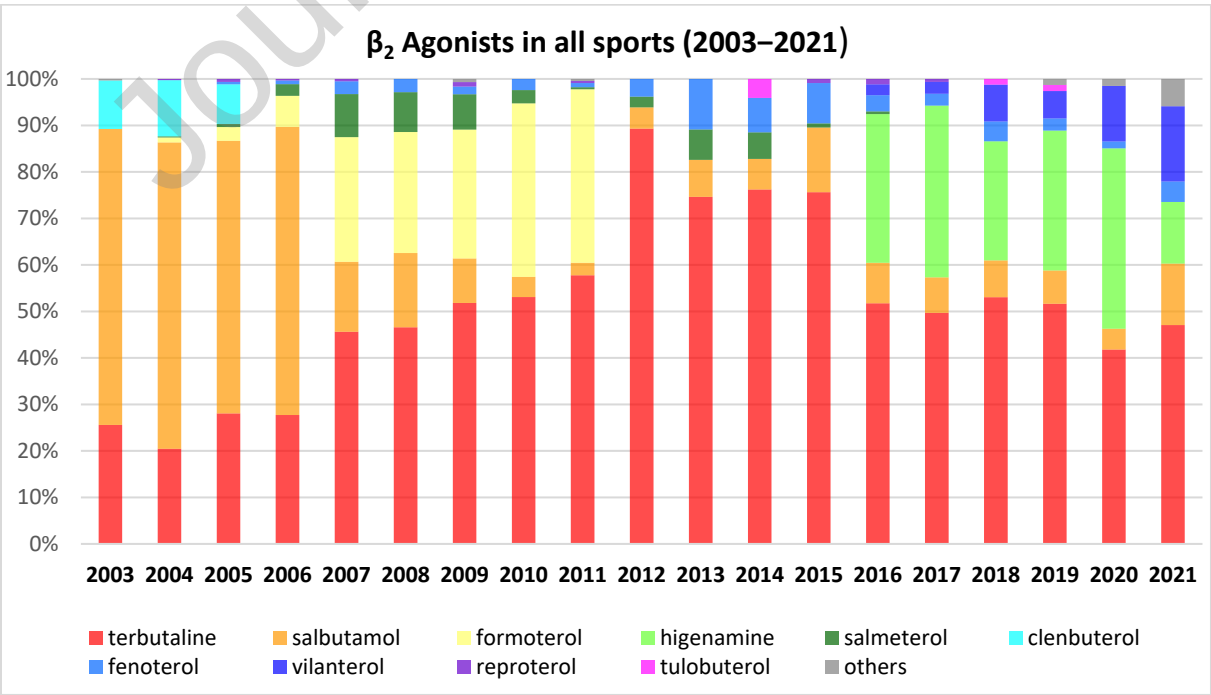
4.4 β_2 Agonists (S3)

β_2 Agonists, also known as β_2-adrenergic receptor agonists, are drugs primarily used to treat respiratory diseases such as bronchial asthma and chronic obstructive pulmonary disease (COPD). β_2 Agonists acting on the β_2-adrenergic receptor, these structures mimic the action of the endogenous catecholamines epinephrine and norepinephrine (Hsu and Bajaj 2022). The activation of β_2-adrenergic receptor results in initiation of signaling pathway involving adenylyl cyclase, which is followed by increase of intracellular cyclic adenosine monophosphate (cAMP) and activation of PKA, which in turn activates additional cellular targets and this results in airway smooth muscle relaxation (Billington et al. 2017). Based on the duration effect and onset of action, β_2 agonists are commonly classified into three classes. Short-acting β_2 agonists (SABAs) are have short therapeutic effect (3–6 hours) and are used for immediate symptomatic relief. Long-acting β_2 agonists (LABAs) exhibit longer therapeutic effect (about 12 hours) and they are used for sustained treatment. Finally, ultra-long-acting β_2 agonists (ultra-LABAs) exhibit even longer duration effect (up to 24 hours) and are also used for sustained treatment. Adverse effects of β_2 agonists involve desensitization of β_2-adrenergic receptor, or a lack of selectivity toward β_2-adrenergic receptor resulting in off-target effects mediated by α_1, α_2, or β_1-adrenergic receptors. The adverse effects of β_2 agonists are commonly associated with cardiac (arrythmia, cardiomyopathy, ischemia), metabolic, or musculoskeletal system (Hsu and Bajaj 2022). In addition to respiratory system, β_2 agonists further act on cardiovascular and metabolic systems. These compounds can also activate nonsteroidal anabolic processes that lead to muscle mass enhancement. β_2 Agonists are considered as potentially powerful doping agents due to their ability to stimulate receptors in the central and peripheral nervous system or tissues controlling a range of functions such as bronchodilation, anabolic processes, and enhancement of the anti-inflammatory effects of corticosteroids (Hackney 2018a). β_2 Agonists (S3) in the 2023 prohibited list are prohibited at all times (in- and out-of-competition) and involve all selective and non-selective β_2 agonists including all optical isomers. It is important to note that the prohibition of β_2 agonists by WADA involves several therapeutic exceptions. Some of the substances belonging to S3 section may be found, without limitation, in medications used for treatment of asthma and other respiratory diseases (WADA 2022d). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, S3 section represents sixth commonly detected section of substances misused as a potential doping in all sports, representing 5.7% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of β_2 agonists (S3) is represented by: terbutaline (43.4%), salbutamol (31.0%), formoterol (10.8%), higenamine (5.0%), salmeterol (3.0%), clenbuterol (2.7%) which is since 2006 reclassified into the section of anabolic agents (S1) to the group of other anabolic agents (S1.2), fenoterol (2.2%), vilanterol (1.0%), reproterol (0.4%), tulobuterol (0.2%), and others (0.2%). Findings of the commonly detected substances of the S3 section are summarized in Table 4, graphically expressed in Chart 4, and their structures are shown in Figure 3.

Journal Pre-proof

Table 4. β_2 Agonists in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: Notes: clenbuterol was previously (2003–2005) classified as β_2 agonist, however since has been reclassified into the section anabolic agents (S1) to the group of other anabolic agents (S1.2). Clenbuterol is characterized in the S1.2 subsection according to novel classification, however the statistics reflects reports from WADA-accredited laboratories within the time period.

	terbutaline	salbutamol	formoterol	higenamine	salmeterol	clenbuterol	fenoterol	vilanterol	reproterol	tulobuterol	others	Total (annually)	Percentage of total (annually)
2003	76	189				31					1	**297**	**6.3%**
2004	78	251	4		1	46			1			**381**	**8.1%**
2005	171	357	18		4	52	3		4			**609**	**13.0%**
2006	175	391	42		16		5		2			**631**	**13.5%**
2007	182	60	107		37		11		2			**399**	**8.5%**
2008	163	56	91		30		10					**350**	**7.5%**
2009	157	29	84		23		5		3		2	**303**	**6.5%**
2010	111	9	78		6		5					**209**	**4.5%**
2011	130	6	84		1		2		1		1	**225**	**4.8%**
2012	117	6			3		5					**131**	**2.8%**
2013	103	11			9		15					**138**	**2.9%**
2014	93	8			7		9			5		**122**	**2.6%**
2015	87	16			1		10		1			**115**	**2.5%**
2016	89	15		55	1		6	4	2			**172**	**3.7%**
2017	78	12		58			4	4	1			**157**	**3.3%**
2018	87	13		42			7	13		2		**164**	**3.5%**
2019	79	11		46			4	9		2	2	**153**	**3.3%**
2020	28	3		26			1	8			1	**67**	**1.4%**
2021	32	9		9			3	11			4	**68**	**1.4%**
Total (2003-2021)	**2,036**	**1,452**	**508**	**236**	**139**	**129**	**105**	**49**	**17**	**9**	**11**	**4,691**	
Percentage of total	**43.4%**	**31.0%**	**10.8%**	**5.0%**	**3.0%**	**2.7%**	**2.2%**	**1.0%**	**0.4%**	**0.2%**	**0.2%**		



Journal Pre-proof

Chart 4. β_2 Agonists in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: clenbuterol was previously (2003–2005) classified as β_2 agonist, however since has been reclassified into the section anabolic agents (S1) to the group of other anabolic agents (S1.2). Clenbuterol is characterized in the S1.2 subsection according to novel classification, however the statistics reflects reports from WADA-accredited laboratories within the time period.

Terbutaline, the most frequently detected representant of β_2 agonists (S3 section), is a synthetic agonist of β_2-adrenergic receptor involving resorcinol motif. In contrast to epinephrine and isoproterenol, the structure of terbutaline bears tertiary butyl group attached to the atom of nitrogen (Gruetter 2007). The structure involves one stereogenic center which provides two enantiomeric forms. Clinically, in the treatment of asthma terbutaline is widely used as a racemic mixture. The higher agonistic effect exhibits (*R*)-(-)-terbutaline in comparison with less active (*S*)-(+)-terbutaline. The use of racemic form exhibits no apparent interactions between the enantiomers (Tang and Zhang 2011). Terbutaline has a fast onset of action (<5 min) with a duration of effect about 4–6 hours, its therapeutic dose is 500 µg (up to 4 times per day) (Billington et al. 2017). Terbutaline belongs to SABAs and it was first introduced in late 1960s. The drug is used for the management of bronchospasm associated with asthma, bronchitis, emphysema, and COPD. Terbutaline has a low solubility, however its permeability is high, which means that it can cross the placenta or blood-brain barrier (BBB). The drug can be administered orally, parenterally, or inhaled (Sultan et al. 2023). In scientific community, there is still an ongoing debate about the effects of terbutaline in athletes. For example, in 2019, Molphy et al. concluded that therapeutic dosing of terbutaline does not lead to an improvement in 3-km running performance (Molphy et al. 2019). However, earlier in 2016, Kalsen et al. reported that inhalation of 15 mg of terbutaline resulted in increase in power output due to increased rates of glycogenolysis and glycolysis in skeletal muscles. Moreover, the author suggested that terbutaline may postpone development of fatigue (Kalsen et al. 2016b). Adverse effects associated with use of terbutaline are represented by headache, nausea, vomiting, anxiety, restlessness, weakness, sweating, chest discomfort and others (Sultan et al. 2023).

Salbutamol, the second commonly detected substance in the S3 section, is a structural derivative of terbutaline in which a motif of resorcinol (*meta*-benzenediol) is replaced with substituted *ortho*-(hydroxymethyl)phenol ring. Similar as terbutaline, structure of salbutamol also involves *N-tert*-butyl group as well as one stereogenic center. The pharmacological activity associates with (*R*)-salbutamol due to its binding activity to β_2-adrenergic receptor, while (*S*)-salbutamol is assumed to be inert in humans or even associated with adverse effects (Marques and Vale 2022). Salbutamol has a fast onset of action (<5 min) with a duration of effect about 3–6 hours, its therapeutic dose is 100–200 µg (up to 4 times per day) (Billington et al. 2017). Salbutamol, introduced in 1968, belongs to SABAs which is used as an alternative reliever in the treatment of asthma. The drug has a potent smooth muscle relaxant properties and it is ranked by the World Health Organization (WHO) as one of the most effective and safest medicines (Marques and Vale 2022). In sport, systematic intake of salbutamol is banned, however its inhalation for treatment of asthma is permitted with dosage restrictions. The exception for inhaled salbutamol in the 2023 prohibited list involves maximum 1600 µg over 24 hours in divided doses not to exceed 600 µg over 8 hours starting from any dose (WADA 2022d). Salbutamol in high doses can be misused in sport as a potential doping due to its effects to boost muscle efficiency and lung capacity, the drug improves the ability of athletes to sprint for short periods and provides a competitive advantage. These facts results in the development of sensitive and rapid detection methods for salbutamol (Lin et al. 2023). Adverse cardiac effects associated with salbutamol includes sinus tachycardia, ventricular and supraventricular tachycardia, and myocardial ischemia (Boucher et al. 2011).

Journal Pre-proof

Formoterol, third commonly detected substance in the S3 section, has a longer side-chain in comparison with terbutaline or salbutamol. The structure involves an acylamino group on the substituted benzene ring (Davis 2007). Moreover, the structure of formoterol involves two stereogenic centers. The drug is commercialized in its racemic form, however it has been found that (*R*,*R*)-(-)-formoterol is about 1000-fold more potent than its enantiomer (*S*,*S*)-(+)-formoterol (Vardanyan and Hruby 2016). Formoterol is a potent and highly selective agonist of β_2-adrenergic receptor. The drug is used for the treatment of asthma and COPD. Formoterol belongs to LABAs with an onset of about 7 minutes and duration effect about 12 hours, its therapeutic dose is 12-24 µg and the drug is used twice per day (Billington et al. 2017). If inhaled, formoterol is 15 times more potent than salbutamol, and up to 50 more potent when administered orally (Lee et al. 2015). Similar to other β_2 agonists, formoterol can also be applied as a potential doping. The drug can increase mean power and counteract development of fatigue in athletes (Kalsen et al. 2016a). The 2023 prohibited list involves an exception for use of formoterol in inhaled form with maximal delivered dose of 54 µg over 24 hours (WADA 2022d). The side effects of formoterol are similar to those reported for other β_2 agonists such as headache, tremor, nausea, vomiting, and palpitation (Aronson 2016).

Higenamine, fourth commonly detected compound in the S3 section, is a natural alkaloid produced by various plants such as *Aconitum carmichaelii*, *Asarum heterotropoides*, *Gnetum parvifolium*, *Nandina domestica*, and others. Structurally, higenamine involves catechol motif which is also present in structures of epinephrine and norepinephrine. Structure of higenamine contains one stereogenic center. The (*S*)-(-)-higenamine exhibits stronger cardiotonic, antiplatelet, and shows better effects in diseases associated with inducible nitric oxide synthase over-expression (Rangelov Kozhuharov et al. 2022). Higenamine is a non-selective agonist of β receptors with various effects on heart, brain, lung, smooth muscle, striate muscle, vasculature, and other. The drug is used for the diagnosis, prevention and treat of coronary artery disease (CAD), myocardial perfusion imaging, myocardial ischemia/reperfusion injury, bradyarrhythmia, diffuse intravascular coagulation, and bronchoconstriction (Wen et al. 2022). Higenamine is present in different commercially available supplements as well as in various nutrients or traditional Chinese medicines and may represent a potential risk for careers of athletes (Hudzik et al. 2021). As seen in Table 4, findings of higenamine have been reported by WADA in recent years. Muniz-Santos et al. investigated dietary higenamine from Annonaceae family fruits as a possible source of unintentional dopoing. Although a single ingestion did not reach the threshold determined by WADA, the authors were able to detect excreted higenamine (Muniz-Santos et al. 2023). Despite of relative low concentration of higenamine in food or dietary supplements, long-term use of these supplements can be potentially detected as unintentional doping (Kozhuharov et al. 2022). For these reasons, position of higenamine in prohibited lists reported by WADA is still an ongoing debate among scientific community.

Salmeterol, fifth commonly detected substance in the S3 section, involves in its structure similar *ortho*-(hydroxymethyl)phenol motif as salbutamol, however it bears long lipophilic side chain with benzene ring. The structure also involves one stereogenic center resulting in two enantiomeric forms of the drug. In general, the biological activity based on the affinity toward β_2-adrenergic receptor resides in (*R*)-salmeterol, while (S)-salmeterol is usually considered to be inert (Čižmáriková et al. 2020). Usually, salmeterol is administered via inhalation as a racemic mixture (Jacobson et al. 2017). The onset of action for salmeterol is about 15 minutes with a duration about 12 hours. Its therapeutic dose is 50–100 µg twice per day. Salmeterol belongs to LABAs (Billington et al. 2017). Similar to other β_2 agonists, salmeterol is used to treat asthma and COPD. Inhalation of salmeterol provides only low concentrations of the drug in urine, while the majority of the drug is eliminated as the α-hydroxysalmeterol metabolite. Both, salmeterol and α-hydroxysalmeterol are predominantly eliminated in the feces. The lipophilic character of salmeterol and its rapid absorption in tissue provides

Journal Pre-proof

sub-ng/mL concentration levels in plasma (Sakellariou et al. 2021). According to the 2023 prohibited list, salmeterol is prohibited by WADA due to its β_2-adrenergic receptor activity. The exception for inhaled salmeterol is maximum 200 µg over 24 hours (WADA 2022d). Usage of salmeterol is also associated with several adverse effects such as upper respiratory infection or inflammation, headache, cough, nausea, vomiting and others (Adams and Nguyen 2022).

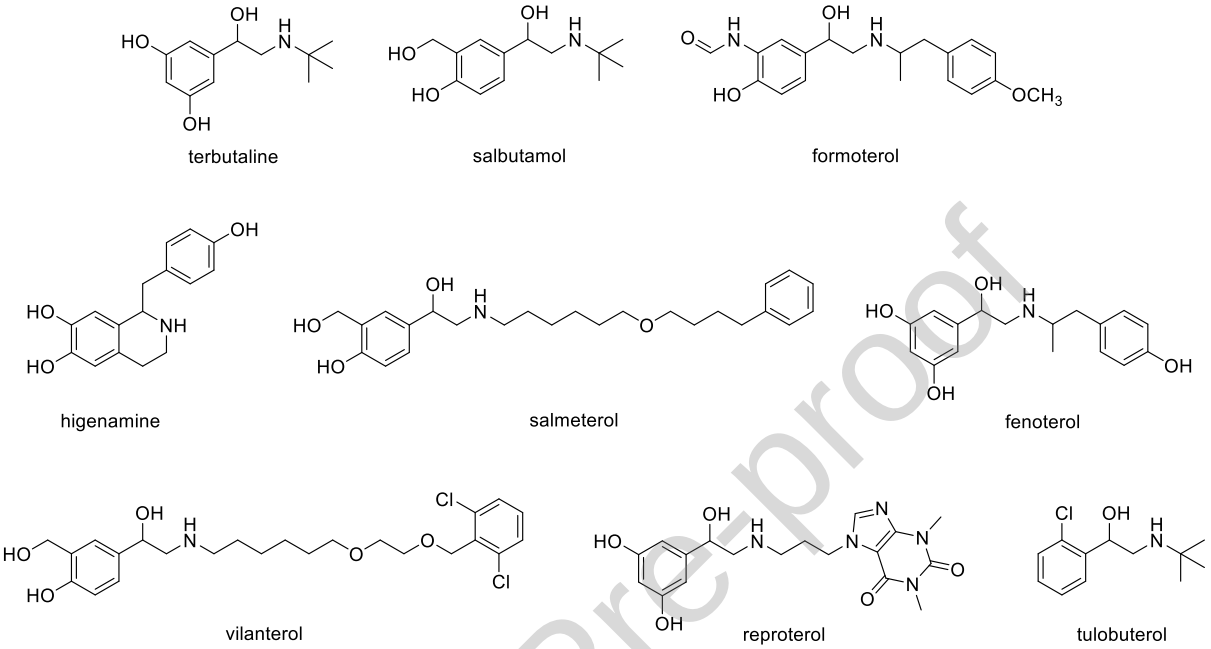


Figure 3. Structures of the most misused β_2 agonists in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.5 Hormone and metabolic modulators (S4)

Hormone and metabolic modulators represent a diverse section of substances incorporated in the 2023 prohibited list reported WADA. The section comprises of four classes: aromatase inhibitors (S4.1), antiestrogenic substances that involve antiestrogens and selective estrogen receptor modulators (SERMs) (S4.2), agents preventing activin receptor IIB activation (S4.3), and metabolic modulators (S4.4). Aromatase inhibitors as well as antiestrogenic substances are commonly utilized as ancillary drugs by AAS users. These compounds are able to counteract the adverse effects of such agents. Agents preventing activin receptor IIB activation incorporates variable compounds that inhibit activin A and myostatin and their role in muscle loss. Metabolic modulators are represented by compounds with varied applications such as insulin which provides therapeutic effects to the diabetic athletes and potential performance-enhancing benefits as a non-therapeutic agent (Chester 2022). In general, various hormone and metabolic modulators bear potential to enhance performance in sport, therefore these drugs are illicitly used as a doping (Gómez-Guerrero et al. 2022). Hormone and metabolic modulators (S4) are prohibited at all times (in- and out-of-competition). Substances in classes S4.1 and S4.2 are specified substances, while those in classes S4.3 and S4.4 are non-specified substances. Some of the substances belonging to S4 section may be found, without limitation, in medications used for treatment of various diseases or conditions such as breast cancer, diabetes, female infertility, or polycystic ovarian syndrome (WADA 2022d). The section was previously (2003–2007) termed as “agents with antiestrogenic activity”. Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S4 section represents eighth commonly detected section of substances misused as a potential doping in all sports, representing 3.5% of overall

Journal Pre-proof

findings (Table 1). As seen in Table 1 as well as in Chart 1, findings of substances from the S4 section increasing more evidently in recent years. The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of hormone and metabolic modulators (S4) is represented by: meldonium (30.3%), tamoxifen (22.3%), clomifene (16.5%), anastrozole (9.6%), letrozole (6.0%), GW1516 and peroxisome proliferator activated receptor δ (PPARδ) agonists (4.8%), trimetazidine (3.3%), 1,4,6-androstatriene-3,17-dione (ATD) (2.7%), formestane (1.0%), 4-androstene-3,6,17-trione (0.7%), exemestane (0.7%), and others (2.3%). Findings of the commonly detected substances of the S4 section are summarized in Table 5, graphically expressed in Chart 5, and their structures are shown in Figure 4.

Table 5. Hormone and metabolic modulators in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: the section was previously termed as "agents with antiestrogenic activity". Notes to findings: 1,4,6-androstatriene-3,17-dione (ATD) – since 2012 the findings are reported as "androstatrienedione"; 4-androstene-3,6,17-trione - during 2006–2009 findings reported as "6-oxo-androstenedione" and since 2012 findings reported as "androstene-3,6,17-trione (6-oxo)"; formestane – includes one finding reported as "the GC/C/IRMS result for formestane consistent with an exogenous origin"; GW1516 and PPARδ agonists – includes 5 findings reported in 2012 and 2014 as "peroxisome proliferator activated receptor δ (PPARδ) agonists" and 18 findings reported in 2021 as "GW501516/GW1516". Abbreviations: ATD, 1,4,6-androstatriene-3,17-dione; PPARδ, peroxisome proliferator activated receptor δ.

	meldonium	tamoxifen	clomifene	anastrozole	letrozole	GW1516 and PPARδ agonists	trimetazidine	1,4,6-androstatriene-3,17-dione (ATD)	formestane	4-androstene-3,6,17-trione	exemestane	others	Total (annually)	Percentage of total (annually)
2003		5										1	**6**	**0.2%**
2004		8											**8**	**0.3%**
2005		11	7	1								2	**21**	**0.7%**
2006		11	4	4	1			1	5	1	1	2	**30**	**1.0%**
2007		6	5		1			2	2			2	**18**	**0.6%**
2008		12	1	4	5			3		2		2	**29**	**1.0%**
2009		19	6	10	5			1	4	2	2	1	**50**	**1.7%**
2010		22	15	11	6			12	6		1	2	**75**	**2.6%**
2011		34	16	8	5			3	3			1	**70**	**2.4%**
2012		38	12	6	7	1		5	4	1			**74**	**2.5%**
2013		40	24	6	5	11		2		2		3	**93**	**3.2%**
2014		42	33	21	8	4	18	3		2	2	12	**145**	**5.0%**
2015		59	37	16	10	6	11	7	1	2	1	5	**155**	**5.3%**
2016	515	61	57	36	19	5	8	10	1	3	2	4	**721**	**24.8%**
2017	79	64	65	40	22	31	6	7		1		6	**321**	**11.1%**
2018	111	71	57	40	24	19	6	12	2	3	2	4	**351**	**12.1%**
2019	79	80	73	38	30	32	5	6			6	13	**362**	**12.5%**
2020	26	28	22	16	17	12	3	1			2		**127**	**4.4%**
2021	70	36	44	23	9	18	38	2				8	**248**	**8.5%**
Total (2003-2021)	**880**	**647**	**478**	**280**	**174**	**139**	**95**	**77**	**28**	**19**	**19**	**68**	**2,904**	
Percentage of total	**30.3%**	**22.3%**	**16.5%**	**9.6%**	**6.0%**	**4.8%**	**3.3%**	**2.7%**	**1.0%**	**0.7%**	**0.7%**	**2.3%**		

Journal Pre-proof

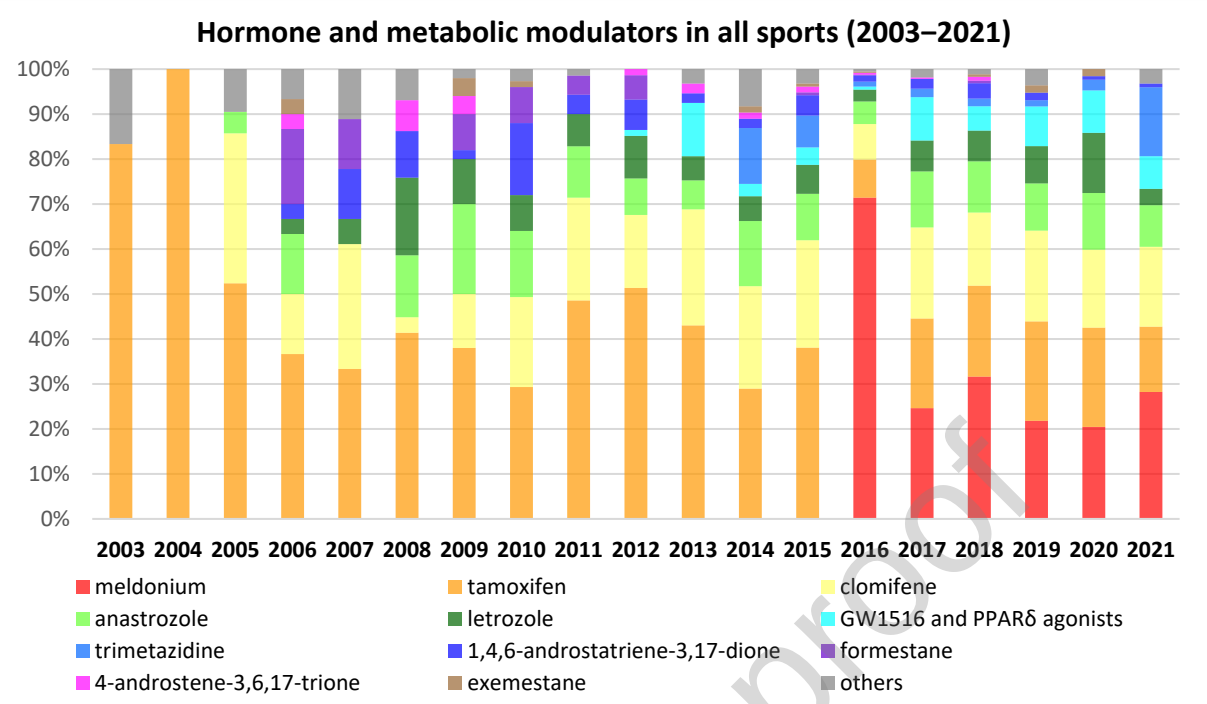


Chart 5. Hormone and metabolic modulators in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: the section was previously termed as "agents with antiestrogenic activity". Abbreviations: ATD, 1,4,6-androstatriene-3,17-dione; PPARδ, peroxisome proliferator activated receptor δ.

4.5.1 Aromatase inhibitors (S4.1)

Aromatase is an enzyme, a monooxygenase from the cytochrome P450 family, associated in the conversion of androgens to estrogens. Aromatase catalyzes aromatization of androgens to estrogens, in particular, formation of estradiol from testosterone and formation of estrone from androstenedione. The enzyme can be found in tissues such as brain, adipose tissue, gonads, bone. Inhibitors of aromatase represent a targeted therapy in treatment of estrogen-dependent cancers in women, for example in breast cancer treatment. In males, aromatase inhibitors can be used off-label in selected cases of male infertility (Vari et al. 2016). These substances can increase production of endogenous testosterone and testosterone serum levels (Schlegel 2012). Therefore, aromatase inhibitors can be misused by athletes as a doping due to their ability to block estrogen production which results in increased secretion of endogenous testosterone as well as corrective medication for side effects caused by administration of exogenous androgens and anabolic steroids. Therapy with aromatase inhibitors is also associated with adverse effects such as increased risk of cardiovascular events, or development of insulin resistance and diabetes (Boszkiewicz et al. 2022). Prohibited aromatase inhibitors incorporated in the 2023 prohibited list reported by WADA includes for example: 1,4,6-androstatriene-3,17-dione, 4-androstene-3,6,17-trione, aminoglutethimide, anastrozole, exemestane, formestane, letrozole, testolactone and others (WADA 2022d).

Anastrozole, fourth commonly detected compound in the S4 section, is a derivative of 1,2,4-triazole with 3,5-bis(2-cyanoprop-2-yl)benzyl moiety attached to N1 position. Anastrozole is highly selective competitive inhibitor of aromatase. The compound exhibits only little to no effects on other steroid hormones (Kelly and Buzdar 2010). Anastrozole is clinically applied as a chemotherapeutic drug to treat breast cancer in postmenopausal women. The condition for a successful treatment is the presence of estrogen receptors in cancer cells (Piotrowska and Piotrowska 2019). Moreover, anastrozole also achieved notable results in the chemoprevention of breast cancer. The mean

Journal Pre-proof

plasmatic half-life of anastrozole is about 50 hours, indicating that single daily dose of the drug is adequate. The compound is metabolized in the liver via *N*-dealkylation, hydroxylation, and glucuronidation reactions providing triazole, hydroxy-anastrozole glucuronide, and anastrozole glucuronide as main metabolites observed in the plasma and urine. Excretion of the main metabolites is primarily through urine. The adverse effects associated with anastrozole use include hot flashes, asthenia, headache, and edema (Barros-Oliveira et al. 2017). As a doping agent, anastrozole is misused to counteract estrogen-like side effects of AAS in men and to increase testosterone concentration in plasma (Weber et al. 2017).

Letrozole, fifth commonly detected substance in the S4 section, is a derivative of 1,2,4-triazole with bis(4-cyanophenyl)methyl moiety attached to N1 position. The drug has been tested since 1986, then in 1996 letrozole was approved in Europe for the treatment of advanced breast cancer and in 1997 also in the USA. Letrozole is applied in treatment of metastatic breast cancer as well as neoadjuvant treatment in breast cancer, it is well-tolerated and effective drug. Letrozole is a highly potent aromatase inhibitor which is highly selective for the aromatase and demonstrated greater potency compared with aminoglutethimide, anastrozole, exemestane, and formestane in several preclinical studies. Orally administered letrozole is rapidly absorbed with bioavailability of 99.9%. The terminal half-life of letrozole is 42 hours (Dellapasqua and Colleoni 2010). The main elimination route is mediated by cytochrome P450, isoenzymes 3A4 (CYP3A4) and 2A6 (CYP2A6), into pharmacologically inactive carbinol metabolite which further undergoes glucuronide conjugation (Vanol et al. 2016). The glucuronide conjugate is then renally excreted. Letrozole has been extensively studied in the past years. Its adverse effects include hepatitis and autoimmune features observed in women with breast cancer treated with letrozole for 3 months (Mukherjee et al. 2022). Other side effects of letrozole may include arthralgias or arthritis (Dellapasqua and Colleoni 2010). Despite its ban, letrozole is misused by athletes as a doping agent due to its inhibitory effect on aromatase. Illicit use of letrozole in sport therefore results in development of novel and specific methods for its detection (Ameline et al. 2021).

4.5.2 Antiestrogenic substances (antiestrogens and SERMs) (S4.2)

Antiestrogens, also called estrogen antagonists, are compounds able to compete with estrogens for binding to the estrogen receptor (Clarke et al. 2003). Antiestrogens either interfere signaling mediated by estrogen receptor or inhibit estrogen production, thereby antiestrogens block biological actions mediated by estrogens (Barakat et al. 2018). The first antiestrogens, such as ethamoxytriphetol and clomiphene, were prepared in the middle of 1950s as fertility agents (Clarke et al. 2003). In 1960s became apparent that *E*- and *Z*-isomers of triphenylethylenes exhibit opposite biological activities. For example, tamoxifen, a *Z*-isomer of substituted 1,1,2-triphenyl-1-butene, in rats acts as an antiestrogen while its *E*-isomer exhibit estrogen activity. Moreover, tamoxifen appeared to exhibit estrogen properties in mice, while in rats acts as a partial agonist with antiestrogenic properties. Furthermore, tamoxifen exhibits antiestrogen properties in breast cancer treatment in humans (Jordan 2003). Despite it was initially developed as an estrogen-antagonist, tamoxifen can be also used to prevent postmenopausal osteoporosis or reduce cholesterol, due to its estrogen-agonist effects (Dhingra 1999). Based on the opposite biological activities, acting as estrogen-agonist or estrogen-antagonist, in different tissues, tamoxifen has been reclassified as a selective estrogen receptor modulator (SERM) (Nardulli 2003). However, significant adverse effects of tamoxifen resulted in the development of other SERMs such as raloxifene. An ideal SERM is a drug exhibiting desired tissue-selective properties, acting as estrogen-agonist or estrogen-antagonist (Dhingra 1999). In sport, antiestrogens can be illicitly used to treat and prevent side effects of AAS misuse (Gerace et al. 2012). Misuse of SERMs, acting as estrogen-agonists or estrogen-antagonists depending on the type of tissue,

Journal Pre-proof

may increase testosterone levels, therefore these compounds are mainly used by male athletes (Hackney 2018b). The S4.2 class in the 2023 prohibited list includes, but it is not limited to, compound such as bazedoxifene, clomifene, cyclofenil, fulvestrant, ospemifene, raloxifene, tamoxifen, and toremifene (WADA 2022d).

Tamoxifen, the second commonly found substance in the S4 section, is a *Z*-configured derivative of 1,1,2-triphenyl-1-butene in which one benzene ring involves *N*,*N*-dimethylaminoethoxy motif incorporated to the *para*-position. The drug is classified as SERM that acts as an estrogen-agonist in the uterus however as an estrogen-antagonist in the breast (Chang 2012). Due to its unique application in both, treatment and chemoprevention of breast cancer, tamoxifen is considered as a pioneering drug. Tamoxifen saving lives of patients with estrogen receptor positive breast cancer worldwide for the past four decades. Its structure is also applied as a scaffold for the development of novel SERMs (Shagufta and Ahmad 2018). The drug is administered orally, reaching peak concentrations within 4–7 hours. Its terminal half-life is 7 days (Dodd-Butera and Broderick 2005). Pharmacological studies in humans revealed its conversion to three active metabolites: *N*-desmethyltamoxifen, *N*-desmethyl-4-hydroxytamoxifen (endoxifen), and 4-hydroxytamoxifen (afimoxifene). These metabolites act as a potent antiestrogens and were used to investigate mechanism of action of the parent drug (Shagufta and Ahmad 2018). Interestingly, afimoxifene and endoxifen have 30–100 times higher affinity estrogen receptors than tamoxifen (Ivanova et al. 2017). Terminal half-life of the main metabolite, *N*-desmethyltamoxifen, is 14 days. Tamoxifen is enterohepatically recirculated and excreted primarily in feces. Acute toxicity of tamoxifen includes nausea, vomiting, hot flashes. Less frequently vaginal bleeding, menstrual irregularities, skin rash. Uncommon, but rarely reported adverse events includes hypercalcemia, edema, depression, and thromboembolic events. Chronic toxicity of tamoxifen in humans is represented by increased risk of two cancer types: endometrial cancer and uterine sarcoma. Further adverse events include increased risk of pulmonary embolism, deep vein thrombosis, and stroke. The drug administered to pregnant women may also harm the fetus (Dodd-Butera and Broderick 2005). Due to its properties, to increase endogenous production of androgens, tamoxifen is often misused by male athletes in sport (Ivanova et al. 2017).

Clomifene, also known as clomiphene that is the third commonly detected substance in the S4 section, is a derivative of chlorotriphenylethylene in which one benzene ring involves *N*,*N*-diethylaminoethoxy motif incorporated to the *para*-position. The drug belongs to SERMs and provides both estrogenic and antiestrogenic effects. Clomifene binds selectively to estrogen receptors in the hypothalamus, endometrium, ovary, and cervix. It exhibits a partial estrogen-agonist effects in the hypothalamus that result in increasing levels of gonadotropins. Clomifene further increases secretion of LH as well as FSH, thus providing increased serum levels of testosterone. Clomifene is used to manage and treat anovulatory or oligo-ovulatory infertility (Mbi Feh and Wadhwa 2023). Clomifene is commonly available as a clomifene citrate which exists as a non-racemic mixture of its two isomers, *Z*-clomifene also termed as zuclomiphene (38%) and *E*-clomifene known as enclomiphene (62%). It is proposed, that enclomiphene exhibits estrogen-antagonist activity while zuclomiphene has an estrogen-agonist activity (Earl and Kim 2019). The drug is administered orally, it has a high binding capacity to plasma proteins, undergoes enterohepatic recirculation and accumulates in fatty tissues, it also has a long elimination half-life of 5 days (Kahyaoğlu et al. 2017). Metabolism of clomifene occurs mainly in the liver as well as in the intestinal mucosa and involves hydroxylation, hydrogenation, *N*-deethylation, formation of *N*-oxide, methoxylation and combination thereof (Ghobadi et al. 2008; Mürdter et al. 2012; Mazzarino et al. 2013). Clomifene was deeply investigated to identify appropriate target metabolites (Ganchev et al. 2011; Lu et al. 2013a; Kröner et al. 2021; Euler et al. 2022). Metabolites provided via 4-hydroxylation such as 4-hydroxyclomifene and 4-hydroxy-*N*-

desethylclomifene are considered to exhibit at least 100-fold higher inhibitory potency at the estrogen receptor than their parent drug (Kim et al. 2018). Common adverse effects associated with clomifene use involve nausea, vomiting, ovarian enlargement, blurred vision, pelvic pain, abnormal uterine bleeding, hypertriglyceridemia. Other adverse effects are headache, dizziness, psychiatric illnesses exacerbation, gynecomastia, testicular tumor. Serious reactions associated with the drug are ovarian hyperstimulation, multiple pregnancies, thrombocytopenia, pancreatitis, increased risk of malignant melanoma, hepatic damage and others (Mbi Feh and Wadhwa 2023). In sport, clomifene is misused as a doping due to its testosterone-increasing properties. In addition, clomifene also increases level of epitestosterone and 4-androstenedione (Barghi et al. 2022). Therefore, clomifene is commonly abused by male athletes. Its misuse is detectable in urine for months following its use. In a study evaluating daily orally administered clomifene citrate for 30 days in a dose of 50 mg can result in its urinary detection for ≥4 months (Miller et al. 2019). Moreover, clomifene may be further abused in the agricultural industry which also represents a hidden risk for athletes. Clomifene can be illicitly used to increase the egg production rate of laying hens. This creates a potential risk for consumers of eggs as well as edible tissues of these hens (Seyerlein et al. 2021). The study from 2022 investigated whether the consumption of clomiphene-contaminated eggs may result in positive clomifene detection in human urine samples. The authors reported that consumption of eggs from laying hens treated with clomifene results in AAFs (Euler et al. 2022).

4.5.3 Agents preventing activin receptor IIB activation (S4.3)

Activin receptor IIB is a transmembrane receptor which is associated to the activin and myostatin signaling pathway (Han 2011). The receptor is involved in the negative regulation of skeletal muscle mass (Pistilli et al. 2011). Myostatin , a member of transforming growth factor-β (TGF-β) superfamily, is a negative regulator of skeletal muscle growth and development (Chen et al. 2021). In addition, activin A also plays an important role in the negative regulation of muscle growth (Solagna et al. 2021). Therefore, compounds affecting activation of activin receptor IIB can be therapeutically used for the treatment of muscle wasting disorders, however these substances could be misused as potential performance-enhancing agents in sports (Thevis et al. 2019). Agents that prevent activation of activin receptor IIB are include to the 2023 prohibited list reported by WADA as S4.3 class. This class includes antibodies neutralizing activin A, activin receptor IIB competitive antagonists, anti-activin receptor IIB antibodies, as well as myostatin inhibitors (WADA 2022d).

4.5.4 Metabolic modulators (S4.4)

In the 2023 prohibited list the class of metabolic modulators (S4.4) involves four subclasses: activators of the 5' adenosine monophosphate-activated protein kinase (AMPK) and PPARδ agonists (S4.4.1); insulins and insulin-mimetics (S4.4.2); meldonium (S4.4.3); trimetazidine (S4.4.4) (WADA 2022d). Activation of the AMPK can provide beneficial effects such as improvement of hyperglycemic state in diabetes or reduction of obesity and inflammation. Moreover, stimulation of AMPK activity can also enhance exercise capacity. In mice, 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR) which is a known activator of the AMPK increased the running capacity without any training. As the result, WADA included certain AMPK activators to the prohibited list (Niederberger et al. 2015). The PPARδ regulates muscle metabolism as well as reprograms muscle fiber types. In mice, the compound referred as GW1516 (also known as GW501516) which acts as an agonist of PPARδ increased running performance in both trained and untrained mice. The drug also promoted mitochondrial fatty acid oxidation and increased fat metabolism in muscle tissue (Chen et al. 2015). Insulin is an essential and versatile peptide hormone which controls glucose levels in the blood, however it also plays an

Journal Pre-proof

important role in protein anabolism. Insulin has been prohibited in sports since 1999 due to its functions to improve performance and muscle strength (Thevis et al. 2010c). Insulin prevents proteolysis, favors protein synthesis and storage of glycogen and nutrients. Therefore, insulin is usually misused by athletes that perform high-strength activities, such as bodybuilders and weightlifters (Heidet et al. 2019).

Meldonium, the most commonly detected substance in the S4 section, is an analog of γ-butyrobetaine in which a CH_2 motif at γ-position is replaced by NH motif. Its structure is similar to two biologically important betaines, L-carnitine and its precursor γ-butyrobetaine, that can be found in the liver, kidney and brain. The biological role of L-carnitine is crucial in transportation of activated long-chain fatty acids as well as other peroxisomal β-oxidation products to the mitochondria. L-Carnitine is further involved in the modulation of acetyl coenzyme A (acetyl-CoA)/CoA ratio, optimalization of the energy storage, and reduction of the toxic effects of poorly metabolized acyl groups. The metabolism of L-carnitine results in the formation of trimethylamine *N*-oxide which participates in pathogenesis of atherosclerosis and hearth failure. Meldonium acts as a competitive inhibitor of the enzyme γ-butyrobetaine hydroxylase (BBOX) which converts γ-butyrobetaine in L-carnitine. Thus, meldonium decreases levels of L-carnitine in cell which protects mitochondria from injury caused by accumulation of toxic long-chain fatty acids intermediates. Due to regulation of fatty acid metabolism, meldonium is used as a cardioprtective drug used mostly in Eastern Europe because the drug is not approved by the FDA (Lippi and Mattiuzzi 2017). Meldonium reaches the maximal blood concentration 2 hours after its oral administration and the half-life excretion in plasma is about 18 hours (Schobersberger et al. 2017). Adverse effects associated with the use of meldonium include headache, tachycardia, agitation, allergic skin reaction, and dyspepsia (Stuart et al. 2016). In sports, meldonium is used as a doping agent due to improved peroxisomal utilization of fatty acids by muscles, decreased lactate production after exercise, improved utilization and storage of glycogen, and prevention of oxidative stress after intensive muscular workload. As the result, meldonium has been banned by WADA since January 2016 (Lippi and Mattiuzzi 2017).

Journal Pre-proof

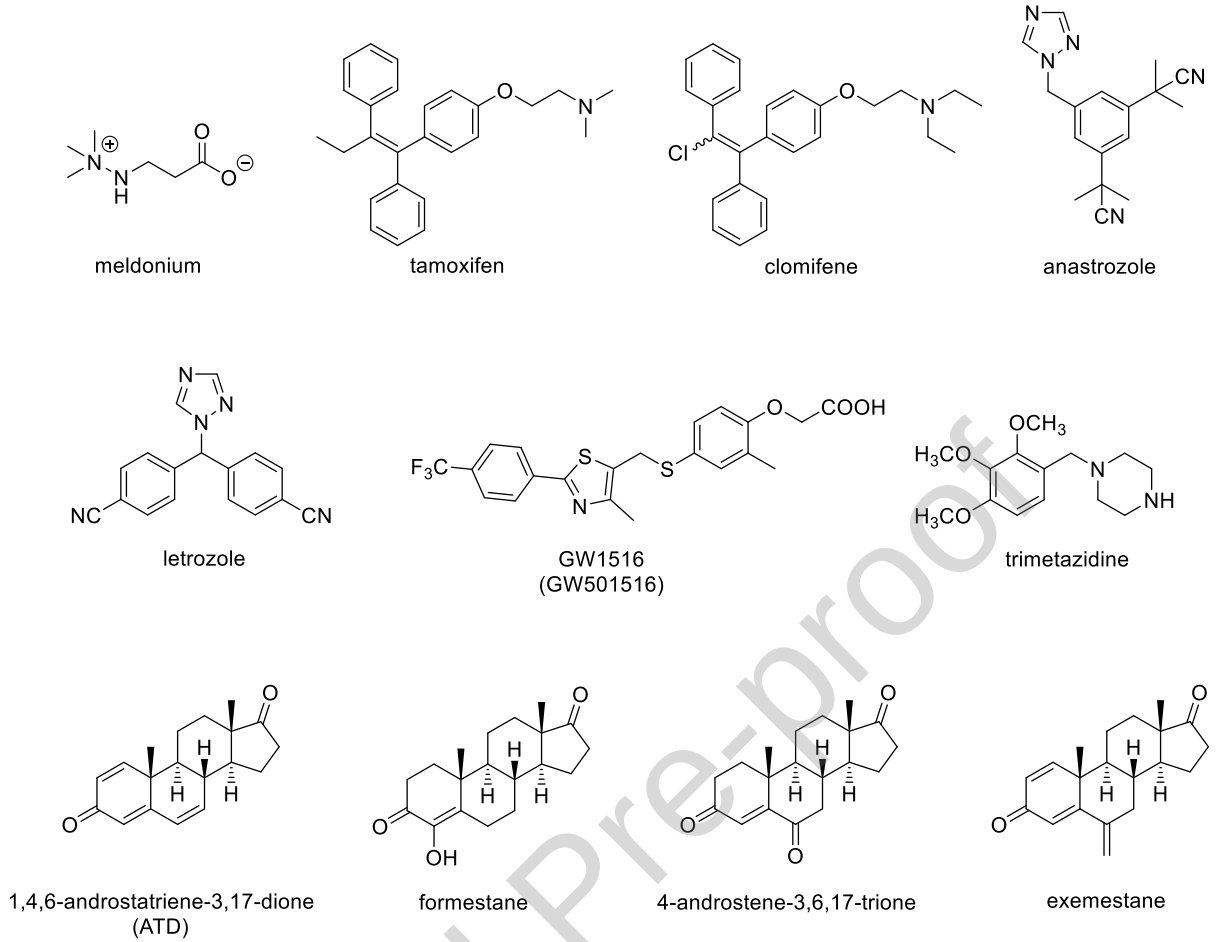


Figure 4. Structures of the most commonly detected hormone and metabolic modulators in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.6 Diuretics and masking agents (S5)

Diuretics represent a diverse group of compounds that are clinically used in the treatment of hypertension, heart failure, renal failure, liver cirrhosis, diseases of kidney and lung, or to reduce adverse effects caused by salts and/or water retention. These compounds cause an increased rate of urine flow and sodium excretion to improve composition and volume of body fluids or to eliminate excess of fluid from tissues. In addition to Na^+ and Cl^-, diuretics can also influence renal absorption and excretion of other cations such as H^+, K^+, Ca^{2+}, Mg^{2+} as well as anions involving HCO_3^-, $H_2PO_4^-$. Diuretics, on the basis of the ability to increase volume of urine, are misused in sports to dilute doping agents and their metabolites in the urine which in turn complicates analytical anti-doping evaluation. Moreover, diuretics are also widespread among athletes who want to lose weight rapidly. Misuse of diuretics to reduce body weight can be found in general sports where weight categories are involved as well as in sports where athletes want to maintain a low body weight. Therefore, these agents are usually detected in boxing, judo, wrestling, weight-lifting as well as gymnasts and ballet dancers (Cadwallader et al. 2010). Other masking agents incorporated on the prohibited list are represented by substances used to reduce concentration of acidic compounds in urine, for example glucuronic conjugates of misused doping agents, such as probenecid; plasma expanders that are used to maintain plasma volume after misuse of EPO or RBCs concentrates, for example represented by intravenous administration of albumin, dextran, hydroxyethyl starch, and mannitol (Ventura and Segura 2010; WADA 2022d). It is worth mentioning that, from 2005 to 2009, substances modulating 5α-reductase

activity were included in the WADA prohibited list as "diuretics and other masking agents" in the S5 section. The activity of 5α-reductase can be modulated by 4-azasteroids that are used for the treatment of benign prostatic hyperplasia or androgenic alopecia. However, 4-azasteroids such as finasteride and dutasteride can be misused in sport as doping agents to modulate excretion of steroids, and thus, to mask the abuse of both pseudoendogenous and exogenous steroids. Analytical determination of finasteride can be accomplished in biological fluids, whereas dutasteride can be determined only in blood samples due to its pharmacokinetic properties. Nevertheless, since 2014, 5α-reductase inhibitors are included in the technical document "Endogenous Anabolic Androgenic Steroids Measurement and Reporting" as confounding factors (WADA 2019a). Diuretics and other masking agents are banned by WADA and involved in the 2023 prohibited list as the S5 section. Diuretics and masking agents are prohibited at all times (in- and out-of-competition). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S5 section represents third commonly detected section of substances misused as a potential doping in all sports, representing 8.7% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of diuretics and masking agents (S5) is represented by: furosemide (32.2%), hydrochlorothiazide (27.7%), canrenone (8.3%), chlorothiazide (5.2%), finasteride (3.0%), triamterene (2.9%), amiloride (2.9%), acetazolamide (2.2%), indapamide (2.2%), dorzolamide (2.1%), and others (11.2%). Findings of the commonly detected substances of the S5 section are summarized in Table 6, graphically expressed in Chart 6, and their structures are shown in Figure 5.

Table 6. Diuretics and masking agents in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: finasteride has been removed from the 2009 prohibited list reported by WADA.

	furosemide	hydrochlorothiazide	canrenone	chlorothiazide	finasteride	triamterene	amiloride	acetazolamide	indapamide	dorzolamide	others	Total (annually)	Percentage of total (annually)
2003	48	42	14		1	10	6	3	1		17	**142**	**2.0%**
2004	62	44	9	1		7	4	4	2		24	**157**	**2.2%**
2005	91	67	14	3	28	9	10	6	4		14	**246**	**3.5%**
2006	90	88	8	1	44	7	10	2	5		35	**290**	**4.1%**
2007	111	103	15	7	53	15	10	4	4		37	**359**	**5.0%**
2008	104	137	15	27	84	7	6	6	12		38	**436**	**6.1%**
2009	92	90	21	12		10	8	8	14	1	17	**273**	**3.8%**
2010	152	120	27	25		8	13	12	7		32	**396**	**5.6%**
2011	123	123	29	36		10	12	8	4		23	**368**	**5.2%**
2012	127	101	24	24		7	6	8	8		17	**322**	**4.5%**
2013	145	125	23	23		15	18	14	6	1	23	**393**	**5.5%**
2014	128	120	38	21		5	14	14	10		39	**389**	**5.5%**
2015	153	125	33	20		9	24	9	9	4	42	**428**	**6.0%**
2016	131	158	51	24		12	17	11	16	7	72	**499**	**7.0%**
2017	215	143	77	38		11	23	7	9	9	82	**614**	**8.6%**
2018	172	127	60	36		26	10	17	14	29	99	**590**	**8.3%**
2019	199	141	69	48		25	14	13	15	50	103	**677**	**9.5%**
2020	61	47	23	12		8	2	8	4	21	27	**213**	**3.0%**
2021	84	72	42	13		8	1	3	12	28	57	**320**	**4.5%**
Total (2003-2021)	**2,288**	**1,973**	**592**	**371**	**210**	**209**	**208**	**157**	**156**	**150**	**798**	**7,112**	
Percentage of total	**32.2%**	**27.7%**	**8.3%**	**5.2%**	**3.0%**	**2.9%**	**2.9%**	**2.2%**	**2.2%**	**2.1%**	**11.2%**		

Journal Pre-proof

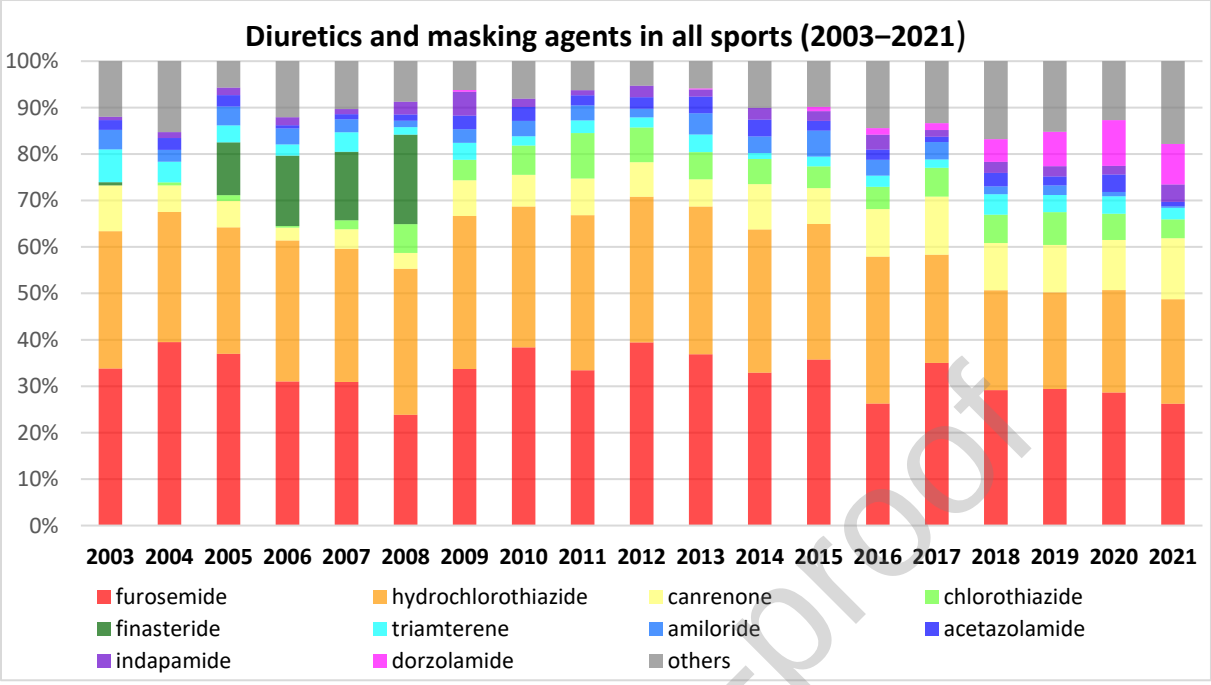


Chart 6. Diuretics and masking agents in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: finasteride has been removed from the 2009 prohibited list reported by WADA.

Furosemide, the most commonly detected compound in the S5 section, is a derivative of 4-chlorobenzoid acid which involves a (furan-2-ylmethyl)amino motif attached to C2 position and a sulfonamide group attached to C5 position. Furosemide is an inhibitor of $Na^+/K^+/2Cl^-$ symporter which acts as a potent short-acting diuretic agent. The inhibitor binds to the Cl^- binding site in the transmembrane domain of $Na^+/K^+/2Cl^-$ symporter, which is found in the thick ascending limb of the loop of Henle. Blocking of the symporter function results in reduced ability of kidney to concentrate urine as well as subsequent increased urinary excretion of Na^+ and Cl^-. In addition, the excretion of Ca^{2+}, Mg^{2+}, and K^+ is also increased. Acute administration of $Na^+/K^+/2Cl^-$ symporter inhibitor causes an increased excretion of uric acid while chronic administration has the converse effect (Cadwallader et al. 2010). The drug is used in the treatment of edema and hypertension unresponsive to usual treatment (Carone et al. 2016). Furosemide can be administered orally (tablets or an oral solution) or intravenously which is twice as potent as the drug administered orally. Orally administered furosemide has an onset of action within the first hour and it takes 1–2 hours to achieve a peak effect (Khan et al. 2023). Metabolism of furosemide occurs primarily in the kidney and provides furosemide glucuronide which also exhibits a diuretic effect. About 50% of the drug is excreted unchanged in urine (Oh and Han 2015). Terminal half-life of the drug is about 2 hours and the total time of its therapeutic effect is 6–8 hours. Adverse effects associated with the use of furosemide involve hepatic encephalopathy in patients with cirrhosis, pancreatitis, severe anaphylactic or anaphylactoid reactions, ototoxicity, paresthesia, acute kidney injury due to fluid loss, hyperglycemia, and others. Furosemide overdose, as the result of extensive diuretic activity, associates with signs and symptoms such as blood volume reduction, dehydration, electrolyte imbalance, and others (Khan et al. 2023). In sports, furosemide as a potent diuretic, can be misused to both masking the presence of other performance-enhancing substances or excreting water for rapid weight lost. However, the study reported in 2021 suggested that its utilization of furosemide as a masking agent is limited (Brinkman et al. 2021). Moreover,

Journal Pre-proof

development of novel methods for detection of furosemide, as well as other diuretic agents, in organic media can provide an effective analytical tool avoiding their misuse in sports (Ksenofontov et al. 2019).

Hydrochlorothiazide, second commonly detected substance in the S5 section, is a substituted 3,4-dihydro-2*H*-1,2,4-benzothiadiazine-1,1-dioxide which involves the chlorine at C6 position and sulfonamide group attached to C7 position. Hydrochlorothiazide is an inhibitor of Na^+/Cl^- symporter. These inhibitors exhibit an optimal diuretic effect in the early distal convoluted tubule and a lesser diuretic effect in the proximal tubule. The inhibition of Na^+/Cl^- symporter also reduces the Na^+ reabsorption. Although inhibitors of Na^+/Cl^- symporter are expected to significantly increase excretion of Na^+ and Cl^-, this effect is moderate because about 90% of filtered Na^+ is reabsorbed before reaching the distal convoluted tubule. Inhibition of Na^+/Cl^- symporter markedly increases excretion of K^+, while excretion of the uric acid is increased upon acute inhibition however decreased upon chronic inhibition. Moreover, this inhibition decreases excretion of Ca^{2+} (Cadwallader et al. 2010). This action promotes diuretic effect and lowers the blood pressure. Hydrochlorothiazide is clinically applied for the treatment of hypertension, however currently its use is replaced by the angiotensin-converting enzyme inhibitors. The drug is also used to treat edema. Hydrochlorothiazide has an onset of action of about 2 hours with the peak effect in 4 hours and its effect duration is 6–12 hours. Majority of the drug is not metabolized and excreted in urine. Hydrochlorothiazide has a variable half-life about 6–12 hours. Adverse effects associated with the use of hydrochlorothiazide can be caused by electrolyte imbalance such as hypokalemia, hyponatremia, hypercalcemia, hypomagnesemia. It can also cause acute and acute-closure glaucoma. Further adverse events include headache, nausea, vomiting, weakness, pancreatitis, gastric irritation and others (Herman and Bashir 2023). Similar to other diuretic agents, hydrochlorothiazide can be misused in sports as a masking agent to complicate detection of other prohibited substances or to reduce weight of an athlete via water-excreting properties. However, a positive doping control to hydrochlorothiazide has been also reported as a result of contamination. In 2014, a doping control of an athlete which used the permitted non-steroidal anti-inflammatory drug ibuprofen revealed a positive finding for hydrochlorothiazide. Subsequent analysis of the tablet confirmed a presence of approximately 2 µg of hydrochlorothiazide per tablet (Helmlin et al. 2016). In another case, a doping control of an athlete who used nutritional supplements also provided a positive finding for hydrochlorothiazide. Subsequent analysis of nutritional supplements revealed a presence of hydrochlorothiazide at the ppm level (Favretto et al. 2019). In 2021, Eichner et al. reported the summary of nine positive findings of athletes identified by United States Anti-Doping Agency (USADA) between 2017 and 2020 caused by generically manufactured prescription medications. The medications were contaminated with a single or multiple diuretics. The most detected contaminant was hydrochlorothiazide (8 instances), followed by torasemide (2 instances), and triamterene (1 instance). Increasing number of reports regarding a contamination of medications or nutrient supplements with diuretics resulted in June 2021 in a new Stakeholder Notice announced by WADA. The document implements a minimum reporting level in potential diuretic contamination cases at or below 20 ng/mL with additional investigation provisions for weight category sports that helps to ensure fair analysis for all athletes (Eichner et al. 2021).

Canrenone, third commonly detected substance in the S5 section, is a compound with steroidal spirolactone scaffold. Canrenone and other structurally related compounds such as spironolactone and eplerenone act as competitive inhibitors of aldosterone which binds to mineralocorticoid receptor. Under regular conditions, aldosterone binds to the receptor and forms aldosterone-mineralocorticoid receptor complex which is translocated to the nucleus to regulate expression of multiple genes products called aldosterone-induced proteins. Canrenone and related compounds that act as antagonists of the mineralocorticoid receptor form similar complex antagonist-mineralocorticoid receptor which is not able to induce synthesis of aldosterone-induced proteins. In

Journal Pre-proof

addition, antagonists of mineralocorticoid receptor affect the urinary excretion (Cadwallader et al. 2010). As an inhibitor of aldosterone, canrenone is used in the treatment of hyperaldosteronism (Rankin 2008). It is also applied in the treatment of hypertension (Derosa et al. 2017). Interestingly, the antiantrogen activity of canrenone is lower in comparison to spironolactone (Armanini et al. 2014). However, spironolactone is extensively metabolized and has a short half-life (about 1.6 hours). An active metabolite of spironolactone is canrenone which has about 10-fold greater half-life. Similarly, potassium canrenoate is a prodrug that is metabolized to active canrenone (Cadwallader et al. 2010). The half-life of canrenone is 16.5 hours and the compound is metabolized via three pathways. Hydrolysis of the γ-lactone ring to canrenoate which can be further hydroxylated or excreted in urine as a glucuronic ester. Hydroxylation to 15α-hydroxycanrenone. Reduction which provides several dihydro, tetrahydro, and hexahydro derivatives (Humans and Cancer 2001). Adverse effects of canrenone involve hyperkalemia, mild acidosis, diarrrhea, cramping, gastric bleeding, ataxia, impotence, gynecomastia and others (Rankin 2008). Canrenone or other structurally similar diuretics that provide canrenone as a metabolite can also be misused in sports due to their diuretic effect (Cadwallader et al. 2010). Detection of canrenone can be observed within the whole summarized time period (2003–2021) od WADA reports (Chart 6).

Chlorothiazide, fourth commonly detected compound in the S5 section, is a 3,4-dehydro derivative of hydrochlorothiazide. Due to a significant structural similarity with hydrochlorothiazide, chlorothiazide also acts as an inhibitor of Na^+/Cl^- symporter which promotes its diuretic effect (Cadwallader et al. 2010). In 1958, the drug has been approved for medicinal use in the USA for the treatment of hypertension (Sismondo and Greene 2015). It is also used to treat various types of edema (Teng et al. 2020). Chlorothiazide administered orally is more effective from low doses than from high doses. The drug has the plasma half-life in the range of 45–120 minutes, but the biological half-life is about 10 hours. Chlorothiazide is not metabolized and it is excreted unchanged by the kidney (Longer et al. 1985). Chlorothiazide such as other thiazide-like diuretic drugs involving hydrochlorothiazide can be misused in sports. Interestingly, chlorothiazide is a minor metabolite of hydrochlorothiazide and could be detected up to 120 hours after administration (Deventer et al. 2009). The findings of chlorothiazide reported by WADA are markedly less frequent, but common, in comparison to hydrochlorothiazide findings (Chart 6).

Finasteride, fifth commonly detected substance in the S5 section, is a synthetic 4-azasteroid compound. The drug is a potent competitive inhibitor of 5α-reductase type 2 but it also exhibits some inhibitory effects on 5α-reductase type 1. Enzyme 5α-reductase plays a crucial role in the conversion of several steroids. The enzyme converts testosterone to the most natural potent androgen dihydrotestosterone as well as acts in the conversion of progesterone to dihydroprogesterone (Rahimi-Ardabili et al. 2006). Finasteride is a FDA-approved drug which was first time used in 1992 to treat benign prostatic hyperplastia and in 1997 it received approval for the treatment of androgenic alopecia (male pattern hair loss). The drug can be further used to treat signs of hyperandrogenism. Finasteride is available in a form of tablets for oral administration (containing 1 or 5 mg of the active compound). Finasteride has a bioavailability of about 65% and its elimination half-life from the serum is 5-6 hours. The drug is extensive metabolized in the liver via CYP3A4, providing two main metabolites with less than 20% of the activity of finasteride. The metabolites are excreted in the feces (57%) and in the urine (39%). Common adverse effects associated with the use of finasteride involves loss of libido, erectile dysfunction, decreased ejaculatory volume, and gynecomastia. It also correlates with orthostatic hypotension (Zito et al. 2023). In sports, finasteride has been prohibited due to its 5α-reductase inhibitory effect. This action influences metabolic pathways of endogenous and synthetic steroids, which complicated the evaluation of steroid profiles in anti-doping analysis (Thevis et al. 2007). However, since 2009, finasteride has been removed from the prohibited list reported by WADA. The

removal was reconsidered on the basis of current scientific reports (Thevis et al. 2010a). Although findings of finasteride, as a prohibited substance, have been reported only during 2003–2008, finasteride findings represent 3.0% of overall findings reported in the S5 section within the whole time period of 2003–2021 (Table 6).

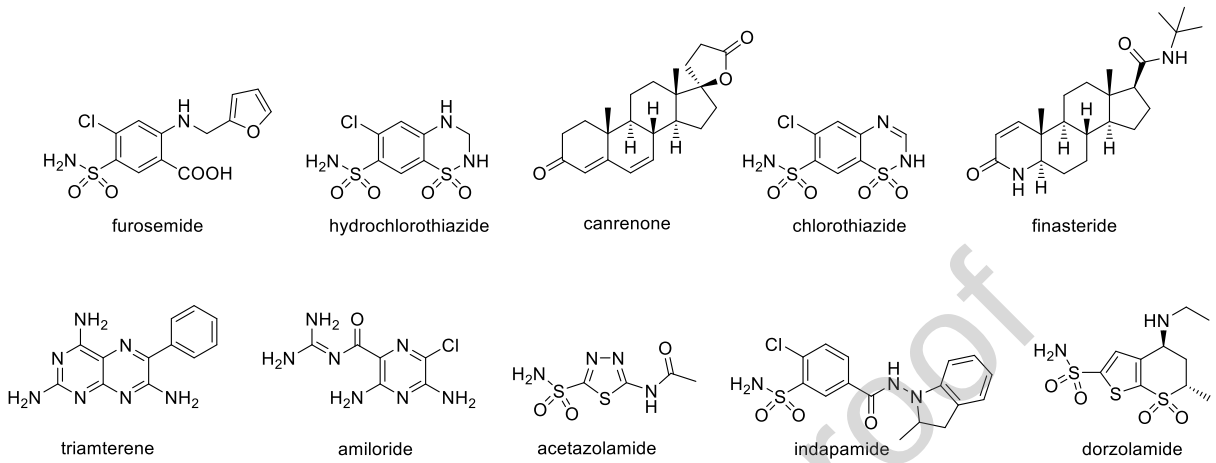


Figure 5. Structures of the most commonly detected diuretics and masking agents in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.7 Stimulants (S6)

Stimulants is a term that involves drugs of diverse classes that increase activity of the CNS. These compounds are used by a high percentage of general population for various reasons such as medical benefits, recreational purposes as well as performance-enhancing drugs. Stimulants include drugs that can be used legally as well as illegal compounds of natural or synthetic origin. In general, their mechanism of action is associated with increased catecholamine levels and increased agonistic activity at adrenergic receptor (Farzam et al. 2023). CNS stimulants have a high potential for addiction. According to definition, addiction is a chronic, relapsing brain disease characterized by compulsive drug seeking and use, despite its harmful consequences. The addiction is considered as a brain disease because drugs change structure and functions of the brain (Ogbonna et al. 2021). Moreover, the use of stimulants is associated with the development of tolerance, which is characterized by applying of increasing doses of a stimulant, over a period of time, to maintain the same response to the drug (Avois et al. 2006). In sports, stimulants are used to reduce tiredness and to increase alertness, competitiveness, and aggression. These substances express a higher potential in competition however they can be also used to increase training intensity (Avois et al. 2006). One of the best known, commonly used, and well-studied CNS stimulants is caffeine, a natural alkaloid presents in various plants which can be found in popular drinks and foods. Caffeine due to its CNS stimulation effect might have a capacity to enhance endurance performance, therefore it is widely used in sports. For this reason, caffeine was listed as a banned substance in sports during 1984–2004 and its use was prohibited in competition. However, since January 2004, WADA removed caffeine from the prohibited list and moved it to WADA's monitoring program (Diel 2020). In 2003, WADA reported 39 caffeine findings. In contrast to all of the previously disclosed sections (S0–S5) involving substances prohibited at all times (in- and out-of-competition), stimulants (S6 section) belong to sections that are prohibited in-competition. The section includes all optical isomers and contains two classes: non-specified stimulants (S6.A) and specified stimulants (S6.B). There are a few exceptions incorporated in the S6 section that include clonidine; imidazoline derivatives for dermatological, nasal, ophthalmic or otic use (brimonidine, indanazoline, oxymetazoline, and others); and stimulants that are included in the 2023

Journal Pre-proof

monitoring program such as caffeine, nicotine, phenylephrine, synephrine, and others. Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S6 section represents second commonly detected section of substances misused as a potential doping in all sports, representing 12.1% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of stimulants (S6) is represented by: amphetamine (22.8%), methylhexanamine (13.1%), cocaine (12.6%), methylphenidate (11.4%), ephedrine (7.7%), pseudoephedrine (2.9%), methamphetamine (2.4%), sibutramine (2.4%), heptaminol (2.2%), mephentermine (2.1%), and others (20.4%). Findings of the commonly detected substances of the S6 section are summarized in Table 7, graphically expressed in Chart 7, and their structures are shown in Figure 6.

Table 7. Stimulants in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: pseudoephedrine has been removed from the 2004 prohibited list by WADA, then again involved to the prohibited list since 2010. Notes to findings: amphetamine – involves findings reported as "amphetamine" and "D-amphetamine"; cocaine – includes findings reported in 2004 and 2005 as "cocaine metabolites"; methamphetamine – includes findings reported as "methamphetamine", "D-methamphetamine", "levomethamphetamine", and "methylamphetamine".

	amphetamine	methylhexanamine	cocaine	methylphenidate	ephedrine	pseudoephedrine	methamphetamine	sibutramine	heptaminol	mephentermine	others	Total (annually)	Percentage of total (annually)
2003	43		48	9	100	189	7		8		112	**516**	**5.2%**
2004	112		75	7	102		8		10		68	**382**	**3.8%**
2005	194		85	17	93		13		11	4	92	**509**	**5.1%**
2006	199		85	32	66		8	3	8	4	85	**490**	**4.9%**
2007	430		101	38	50		21	20	11	1	121	**793**	**8.0%**
2008	166	1	77	40	54		7	17	6	7	97	**472**	**4.7%**
2009	35	31	60	31	44		12	11	8	11	82	**325**	**3.3%**
2010	117	123	65	73	32	17	12	25	12	32	66	**574**	**5.8%**
2011	135	283	40	59	33	9	31	14	7	10	97	**718**	**7.2%**
2012	64	320	59	47	12	13	34	14	5	12	117	**697**	**7.0%**
2013	53	169	52	66	10	13	8	20	6	6	127	**530**	**5.3%**
2014	76	76	46	71	25	8	17	18	6	13	118	**474**	**4.8%**
2015	100	56	70	96	19	9	15	15	9	9	133	**531**	**5.3%**
2016	128	69	65	95	29	13	9	14	8	20	118	**568**	**5.7%**
2017	108	50	69	108	26	6	8	25	16	15	146	**577**	**5.8%**
2018	109	72	87	91	30	4	7	12	35	17	141	**605**	**6.1%**
2019	102	35	77	133	22	7	9	13	26	30	157	**611**	**6.1%**
2020	31	11	33	49	7		2	3	13	11	65	**225**	**2.3%**
2021	67	8	64	70	13	4	8	11	10	7	86	**348**	**3.5%**
Total (2003-2021)	**2,269**	**1,304**	**1,258**	**1,132**	**767**	**292**	**236**	**235**	**215**	**209**	**2,028**	**9,945**	
Percentage of total	**22.8%**	**13.1%**	**12.6%**	**11.4%**	**7.7%**	**2.9%**	**2.4%**	**2.4%**	**2.2%**	**2.1%**	**20.4%**		

Journal Pre-proof

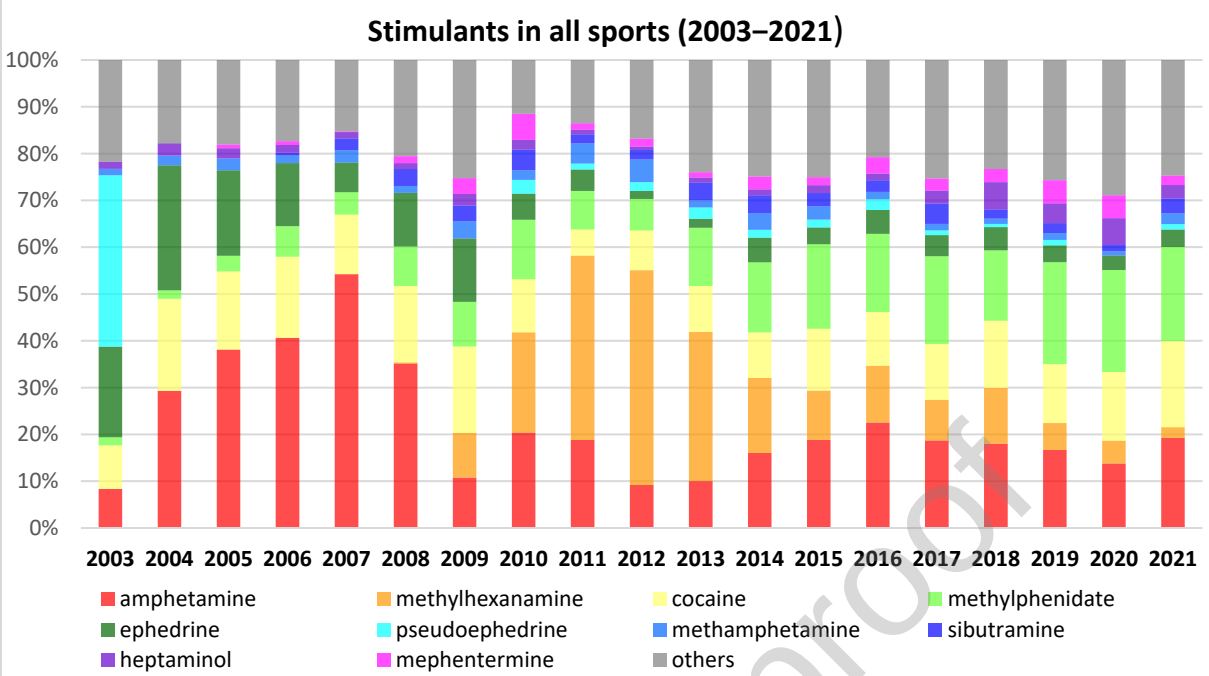


Chart 7. Stimulants in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes: pseudoephedrine has been removed from the 2004 prohibited list by WADA, then again involved to the prohibited list since 2010.

4.7.1 Non-specified stimulants (S6.A)

Since 2009, the section of stimulants is subdivided into non-specified substances (S6.A) and specified substances (S6.B). Based on the List Expert Group classification, stimulants are divided to the both categories using some of the following criteria: the ability to enhance performance in sports; the risk to health; general use in medicinal products; legitimate marked availability; illicit use of the stimulants; legal/controlled status; history and potential of abuse in sports; breakdown pattern of the stimulants; the likelihood of approval of a therapeutic use exemption (Kinahan et al. 2017). The 2023 prohibited list includes 30 substances that are classified as non-specified stimulants. Other substances that are not expressly listed in the class of non-specified stimulants belong to the class of specified stimulants. Following selected compounds are listed as an example of non-specified substances: amphetamine, cocaine, methamphetamine, mephentermine, phentermine, and others (WADA 2022d).

Amphetamine, the most commonly detected substance in the S6 section, is a structure which name is contracted from α-methylphenethylamine. The compound belongs to the group of phenethylamine analogs and it has one stereogenic center. Thus, amphetamine exists in a form of two enantiomers: (*S*)-(+)-amphetamine which is usually referred as D-amphetamine or dextroamphetamine; and (*R*)-(-)-amphetamine which is also known as L-amphetamine or levoamphetamine (Losacker et al. 2021). Amphetamine acts as a stimulant of the CNS. The mode of action is mediated by inhibition of transporters for dopamine, norepinephrine, and serotonin. This results in the increased levels of monoamine neurotransmitters (dopamine, norepinephrine, and serotonin) in the synaptic cleft followed by an increased monoaminergic neurotransmission. Amphetamine enters the presynaptic axon terminal, via diffusion or uptake by the monoamine transporters, and then increases levels of monoamine neurotransmitters in the cytosol by inhibiting of the vesicular monoamine transporter 2 (VMAT2) as well as by disrupting of electrochemical gradients that are necessary for the function of the vesicular transporter. In addition, amphetamine inhibits the

Journal Pre-proof

metabolism of monoamine neurotransmitters via inhibition of monoamine oxidase (MAO). Amphetamine is approved by the FDA for the treatment of attention deficit hyperactivity disorder (ADHD) and narcolepsy (Martin and Le 2023). Amphetamine was initially used as a racemic mixture, however later it has been supplanted by enantiopure dextroamphetamine or by enantioriched mixture of dextroamphetamine and levoamphetamine (75:25). Comparison of the both enantiomers employing various approaches revealed that dextroamphetamine is about 3–7 times more potent at the dopamine synapse and about 1–2 times more potent at the norepinephrine synapse. After oral administration, most of the amphetamine dose (>90%) is absorbed (Markowitz and Patrick 2017). The drug reaches the maximum plasma concentration within 4 hours after ingestion. The both enantiomers exhibit similar protein binding and distribution volume, whereas the elimination half-life of dextroamphetamine is shorter than of the levoamphetamine due to faster metabolism of the more biologically active enantiomer (de la Torre et al. 2004). Metabolism of amphetamine involves two main oxidative pathways. First, amphetamine undergoes an aromatic hydroxylation at the *para*-position of the benzene ring to form pharmacologically active metabolite 4-hydroxyamphetamine. Second, is oxidative deamination which provides phenylacetone as a metabolite. In addition, β-hydroxylation may occur and provide norephedrine as an active metabolite. Both active metabolites, 4-hydroxyamphetamine and norephedrine, are further metabolized via oxidation (Markowitz and Patrick 2017). Despite of hepatic metabolism, a significant amount of amphetamine is excreted in the urine as the unchanged parent drug (Jones and Karlsson 2005). Chronic use of amphetamine is associated with addiction to the drug and tolerance. Adverse effects associated with a chronic use of amphetamine include myocardial pathology and growth retardation in adolescents. Other side and negative effects of the drug are anxiety, confusion, delirium, hypertension, tachycardia, insomnia, and others. In sports, performance-enhancing effects of amphetamine were first investigated in 1959. Amphetamine and other amphetamines are banned by the WADA because these compounds may improve reaction time when fatigue, increase muscular strength and endurance, increase acceleration, raise lactic acid at maximal exercise, increase capacity of aerobic endurance, and stimulate metabolism by inducting a loss of body fat (Avois et al. 2006).

Cocaine, third commonly detected substance in the S6 section, is an ester of benzoic acid and methylecgonine. Cocaine is a natural tropane alkaloid mainly obtained from the leaves of the two Coca species, *Erythroxylum coca* and *Erythroxylum novogranatense*, that are indigenous to South America (Trifilieff and Martinez 2014). It is a potent CNS stimulant acting which action is mediated via inhibition of the reuptake of monoamine neurotransmitters (dopamine, norepinephrine, and serotonin) that is associated with increased monoaminergic neurotransmission (Pomara et al. 2012). Moreover, cocaine also exhibits local anesthetic effects via blockade of sodium channel and interference with action potential propagation. Cocaine has been recently approved by FDA as a nasal anesthetic. Administration of cocaine involves topical application, the drug can be also snorted, smoked, swallowed or injected by users. (Richards and Laurin 2023). Plasmatic half-life of cocaine is relatively short 0.5–1.5 hours (Pomara et al. 2012). Metabolism of cocaine, in addition to spontaneous hydrolysis, involves two main routes. One route, represented by methyl ester hydrolysis, provides benzoylecgonine via carboxylesterase hydrolases in the liver. Second route, represented by hydrolysis of benzoyl ester, is mediated by butyrylcholinesterase (BChE) to form ergonine methyl ester. In addition to these routes, cocaine also undergoes *N*-demethylation mediated by BChE which produces norcocaine (Connors and Hoffman 2013). Cocaine is a highly addictive drug and it is commonly misused as a recreational drug which is associated with a significant morbidity and mortality (Connors and Hoffman 2013). The death may be caused by respiratory failure or by asphyxiation if concomitant emesis is present. Other adverse and side effects associated with the use of cocaine include hyperreflexia, headache, mydriasis, abdominal pain, anxiety, restlessness, and excitement (Richards

Journal Pre-proof

and Laurin 2023). In sports, the first use of cocaine as a potential performance-enhancing compound, according to modern regulations, can be associated to racewalking competitions in 18th century. In the late 19th century, the use of cocaine has been frequently mentioned and its properties were associated with performance enhancement as well as fatigue reduction (Thevis et al. 2010b). A number of fatalities associated with coronary occlusion have occurred in athletes that misused cocaine. Mainly in athletes who intensively exercised after the drug administration. Combining of cocaine with alcohol or anabolic steroids may be also dangerous. More specifically, combining use of cocaine and alcohol is extreme cardiotoxic. These practices may lead to an increased risk of sudden death caused by cardiac arrest or seizures followed by respiratory arrest (Avois et al. 2006).

4.7.2 Specified stimulants (S6.B)

Specified stimulants, based on the 2023 prohibited list reported by WADA, are substances classified as stimulants that are not expressly listed in the class of non-specified stimulants (S6.A). This class includes, but is not limited to compounds such as methylhexanamine, methylphenidate, ephedrine, pseudoephedrine, sibutramine, heptaminol, strychnine, and others (WADA 2022d).

Methylhexanamine, second commonly detected substance in the S6 section, is an aliphatic primary amine which is also known under various names such as 2-amino-4-methylhexane, 1,3-dimethylpentylamine, 1,3-dimethylamylamine, or geranamine (Ocaña-Rios et al. 2022). Structure of the compound involves two chiral centers that provides four stereoisomers. The compound was previously claimed to be of natural origin and present in *Pelargonium* and *Geranium* species, however subsequent studies did not confirm presence of the compound in natural specimens (ElSohly et al. 2015; Kerpel dos Santos et al. 2019). Methylhexanamine acts as a CNS stimulant due to mimicking effects of epinephrine (Ocaña-Rios et al. 2022). The drug is usually administered orally. A single dose of 25 mg of the drug reaching peak concentration within 3–5 hours after ingestion and has a terminal half-life of 8.45 ± 1.9 hours (Schilling et al. 2013). Methylhexanamine has been previously added to dietary supplements due to its effects for weight loss and to accelerate various metabolic functions (Ocaña-Rios et al. 2022). In 2013, methylhexanamine has been banned by the FDA however, despite the prohibition, methylhexanamine has been detected in various dietary supplements (Cohen et al. 2018). Therefore, the drug may also represent danger to athletes resulting in positive finding. The drug has been associated with several deaths such as the deaths of two soldiers during a military training exercise in 2010 caused by heart attacks or a female runner during the 2012 London Marathon who collapsed and died near the finish (Derave and Tipton 2014).

Methylphenidate, fourth commonly detected substance in the S6 section, is a derivative of phenethylamine with incorporated piperidine ring. It is a chiral compound which involves two chiral centers. Originally, the drug has been marketed as a mixture of two racemates, consisting 80% of DL-*erythro*-methylphenidate and 20% of DL-*threo*-methylphenidate. Further pharmacological evaluation revealed that the stimulant activity resides only in DL-*threo*-methylphenidate however, both racemates exhibit similar hypertensive effects and lethal risk. In the present, DL-*threo*-methylphenidate is the only racemic form of methylphenidate in use. In the mixture, D-*threo*-methylphenidate which is a (*R*,*R*)-isomer is much more active than its enantiomer L-*threo*-methylphenidate with (*S*,*S*)-configuration. Thus, pharmaceutical formulations containing pure D-*threo*-methylphenidate are also available nowadays (Dinis-Oliveira 2017). Mode of action of methylphenidate includes blocking of the reuptake of neurotransmitters (dopamine and norepinephrine), agonist activity at serotonin type 1A receptor, as well as redistribution of the VMAT-2 (Faraone 2018). The drug is FDA-approved for treatment of ADHD in children and adults. Methylphenidate is also used as a second-line treatment for narcolepsy in adults. The drug is usually administered orally or applied as a transdermal patch. The peak plasma

Journal Pre-proof

concentration is reached in 1.9 hours for immediate-release tablets or in 4.7 hours for extended-release tablets. The major metabolic route is represented by ester hydrolysis resulting in pharmacologically inactive ritalinic acid. The elimination is mediated mainly by excretion in the urine, while small amounts are excreted in feces. Typical adverse effects of the drug are insomnia and nervousness. Long-term use of the drug causes a growth retardation in children. Other frequently observed side effects were observed in the CNS (dizziness, headache, tics), gastrointestinal (nausea and vomiting, dry mouth, decreased appetit) and cardiovascular (tachycardia and palpitations) systems (Verghese and Abdijadid 2023). In sports, methylphenidate demonstrated an effect on measures of physical performance in terms of power output and exercise performance in timed trials (Berezanskaya et al. 2022).

Ephedrine, fifth commonly detected substance in the S6 section, is a dimethyl derivative of phenylethanolamine which can be found in plants belonging to the genus *Ephedra* such as *Ephedra sinica* and *Ephedra equisetina* (Limberger et al. 2013). The compound has two chiral centers and thus exists in a form of four stereoisomers, specifically (1*R*,2*S*)-(-)-ephedrine, (1*S*,2*R*)-(+)-ephedrine, (1*S*,2*S*)-(+)-pseudoephedrine and (1*R*,2*R*)-(-)-pseudoephedrine. It is assumed that only two stereoisomers, (-)-ephedrine and (+)-pseudoephedrine, can be found in the natural sources. Stimulant properties of (-)-ephedrine are much higher than those of (+)-pseudoephedrine (Rice et al. 2018). Ephedrine acts as an inhibitor of neuronal norepinephrine reuptake which results in longer norepinephrine presence in the synaptic cleft. It also acts as an agonist of α and β-adrenergic receptors. The drug is FDS-approved and used for treatment of clinically significant hypotension. Ephedrine has also been used for treatment of asthma however, due to side cardiac effects it is only rarely used in modern medicine and replaced by more appropriate drugs. After oral administration of ephedrine its plasma elimination half-life is about 6 hours. The drug is metabolized to norephedrine and both, the metabolite and the parent drug, are excreted in the urine. Adverse effects of ephedrine includes palpitations, headache, dizziness, nausea, vomiting, restlessness, anxiety, and others (Statler et al. 2023). In sports, ephedrine is misused due to its stimulating effects. The drug is prohibited by WADA however, its prohibition includes exception. Ephedrine and methylephedrine are prohibited when the concentration of either in urine is greater than 10 µg/mL (WADA 2022d).

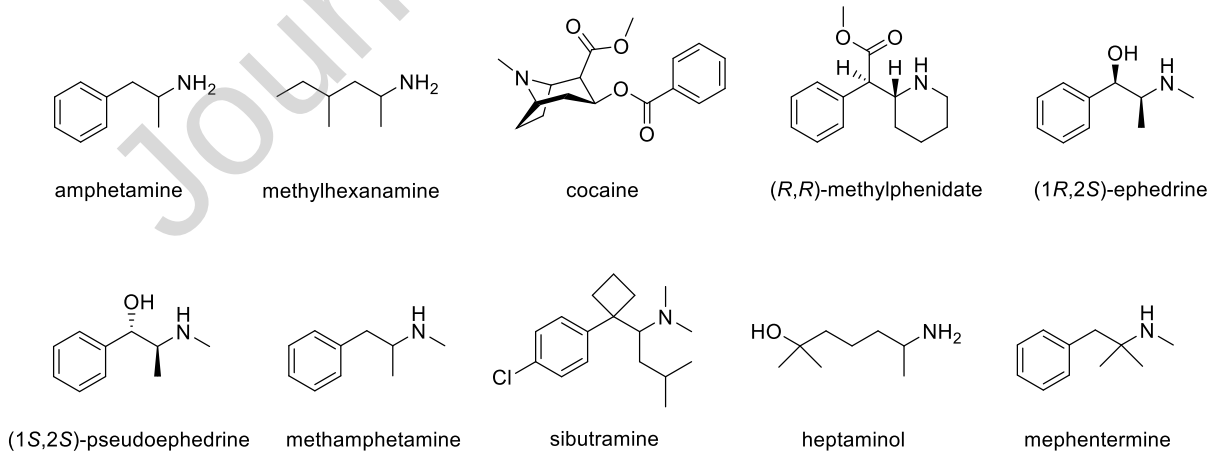


Figure 6. Structures of the most commonly detected stimulants in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

Journal Pre-proof

4.8 Narcotics (S7)

Narcotics or narcotic drugs was the term previously referred to any substance to dull senses and to relieve pain. However, nowadays the term refers to opioids (CDC 2021). In this context, term "opioid" encompasses compounds of natural, semi-synthetic, and synthetic origin. Although, natural and semi-synthetic opioids share a morphine-like structure, synthetic opioids include a variety of chemical classifications. Despite the differences in chemical structures, opioids exhibit similar mode of action (Kerrigan and Goldberger 2020). These compounds act via mimicking the actions of endogenous opioid peptides by interacting with μ-opioid peptide (MOP), δ-opioid peptide (DOP), or κ-opioid peptide (KOP) receptors. MOP receptors are associated with most of the clinical and adverse effects of opioids such as analgesia, sedation, euphoria, constipation, and respiratory distress. KOP receptors after activation also mediate analgesia, sedation, and dyspnea. DOP receptors, are less explored. In general, opioid receptors are G protein-coupled receptors and opioids can act as their agonists, partial agonists, or antagonists. Opioids acting as agonists produce a maximal response from the receptor such as analgesia mediated by morphine. Activation of opioid receptors initiates a cascade which activates G proteins and inhibits adenylyl cyclase that leads to alterations in gene expression. Opioid receptors mediated their effects on both presynaptic and postsynaptic neurons. On presynaptic neurons, opioids close voltage-gated Ca^{2+} channels which leads to decreased release of pain neurotransmitters such as glutamate and substance P. This action results in the reduction of pain signals. In the ventral tegmental area, presynaptic opioid receptors inhibit release of γ-aminobutyric acid (GABA) which causes dopaminergic neurons to fire more frequently and provides feeling of euphoria. On postsynaptic neurons, opioid receptors open K^+ channels which leads to cell hyperpolarization that inhibits cell signaling. Opioids acting as partial agonists bind to the receptors however, they provide only a partial functional response no matter the amount of administered drug is, an example of the partial agonist is buprenorphine. Opioids acting as antagonists bind to the receptor without producing of a functional response, while at the same time also prevent binding of an agonist to the receptor, an example is opioid naloxone (Pathan and Williams 2012). Naloxone, a competitive antagonist of opioid receptors with a high affinity toward MOP receptor, is approved by the FDA for treatment of opioid overdose (Jordan and Morrisonponce 2023). Several opioids are medicinally used to manage and treat pain, cough, or diarrhea. Common adverse effects associated with use of narcotics include sedation or CNS depression, dizziness, nausea, vomiting, or constipation. Chronic use of opioids causes a physical dependence which often includes withdrawal symptoms when administration of opioids is discontinued or decreased. Withdrawal symptoms include rhinorrhea, lacrimation, hyperventilation, hyperthermia, vomiting, diarrhea, muscle pain, and anxiety. Chronic use of opioids is further associated with tolerance, when a person which repeated doses of opioids requires an increased meditation to experience the same analgesic effects. Other adverse effect related to use of opioids is respiratory depression, which in patients suffering from respiratory diseases may result in death (Grewal and Huecker 2023). In sports, narcotics are used due to their analgesic effects to overcome pain caused by strenuous exercise during training or in competition (Thuyne and Delbeke 2008). Narcotics have been included to the list of prohibited products reported by the IOC in 1968. In the 2023 prohibited list reported by WADA, narcotics (S7 section) are incorporated as substances prohibited in-competition. The section includes 12 compound or their derivatives as well as all optical isomers of the substances. However, it is worth mention that from 1st January 2024, tramadol, an opioid pain modulator used to treat moderate to moderately severe pain, was included to the 2024 prohibited list into the S7 section. Tramadol is prohibited in competition for a suspicion of its potential misuse across multiple sports in order to reduce exertional pain and allow an athlete to work harder. It is proposed that tramadol is being misused to provide a performance advantage to an athlete (WADA 2019b, 2023g). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S7 section represents ninth commonly detected section of substances misused as a

Journal Pre-proof

potential doping in all sports, representing only 0.6% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of narcotics (S7) is represented by: morphine (58.1%), oxycodone (14.3%), methadone (10.3%), buprenorphine (4.2%), fentanyl and fentanyls (4.2%), hydromorphone (3.8%), oxymorphone (1.9%), pethidine (1.5%), pentazocine (0.8%), heroin (0.6%), dextromoramide (0.2%), and hydrocodone (0.2%). Findings of the commonly detected substances of the S7 section are summarized in Table 8, graphically expressed in Chart 8, and their structures are shown in Figure 7.

Table 8. Narcotics in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes to findings: heroin – includes one finding reported in 2010 as "diamorphone".

	morphine	oxycodone	methadone	buprenorphine	fentanyl and fentanyls	hydromorphone	oxymorphone	pethidine	pentazocine	heroin	dextromoramide	hydrocodone	Total (annually)	Percentage of total (annually)
2003	22		1			1		2					**26**	**5.0%**
2004	10	1	2			1					1		**15**	**2.9%**
2005	15		1			1							**17**	**3.2%**
2006	11		3			1		1					**16**	**3.0%**
2007	19		1			1							**21**	**4.0%**
2008	21	1	2			2		1		1			**28**	**5.3%**
2009	17	4	1	1		1							**24**	**4.6%**
2010	6	3	2	2	1	2	1		1	1		1	**20**	**3.8%**
2011	11	2	6			1							**20**	**3.8%**
2012	14	6	1		5								**26**	**5.0%**
2013	25	6	5	2	4			1					**43**	**8.2%**
2014	17	6	2	4	3	1		1					**34**	**6.5%**
2015	13	7	1	2	1		3						**27**	**5.1%**
2016	31	8	6	3	2	1	2		1				**54**	**10.3%**
2017	51	7	4	3	3	5							**73**	**13.9%**
2018	8	8	4	1			3						**24**	**4.6%**
2019	6	7	8	4	2	2			1				**30**	**5.7%**
2020	1	3											**4**	**0.8%**
2021	7	6	4		1		1	2	1	1			**23**	**4.4%**
Total (2003-2021)	**305**	**75**	**54**	**22**	**22**	**20**	**10**	**8**	**4**	**3**	**1**	**1**	**525**	
Percentage of total	**58.1%**	**14.3%**	**10.3%**	**4.2%**	**4.2%**	**3.8%**	**1.9%**	**1.5%**	**0.8%**	**0.6%**	**0.2%**	**0.2%**		

Journal Pre-proof

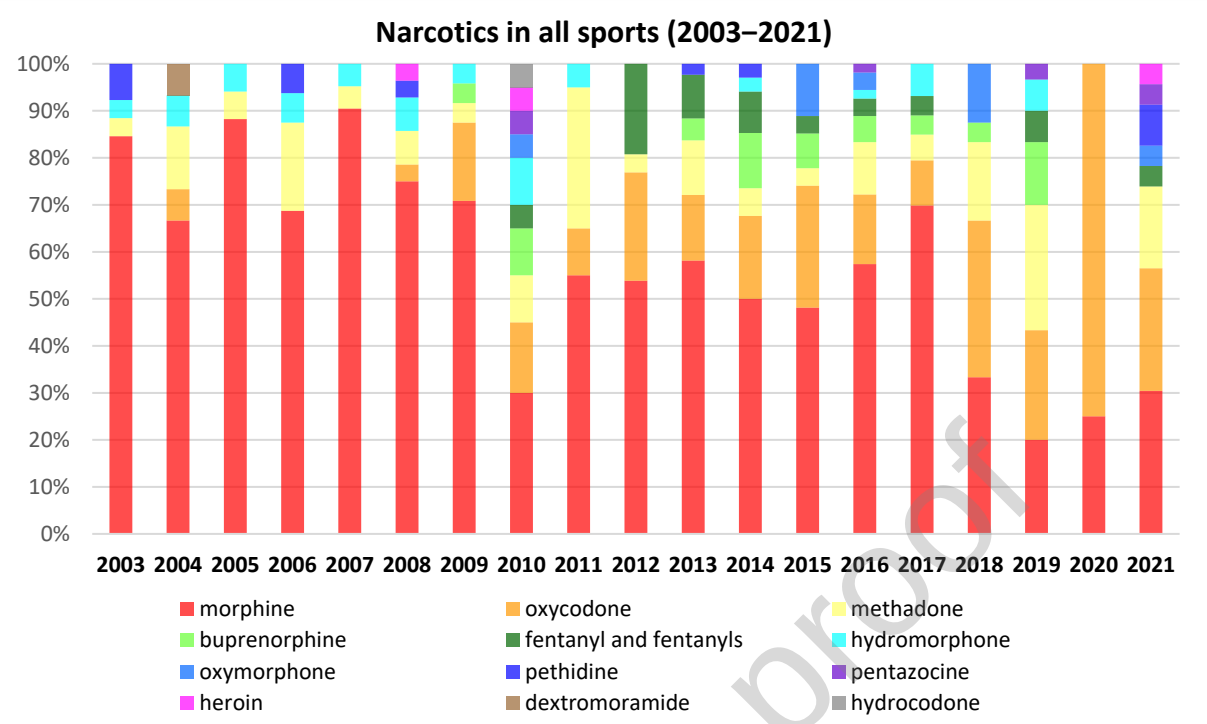


Chart 8. Narcotics in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021.

Morphine, the most commonly found substance in the S7 section, is a natural alkaloid which is the most abundant opiate found in *Papaver somniferum*. Structure of morphine contains pentacyclic scaffold with five chiral centers. Morphine exhibits affinity to MOP, DOP, and KOP receptors. Its analgesic effect is mediated mainly by binding and activating of MOP receptors. Due to an analgesic effect, morphine is approved by the FDA to manage moderate to severe pain that may be both, acute or chronic. Morphine is considered as the classic opioid analgesic with which other painkillers are compared. Administration of the drug usually includes oral, intravenous, epidural, and intrathecal route (Murphy et al. 2023). Peak plasma concentration of morphine is achieved within 15–20 minutes after intramuscularly or subcutaneously administered drug, and within 30–90 minutes after oral administration. Orally administered morphine undergoes extensive first-pass metabolism in the liver. Morphine has the elimination half-life about 2 hours (ranging 1–8 hours) which is independent from the administration route (Lugo and Kern 2002). Metabolism of morphine occurs mainly in the liver by uridine-5'-diphosphate glucuronosyltransferase with specific affinity for the UGT2B7 isoenzyme. Morphine is transferred into two main metabolites, both are the products of glucuronidation. The major metabolite (90%), morphine-3-glucuronide, is analgesically inactive, however it antagonizes morphine and produces stimulatory effects such as myoclonus, seizure, and allodynia. The minor metabolite (10%), morphine-6-glucuronide, exhibits much higher analgesic effect than the parent drug (De Gregori et al. 2012). Common adverse effects associated with morphine are constipation, CNS depression, nausea, vomiting, and urinary retention. The use of morphine is further associated with respiratory depression (Murphy et al. 2023). Chronic morphine abuse leads to physical and psychological dependence (Listos et al. 2019). In sports, morphine acting as a potent analgesic is prohibited by WADA due to its effects to increase tolerance to pain, thereby improving physical performance (Benedetti et al. 2007). Interestingly, in contrast to morphine, codeine which is also known as 3-methylmorphine is not classified as a prohibited substance. However, codeine can undergo a minor metabolic route in the liver via cytochrome P450, isoenzyme 2D6 (CYP2D6), providing

morphine as a metabolite. To determine origin of morphine found in a doping control, morphine/codeine ratio can be used to distinguish the consumption of codeine from morphine abuse (Seif-Barghi et al. 2015).

Oxycodone, second commonly detected substance in the S7 section, is a semi-synthetic opioid with morphine-like structure. Similar as morphine, oxycodone also acts as an agonist of MOP, KOP, and DOP receptors, with the higher activity for MOP receptors. Oxycodone causes hyperpolarization and reduced excitability of neurons in the CNS. As the drug, oxycodone is approved by the FDA to manage moderate to severe pain that can be acute or chronic. The drug is available in the form of tablets, capsules, and oral solution formulations. Its onset of action is 10–30 minutes for the immediate-release formulations, or about 1 hour for the controlled-release formulations. Duration of the action is 3–6 hours (immediate-release) or 12 hours (controlled-release). Oxycodone has a plasmatic half-life of 3–5 hours (Sadiq et al. 2023). Metabolism of oxycodone occurs in the liver and is mediated by CYP3A4 and cytochrome P450 isoenzyme 3A5 (CYP3A5) that produce metabolite noroxycodone via *N*-demethylation (45%), and CYP2D6 which produces oxymorphone via *O*-demethylation (19). Whereas, noroxycodone exhibits lower affinity (about 1/3) for MOP receptors and its potency is 5–10 times lower in comparison with the parent drug, oxymorphone has 10–45 times higher affinity for MOP receptor and is about 8–30 times more potent than oxycodone. The both metabolites are further metabolized to noroxymorphone, which exhibits 2–4 times higher affinity for MOP receptor and is 2 times more potent that oxycodone. Noroxymorphone does not cross the BBB, thus it is unlikely that the metabolite contributes to the analgesic effect. Oxycodone and its metabolites are excreted mainly via the kidney as follows: oxycodone 9–11%, noroxycodone 23%, oxymorphone 10%, noroxymorphone 14%, and reduced metabolites ≤18% (Kinnunen et al. 2019). Common adverse effects for oxycodone use are asthenia, constipation, dizziness, dry mouth, headache, nausea, vomiting, pruritus. Other side effects are bradycardia, hypotension, confusion, hallucinations, sedation, seizures, respiratory depression. The drug also has a potential for addiction (Sadiq et al. 2023). In sports, oxycodone is increasingly used among athletes as well as young students for non-medicinal and recreational use, due to its action on opioid receptors (Adami et al. 2022).

Methadone, third commonly detected substance in the S7 section, is structurally 6-(dimethylamino)-4,4-diphenylheptan-3-one which is a potent synthetic opioid with analgesic activity. Its structure significantly differs from morphine-like opioids. It is a chiral compound involving one chiral center. Methadone acts as a MOP receptor agonist, a serotonin and norepinephrine reuptake inhibitor, and a non-competitive antagonist of *N*-methyl-D-aspartate (NMDA) receptor. Methadone is available as a racemic mixture and the both enantiomers exhibit different pharmacokinetics. (*R*)-Methadone is primarily responsible for the opioid effect with minor antagonism on NMDA receptors, whereas (*S*)-methadone causes an inhibition of serotonin and norepinephrine reuptake as well as antagonism of the NMDA receptor. Mean bioavailability of the drug is about 75%. After intake, the drug undergoes first-pass metabolism. The plasma peak concentration is reached within 4.4–6 hours in patients, while in healthy volunteers the concentration is reached in 2.8 hours. Methadone is a lipophilic drug which is further distributed into the brain, liver, kidney, lung, and muscles. The drug is mainly metabolized in the liver, primarily by CYP3A4 which shows no enantioselectivity. In addition, it is suggested that also other enzymes are significantly involved in hepatic metabolism of the drug, such as cytochrome P450 isoenzymes 2B6 (CYP2B6) and 2C19 (CYP2C19), that exhibit enantioselective metabolism of the drug. (*S*)-Methadone is metabolized mainly by CYP2B6, while metabolism of (*R*)-methadone is preferred by CYP2C19. The clearance of unbound methadone is stereoselective. (*S*)-Methadone is cleared faster than its (*R*)-enantiomer. The major metabolite of methadone, *N*-demethyl methadone, is rapidly converted into 2-ethylidene-1,5-dimethyl-3,3-diphenylpyrrolidine (EDDP) (Guevara et al. 2018). Adverse effects of methadone involve diaphoresis, pruritis, nausea, dry mouth, constipation, sedation,

lethargy, respiratory depression, and others (Durrani and Bansal 2023). In sports, can be misused by athletes due to its opioid effects. In the XV Paralympic Games, held in Rio de Janeiro in 2016, methadone has been reported as the major AAFs found in the Paralympic samples, representing 4 of 12 AAFs. Curiously, the four athletes declared the use of methadone in the doping control forms (Pereira et al. 2017).

Buprenorphine, fourth commonly detected substance in the S7 section, is a semi-synthetic opioid derived from thebaine. Buprenorphine acts as a partial MOP receptor agonist, resulting only in partial activation of the receptor. Furthermore, it is a weak KOP receptor antagonist and DOP receptor agonist. Its mode of action results in many unique properties of the drug, such as its analgesic effects plateau at higher doses, and then its effects become antagonistic. Buprenorphine also exhibits ceiling effects on respiratory depression, thus it is safer than methadone for agonist substitution treatment in addiction. Buprenorphine is approved by the FDA for manage of acute and chronic pain as well as opioid dependence. Administration of buprenorphine include oral, sublingual, transdermal, intravenous, and intramuscular route (Kumar et al. 2023). Orally administered buprenorphine undergoes rapid first-pass metabolism however, the drug administered sublingually represents a more feasible route of administration. Sublingual administration results in the maximal plasma concentrations in a range from 40 minutes to 3.5 hours. Metabolism is mediated mainly by the CYP3A4 via *N*-dealkylation to produce norbuprenorphine. The terminal elimination half-life of the drug is long and in a variable range of 3–44 hours. The drug is excreted mainly in the feces and about 30–40% of the dose is excreted in the urine (Elkader and Sproule 2005). Adverse effects of the drug include CNS depression, hypotension, lower seizures threshold, nausea, vomiting, drowsiness, dizziness, headache, and others. Buprenorphine also has a mild addictive potential (Kumar et al. 2023). In sports, buprenorphine can be used for pain control, similar as other opioids. Due to a low therapeutic dose as well as a long elimination period of buprenorphine, the drug has a low concentration in the urine. Therefore, immunoanalytical methods are usually used for its detection (Malve 2018).

Fentanyl and its derivatives often termed as fentanyls, fifth commonly detected substances in the S7 sections, represent a group of compounds acting as potent opioids. Fentanyl is a synthetic opioid, a representant of the 4-anilidopiperidine class of synthetic opioids (Vuckovic et al. 2009). Fentanyl acts as a potent MOP receptor agonist. It also exhibits lowered affinity to KOP and DOP receptors. Fentanyl produces a potent analgesic effect, it exhibits about 100-fold higher potency than morphine. Fentanyl is clinically used as a sedative in intubated patients, to treat severe pain in patients with renal failure due to its primarily hepatic elimination. The drug can be also used to treat chronic pain in patients who have developed tolerance to opiates. Administration of fentanyl involves intravenous, intramuscular, transdermal, intranasal, and intrathecal route (Ramos-Matos et al. 2023). Sublingual administration of fentanyl has a calculated bioavailability of 54%. The peak plasma concentration ranges from 22.5 to 240 minutes. The elimination half-life is about 7 hours. Metabolism of fentanyl is mediated in the liver, via *N*-dealkylation by CYP3A4 and CYP3A5, which produces norfentanyl as an inactive and non-toxic metabolite. Small amount (<1%) of the drug can undergoes alternative metabolic routes leading to inactive metabolites such as hydroxynorfentanyl and despropionylfentanyl. Excretion occurs mainly in the urine, further it can be excreted in the feces or by exhalation (Saiz-Rodríguez et al. 2019). Adverse effects associated with the use of fentanyl include euphoria, confusion, respiratory depression which may lead to respiratory arrest, drowsiness, nausea, hallucinations, delirium, and others (Ramos-Matos et al. 2023). In 2017, the US Department of Health and Human Services declared the opioid crisis in the USA to be a public health emergency (Irvine et al. 2022). Fentanyl has become a primary driver of the deaths in opioid crisis in the USA, consequently affecting social, economic, and health systems worldwide (Han et al. 2022). However, in sports, the

Journal Pre-proof

findings for fentanyl and other fentanyls reported by WADA during 2003–2021 are not associated with an increased misuse of the substances (Table 8).

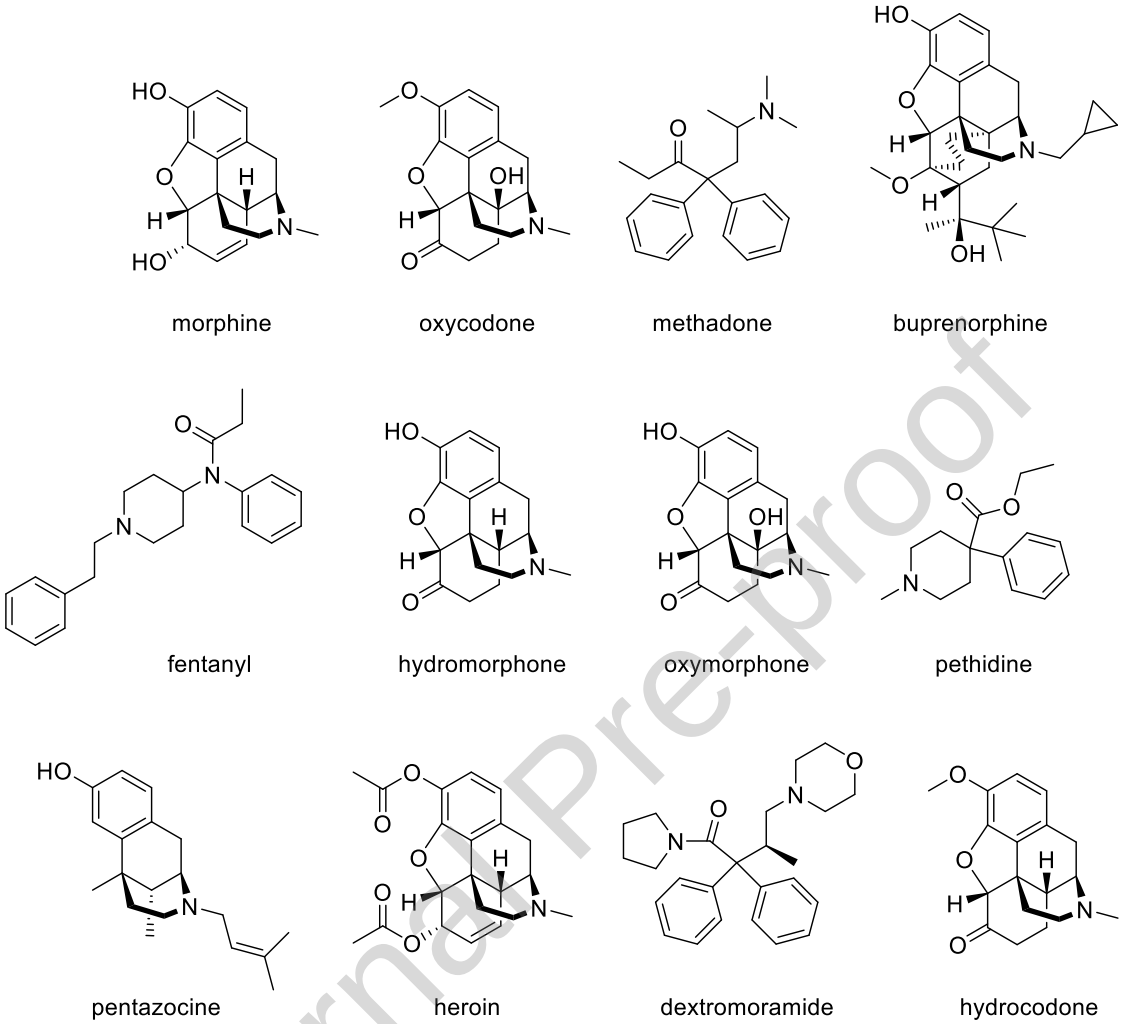


Figure 7. Structures of the most commonly detected narcotics in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.9 Cannabinoids (S8)

Cannabinoids are in general bioactive compounds that bind to cannabinoid receptors. Cannabinoids can be classified into three classes: phytocannabinoids are compounds that occur naturally in plants; endocannabinoids are endogenously produced neurotransmitters that act in the central and peripheral nervous system; synthetic cannabinoids are synthetic derivatives of phytocannabinoids or endocannabinoids. In human, cannabinoids act mainly on the cannabinoid system which consists of cannabinoid receptors, endocannabinoids, and associated enzymes. There are two types of cannabinoid receptors: cannabinoid receptor type 1 (CB_1) and type 2 (CB_2), that belong to the G protein-coupled receptors. The CB_1 receptor is expressed mainly in the CNS; in areas regulating appetite, memory, fear extinction, motor responses; but it is also present in gastrointestinal, liver, adipocyte, and skeletal muscle cells. The CB_2 receptor is common to the periphery with significant expression in the immune system (Kwasnik et al. 2020). Endocannabinoids acts as neuro-regulatory modulators responsible for retrograde neurotransmission. Endocannabinoids released from the postsynaptic neuron bind predominantly to CB_1 receptors on the presynaptic neuron. This binding

Journal Pre-proof

results in the inhibition of neurotransmitter release from the presynaptic neuron. If the presynaptic neurotransmitters have mainly an inhibitory effect, such as GABA, then the final effect is excitatory and *vice versa*. Adverse effects of cannabinoids range from mild to lethal. The most commonly side effects include euphoria, anxiolysis, tachycardia, visuotemporal distortion. Further, more severe symptoms that can be associated with the use of cannabinoids are panic attacks, psychosis, myoclonus, bronchospasm, and others. Chronic abuse of cannabinoids at higher doses may be associated with addiction, cognitive impairment in adolescent, chronic bronchitis, lung cancer, increased risk of myocardial infarction, stroke, thromboembolic events, and others (Sheikh and Dua 2023). In sports, cannabinoids can reduce anxiety but they do not have ergogenic potential in activities. In athletes, cannabis products have been observed to exhibit an ergolytic activity in different sport categories (Campos et al. 2003). Cannabinoids are included in the 2023 prohibited list reported by WADA, as the S8 section. Cannabinoids are prohibited in-competition and the section includes all natural and synthetic cannabinoids, for example in cannabis and cannabis products, natural and synthetic tetrahydrocannabinols (THCs), and synthetic cannabinoids that mimic the effects of tetrahydrocannabinol (THC) (WADA 2022d). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S8 section represents fourth commonly detected section of substances misused as a potential doping in all sports, representing 7.1% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of cannabinoids (S8) is represented mainly by: THC or its metabolites (99.5%), then followed by JWH-018 (0.3%) and JWH-073 (0.1%). Each of other representants such as 1-(5-fluoropentyl)-1*H*-indole-3-carboxylic acid, 11-nor-9-carboxy-Δ^8-THC, AB-CHMINACA, cannabimimetics "Spice", STS-135 metabolites, and UR-144-*N*-pentanoic acid, each representing <0.1% of total findings in the S8 section. Findings of the commonly detected substances of the S8 section are summarized in Table 9, graphically expressed in Chart 9, and their structures are shown in Figure 8.

Table 9. Cannabinoids in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes to findings: cannabimimetics "Spice" - one finding reported in 2010 as "cannabimimetics (Spice)" without additional determination; THC or its metabolites – includes findings reported during 2003–2005 as "cannabis", during 2006–2009 reported as "cannabinoids", and since 2010 reported as "carboxy-THC (11-nor-9-carboxy-Δ^9-tetrahydrocannabinol)". Abbreviations: 11-nor-9-carboxy-Δ^8-THC, 11-nor-9-carboxy-Δ^8-tetrahydrocannabinol; STS-135, *N*-(adamantan-1-yl)-1-(5-fluoropentyl)-1*H*-indole-3-carboxamide; THC, tetrahydrocannabinol.

	THC or its metabolites	JWH-018	JWH-073	1-(5-fluoropentyl)-1*H*-indole-3-carboxylic acid	11-nor-9-carboxy-Δ^8-THC	AB-CHMINACA	cannabimimetics "Spice"	STS-135 metabolites	UR-144-*N*-pentanoic acid	Total (annually)	Percentage of total (annually)
2003	378									**378**	**6.4%**
2004	518									**518**	**8.8%**
2005	503									**503**	**8.6%**
2006	553									**553**	**9.4%**
2007	576									**576**	**9.8%**
2008	496									**496**	**8.5%**
2009	399									**399**	**6.8%**
2010	532						1			**533**	**9.1%**
2011	442	3								**445**	**7.6%**
2012	398	6	2							**406**	**6.9%**
2013	175	8	4					1		**188**	**3.2%**
2014	72								1	**73**	**1.2%**
2015	124			2		1				**127**	**2.2%**

Journal Pre-proof

2016	110									**110**	**1.9%**
2017	154									**154**	**2.6%**
2018	141									**141**	**2.4%**
2019	130									**130**	**2.2%**
2020	42									**42**	**0.7%**
2021	91				1					**92**	**1.6%**
Total (2003-2021)	**5,834**	**17**	**6**	**2**	**1**	**1**	**1**	**1**	**1**	**5,864**	
Percentage of total	**99.5%**	**0.3%**	**0.1%**	**0.0%**	**0.0%**	**0.0%**	**0.0%**	**0.0%**	**0.0%**		

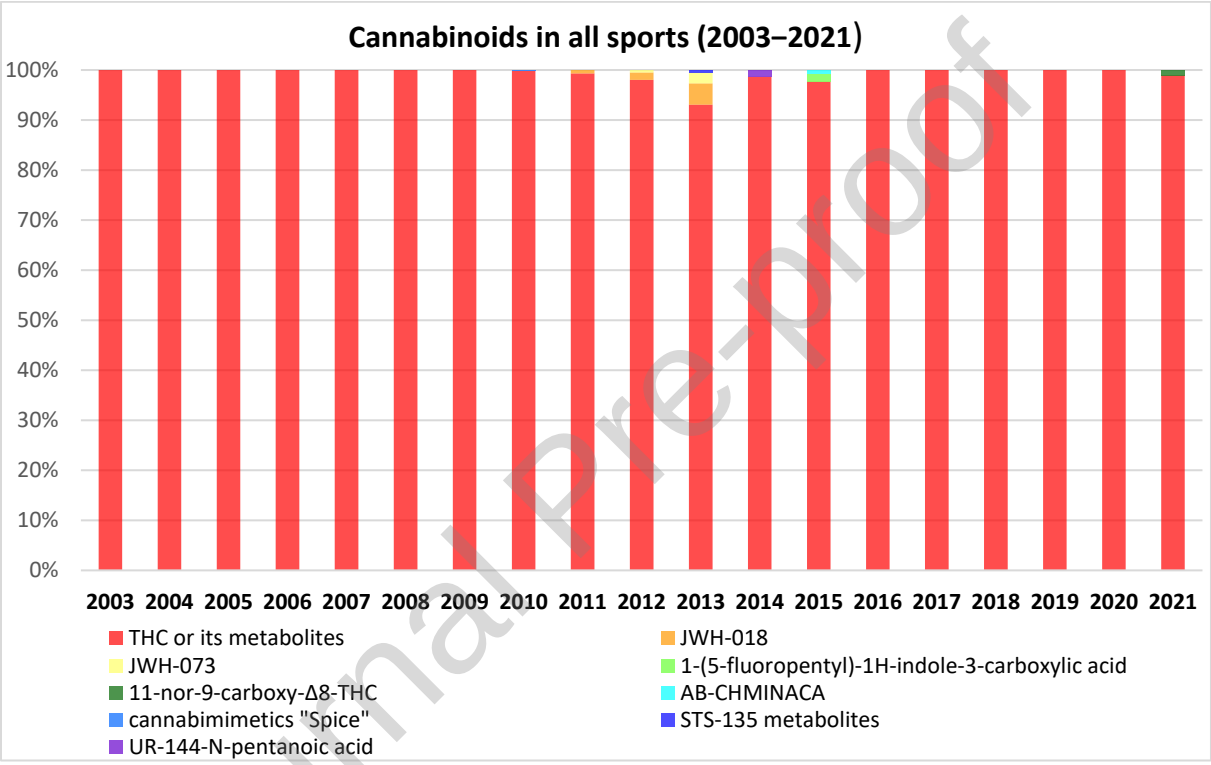


Chart 9. Cannabinoids in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Abbreviations: 11-nor-9-carboxy-Δ^8-THC, 11-nor-9-carboxy-Δ^8-tetrahydrocannabinol; STS-135, *N*-(adamantan-1-yl)-1-(5-fluoropentyl)-1*H*-indole-3-carboxamide; THC, tetrahydrocannabinol.

Journal Pre-proof

Tetrahydrocannabinol (THC), also known as Δ^9-tetrahydrocannabinol, or its metabolites represent the most frequently detected substance in the S8 section. THC, a tricyclic phytocannabinoid, is the main psychoactive compound in *Cannabis sativa* (Alves et al. 2020). Mode of action is based on the binding to CB_1 and CB_2 receptors. Effects of THC on emesis, appetite, and pain are attributed to its binding to CB_1 receptors in the CNS. THC is approved by the FDA to manage chemotherapy-induced nausea and vomiting, and for stimulation of appetite in patients with acquired immunodeficiency syndrome (AIDS)-related anorexia (Ng and Gupta 2023). Pharmacokinetics of THC differ according to route of administration. THC can be smoked, consumed, or it can also be administered intravenously. Orally administered THC has a limited bioavailability (10–20% when ingested from gelatin capsules or 3–9% when ingested from chocolate cookie). Bioavailability of smoked THC is highly variable of 2–56%, mean 12–24%. The peak concentration of THC is reached within 1–3 hours after oral intake, while after smoking it is reached only within minutes. Metabolism of THC in humans is mediated mainly by cytochrome P450 isoenzyme 2C9 (CYP2C9) producing 11-hydroxy-Δ^9-THC which is even more

pharmacologically active than the parent drug. The metabolite is further metabolized to 11-nor-9-carboxy-Δ^9-THC which is not a potent cannabinoid. Both metabolites are further conjugated with glucuronic acid. The minor metabolic pathway of THC is mediated by CYP3A4 to form 8-hydroxy-Δ^9-THC (Grant et al. 2018). Adverse effects of THC are dysphoria, hallucinations, paranoia, sedation, confusion, headache, dry mouth, hypotension, seizures, and other. Use of marijuana is also associated with an increased risk for psychiatric conditions such as psychosis, depression, anxiety, or schizophrenia (Ng and Gupta 2023). In sports, findings of THC or its metabolites represent the majority of findings reported in the S8 section (Table 9). In a systematic review, recently reported by Docter et al., the authors reviewed use of cannabis in sports and concluded that does not appear to positively affect performance in athletes. Due to variability in regulation of cannabis use in different sport types and competitions, as well as the growing number of states that decided to legalize recreational use of cannabis, positive findings of THC and its metabolites may be further detected in future years (Docter et al. 2020).

JWH-018, JWH-073, and 1-(5-fluoropentyl)-1*H*-indole-3-carboxylic acid; representing second, third, and fourth commonly detected substances in the S8 section, respectively. JWH-018 are JWH-073 synthetic cannabinoids belonging to the naphthoylindole class (Chimalakonda et al. 2011). 1-(5-fluoropentyl)-1*H*-indole-3-carboxylic acid, also known as 5F-PB-22 3-carboxyindole, is one of the main metabolites of quinolin-8-yl 1-(5-fluoropentyl)indole-3-carboxylate (5F-PB-22), a synthetic cannabinoid having quinolinyl carboxylate structure with incorporated indole ring (Franz et al. 2016). JWH-018 is a potent agonist of the both types of cannabinoid receptors, CB_1 and CB_2, exhibiting at least 3-fold higher binding affinity to the both receptors than THC (Elmore and Baumann 2018). JWH-073 also exhibits a high binding affinity to both cannabinoid receptors, CB_1 and CB_2. JWH-073 is structurally similar and produces qualitatively similar effects as JWH-018, but it is about 50% less potent than JWH-018. However, JWH-073 has a prolonged effect compared to JWH-018, due to its extended half-life and less receptor internalization. JWH-073 was introduced on the illicit market after recognition addictive effects of JWH-018 and its following prohibition in 2008 (Barbieri et al. 2022). 5F-PB-22 acts as an agonist on the both, CB_1 and CB_2, receptors (Banister et al. 2015). 5F-PB-22 first appeared in 2013 and later it has been detected in products obtained in smoke shops, online vendors, and retail outlets (Alkaseem and Baron 2018). In 2014, the use of 5F-PB-22 was associated with 4 deaths (Behonick et al. 2014). Since May 2016, the sale, distribution, import, and purchase of the drug are banned in the United Kingdom (UK) (Alkaseem and Baron 2018).

11-nor-9-carboxy-Δ^8-THC is the major metabolite of Δ^8-THC (Sim et al. 2023), a phytocannabinoid found in *Cannabis sativa* which is an isomer of Δ^9-THC (THC) (Shaker et al. 2023). Pharmacology of the both, Δ^8-THC and Δ^9-THC, is very similar in humans, however Δ^8-THC exhibits weaker potency than Δ^9-THC in different assays. Metabolism of the both phytocannabinoids is also similar (Tagen and Klumpers 2022).

Journal Pre-proof

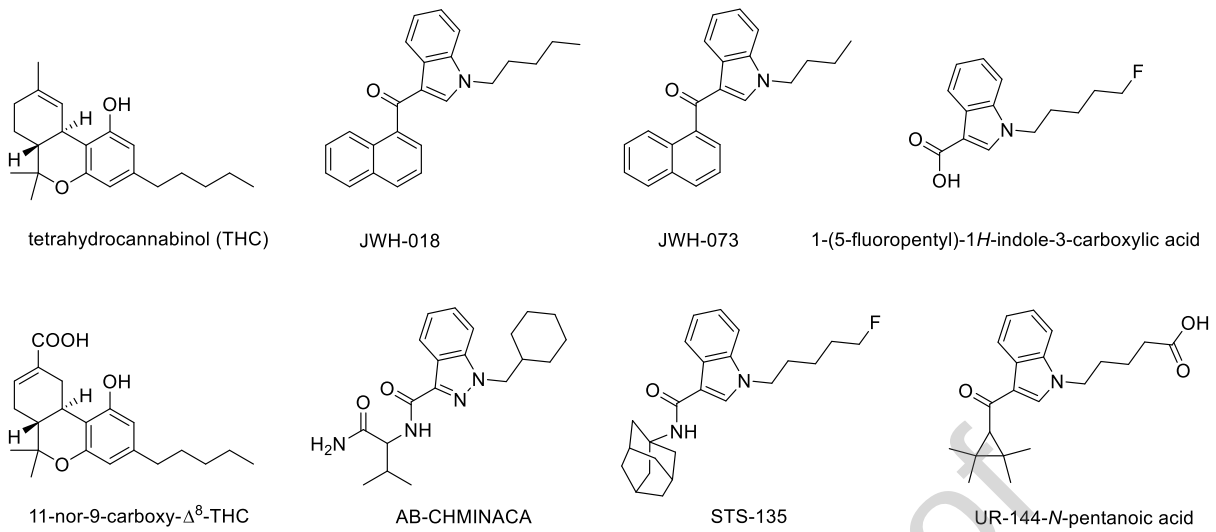


Figure 8. Structures of the most commonly detected cannabinoids in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.10 Glucocorticoids (S9)

Glucocorticoids are steroid hormones that are produced and secreted by the adrenal gland that have diverse effects on many physiological functions (Chourpiliadis and Aeddula 2023). In the human body, cortisol is the main glucocorticoid which regulates various functions such as metabolism, inflammatory response, immune function, it also mediates the stress response. Cortisol, an active glucocorticoid, is converted by 11β-hydroxysteroid dehydrogenase type 2 (11β-HSD2) in the kidney and pancreas to its inactive form cortisone. On the contrary, the activation of inactive cortisone is mediated by 11β-hydroxysteroid dehydrogenase type 1 (11β-HSD1) in most tissues to cortisol (Thau et al. 2022). Glucocorticoids play a role in metabolic, inflammatory, cardiovascular, and behavioral processes. Their mode of action is mediate via glucocorticoid receptors (GRs), members of nuclear receptor superfamily. Activation of the GRs controls expression of thousands of genes. In the absence of glucocorticoids, glucocorticoid receptor (GR) is located mainly in the cytoplasm, inside a multi-protein complex. After the binding of glucocorticoids, GR dissociates from the complex and is translocated into the nucleus, binds to glucocorticoid-responsive elements and regulates expression of target genes (Akalestou et al. 2020). Glucocorticoids have anti-inflammatory, anti-allergic, immunosuppressive functions that are significant for medicinal purposes (Chourpiliadis and Aeddula 2023). The use of glucocorticoids, especially at higher doses and longer duration of treatment, are associated with adverse effects such as diabetes mellitus/glucose intolerance, hypertension, osteoporosis, obesity, infections, and others (Spies et al. 2011). Interestingly, in contrast to anabolic steroids that are misused by athletes to increase muscle mass, glucocorticoids are catabolic steroids that cause peripheral muscle breakdown (Chourpiliadis and Aeddula 2023). In sports, glucocorticoids are prohibited in-competition. Although there is no evidence that short-term use of glucocorticoids enhances performance in sports, it is suggested that use of orally administered glucocorticoids at high doses in week-long courses may improve submaximal intensity exercise performance in athletes. However, these dosages would be easily detected in anti-doping analysis, if taken in-competition. The exact mechanism of this effect remains unclear, but it is suggested that results from combined actions on metabolism, muscle, inflammation, and the nervous system (Vernec et al. 2020). As reported in the 2023 prohibited list by WADA, all glucocorticoids (incorporated as the S9 section) are prohibited when administered by any injectable, oral (including oromucosal, for example buccal, gingival, sublingual), or rectal route. Other routes of administration are not prohibited when used within the manufacturer's

Journal Pre-proof

licensed doses and therapeutic indications. The non-prohibited routes include inhalation and topical administration, including dental-intracanal, dermal, intranasal, ophthalmological, otic, and perianal application. Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the S9 section represents fifth commonly detected section of substances misused as a potential doping in all sports, representing 6.2% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of glucocorticoids (S9) is represented by: budesonide (26.4%), prednisolone (16.3%), triamcinolone acetonide (14.5%), betamethasone (14.4%), prednisone (12.1%), dexamethasone (5.1%), methylprednisolone (5.0%), prednisolone/prednisone (3.5%), 16α-hydroxyprednisolone (0.8%), triamcinolone (0.7%), and others (1.0%). Findings of the commonly detected substances of the S9 section are summarized in Table 10, graphically expressed in Chart 10, and their structures are shown in Figure 9.

Table 10. Glucocorticoids in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes to findings: prednisolone – includes three findings reported as "the GC/C/IRMS result for prednisolone is consistent with an exogenous origin"; prednisolone/prednisone – includes one finding reported in 2018 as "the GC/C/IRMS result for prednisone and prednisolone is consistent with an exogenous origin".

	budesonide	prednisolone	triamcinolone acetonide	betamethasone	prednisone	dexamethasone	methylprednisolone	prednisolone/prednisone	16α-hydroxyprednisolone	triamcinolone	others	Total (annually)	Percentage of total (annually)
2003	4	52	131	80		7	12					**286**	**5.6%**
2004	50	61	246	121	28	17	25					**548**	**10.7%**
2005	116	35	25	47	23	13	22		42		2	**325**	**6.4%**
2006	142	30	18	45	20	9	9	9				**282**	**5.5%**
2007	154	16	18	42	7	6	10	35				**288**	**5.6%**
2008	151	37	16	39	28	14	14	13		4		**316**	**6.2%**
2009	120	16	1	39	3	17	9	41	1	12	6	**265**	**5.2%**
2010	111	16	7	27	9	8	7	39		6	4	**234**	**4.6%**
2011	113	19	16	25	19	21	16	40		2	3	**274**	**5.4%**
2012	157	67	16	30	60	18	15			1	1	**365**	**7.2%**
2013	135	58	12	35	55	18	14			1	2	**330**	**6.5%**
2014	74	56	16	34	44	12	14			1	1	**252**	**4.9%**
2015	7	60	17	31	52	19	19			1	9	**215**	**4.2%**
2016	1	51	18	29	48	13	11			2	11	**184**	**3.6%**
2017	2	71	31	23	57	20	13			1	7	**225**	**4.4%**
2018	5	79	72	34	70	15	8	2				**285**	**5.6%**
2019	5	61	44	23	57	11	20			5	6	**232**	**4.5%**
2020		16	16	14	13	6	4					**69**	**1.4%**
2021	1	28	20	18	26	18	11			2	1	**125**	**2.5%**
Total (2003-2021)	**1,348**	**829**	**740**	**736**	**619**	**262**	**253**	**179**	**43**	**38**	**53**	**5,100**	
Percentage of total	**26.4%**	**16.3%**	**14.5%**	**14.4%**	**12.1%**	**5.1%**	**5.0%**	**3.5%**	**0.8%**	**0.7%**	**1.0%**		

Journal Pre-proof

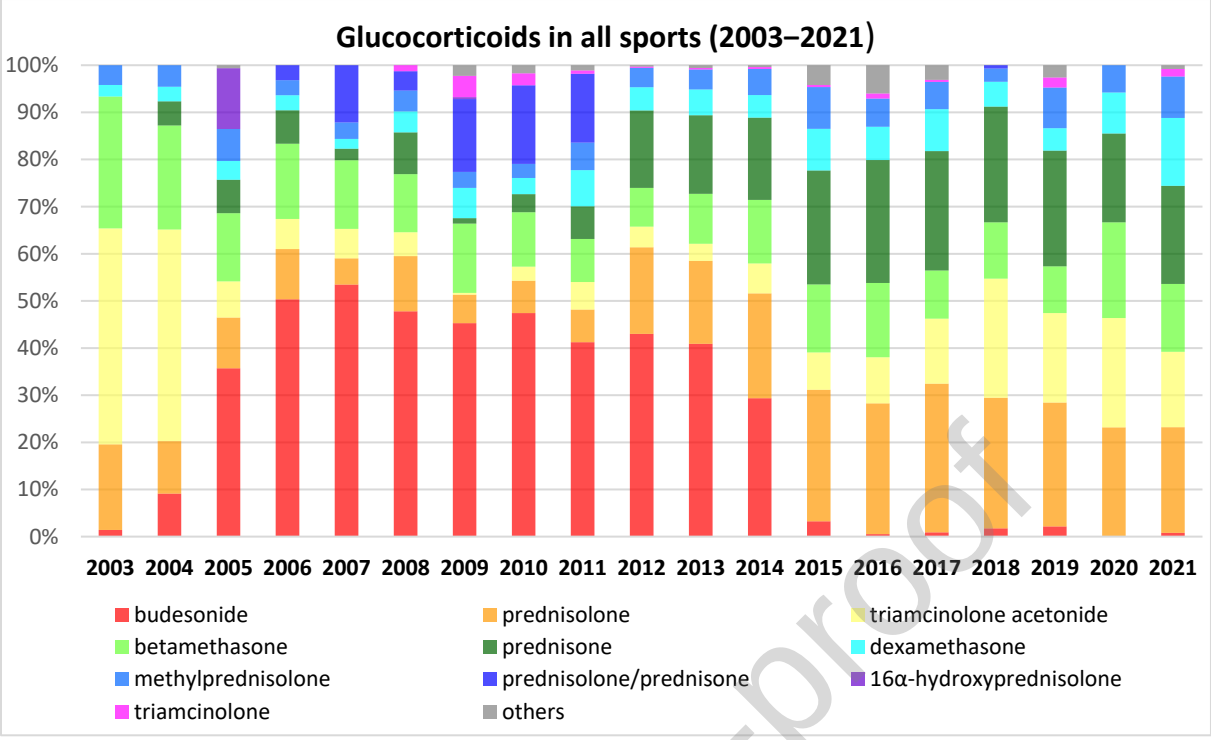


Chart 10. Glucocorticoids in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021.

Budesonide, the most commonly detected compound in the S9 section, is a synthetic glucocorticoid with steroidal scaffold. Budesonide binds to the GR with a higher affinity than cortisol and prednisolone (Zadik et al. 2017). It is used for the treatment of asthma, rhinitis, and inflammatory bowel disease. Budesonide is used as a mixture of (22*R*)- and (22*S*)-epimers (Lu et al. 2013b). (22*R*)-Budesonide is about 2–3 times more potent than its counterpart, it also has a higher distribution volume and clearance. Budesonide, in its native form is rapidly absorbed in the proximal gastrointestinal tract and cleared by the liver. Budesonide undergoes metabolism via CYP3A4 and CYP3A5 (Abdalla and Herfarth 2016). Metabolism of budesonide provides two main metabolites: 16α-hydroxyprednisolone which is produced only from (22*R*)-budesonide via cleavage of acetal group and 6β-hydroxybudesonide which is produced from the both epimers via hydroxylation (Matabosch et al. 2012). The both metabolites have negligible activity in comparison to the parent drug. The metabolites are cleared mainly via kidneys however, small fraction is conjugated and excreted in the bile (Abdalla and Herfarth 2016). Budesonide is generally well tolerated and it seems to cause fewer side effects than traditional glucocorticoids (O'Donnell and O'Morain 2010). In sports, budesonide is the most commonly detected substance in the S9 section however, since 2014, a significant decrease in reported findings is apparent (Table 10). This is associated with the specific regulation reported by WADA and effective from 1st September 2014. The regulation declared that AAFs for budesonide, the laboratories shall determine the presence of its metabolite, 6β-hydroxybudesonide, at levels higher than 30 ng/mL (WADA 2014). However, in the technical document reported by WADA, effective from 1st January 2022, the minimum required performance level (MRPL) for 6β-hydroxybudesonide was modified to 45 ng/mL (WADA 2022e).

Prednisolone, second commonly detected substance in the S9 section, and prednisone, fifth commonly detected substance in the S9 section, are structural related synthetic glucocorticoids that differ only in the substituent attached to C11 position. Prednisone, a C11-keto derivative, is an inactive

Journal Pre-proof

prodrug which is activated, via conversion mediated by 11β-HSD1, to an active prednisolone, a C11-hydroxyl derivative. Prednisone is approved by the FDA to treat a broad range of diseases, such as immunosuppressive/endocrine, dermatologic, gastrointestinal, ophthalmic, hematologic, and other diseases, it is also used as an anti-inflammatory and antineoplastic agent (Puckett et al. 2022). Prednisolone is therapeutically used in various clinical forms due to its anti-inflammatory and immunomodulatory effects. Prednisolone is used to treat various acute or chronic conditions including asthma, hepatitis, arthritis, systemic lupus, allergic dermatitis, Crohn's disease, ocular inflammations, and others (Furman 2019). After oral administration the both, prednisolone and prednisone, are rapidly absorbed and reach peak plasma concentration within 1–3 hours. The plasmatic half-life of prednisone is slightly longer, about 3.4–3.8 hours, than the half-life of prednisolone which is about 2.1–3.5 hours (Pickup 1979). Prednisolone is primarily metabolized via oxidation of hydroxyl group at C11 position, mediated by 11β-HSD2, to form an inactive metabolite prednisone. However, via counteraction mediated by 11β-HSD1 which converts prednisone to an active prednisolone, this reversible reaction results in creation of an equilibrium between the both glucocorticoids. The both, prednisolone and prednisone, are hydroxylated at C6 position. Prednisolone further undergoes reduction of ketone at C20 position (Skauby et al. 2021), which also represents a metabolic route for prednisone (Matabosch et al. 2015b). Adverse effects of prednisolone are usually associated with large dosages or with a prolonged use. The adverse effects include anxiety, increased appetite, hypertension, osteoporosis, sleep disorders, adrenal suppression, cognitive impairment, growth retardation in children, impaired diabetic control, headache, nausea, and others (Furman 2019). For prednisone, adverse effects include hyperglycemia, hypertension, insomnia, osteoporosis, adrenal suppression, cataracts, and others (Puckett et al. 2022). In sports, taking together the findings reported by WADA during 2003–2021 for prednisolone, prednisone, and prednisolone/prednisone represent 31.9% of overall findings in the S9 section (Table 10). Thus, combined findings for prednisolone and prednisone are even higher than those reported for budesonide (26.4%), the most commonly detected substance. However, in a study recently reported by Tacey et al., the authors found that a single dose of prednisolone, an active glucocorticoid, decreases work performance during high-intensity interval cycling. The results suggest that acute administration of glucocorticoids does not act as an ergogenic aid and in fact may reduce exercise performance (Tacey et al. 2019).

Triamcinolone acetonide, third commonly detected substance in the S9 section, is a synthetic glucocorticoid which is a 16,17-acetonide derivative of triamcinolone. Interestingly, the structure has an atom of fluorine incorporated to the C9 position (Narayanan and Kuppermann 2010). Triamcinolone acetonide is used as a topical preparation or as an injectable formulation in various ophthalmic, rheumatologic, and orthopedic indications (Sharma et al. 2022). Orally administered triamcinolone acetonide has a relatively rapid absorption and reaches the maximum plasma concentration about 1.5–2 hours after ingestion. After oral administration, the drug undergoes rapid metabolism and excretion. After 24 hours from administration, the parent drug in the plasma is not detected. Triamcinolone acetonide produces three metabolites including 6β-hydroxytriamcinolone acetonide, 21-carboxytriamcinolone acetonide, and 21-carboxy-6β-hydroxytriamcinolone acetonide. These metabolites are less active than the parent compound. Excretion occurs in the feces and in the urine (Jermak et al. 2007). Common adverse effects of intranasally administered triamcinolone acetonide include sneezing, dry mucosa, nasal irritation, sinus discomfort, throat discomfort, epistaxis, and headache (Gawchik and Saccar 2000). In sports, triamcinolone acetonide is widely used by athletes to treat articular sprains, pain, and injuries. Although, local injections of glucocorticoids are not prohibited by WADA, only a single local injection of triamcinolone acetonide could exceed the MRPL (previously set to the level of 30 ng/mL) for the detection and identification of glucocorticoids. Therefore, the athletes should be cautious when considering triamcinolone acetonide injection during

Journal Pre-proof

a competition period (Chang et al. 2014). Interestingly, in the 2022 technical document reported by WADA, the limit for triamcinolone acetonide has been lowered to 15 ng/mL (WADA 2022e).

Betamethasone, fourth commonly detected substance in the S9 section, is a synthetic glucocorticoid with incorporated fluorine atom at C9 position. Betamethasone is used locally to treat dermatitis and eczema, it is also used as an anti-itching agent (Vardanyan and Hruby 2006a). Orally administered betamethasone phosphate reaches the peak plasma concentration within about 1.5 hours. Its terminal half-life is 13.9 ± 7.5 hours (Jobe et al. 2020). Investigation of betamethasone metabolism, after intramuscular administration of the drug, revealed an oxidation at C11 position producing 11-oxobetamethasone, hydroxylation at C6 position producing 6β-hydroxybetamethasone and 6α-hydroxybetamethasone, reduction of keto grout at C20 position and other (Matabosch et al. 2015a). Common side effects for betamethasone are headache and insomnia (Pohl et al. 2015). In sports, betamethasone is prohibited in competition. Previously, the reporting level of the drug and its metabolites in urine, used to distinguish allowed and prohibited administration, was 30 ng/mL. However, an intranasal treatment with betamethasone may result in the detection of similar or even higher urinary concentrations (Coll et al. 2021). In the 2022 technical document, the reporting level for betamethasone has been increased to 60 ng/mL (WADA 2022e).

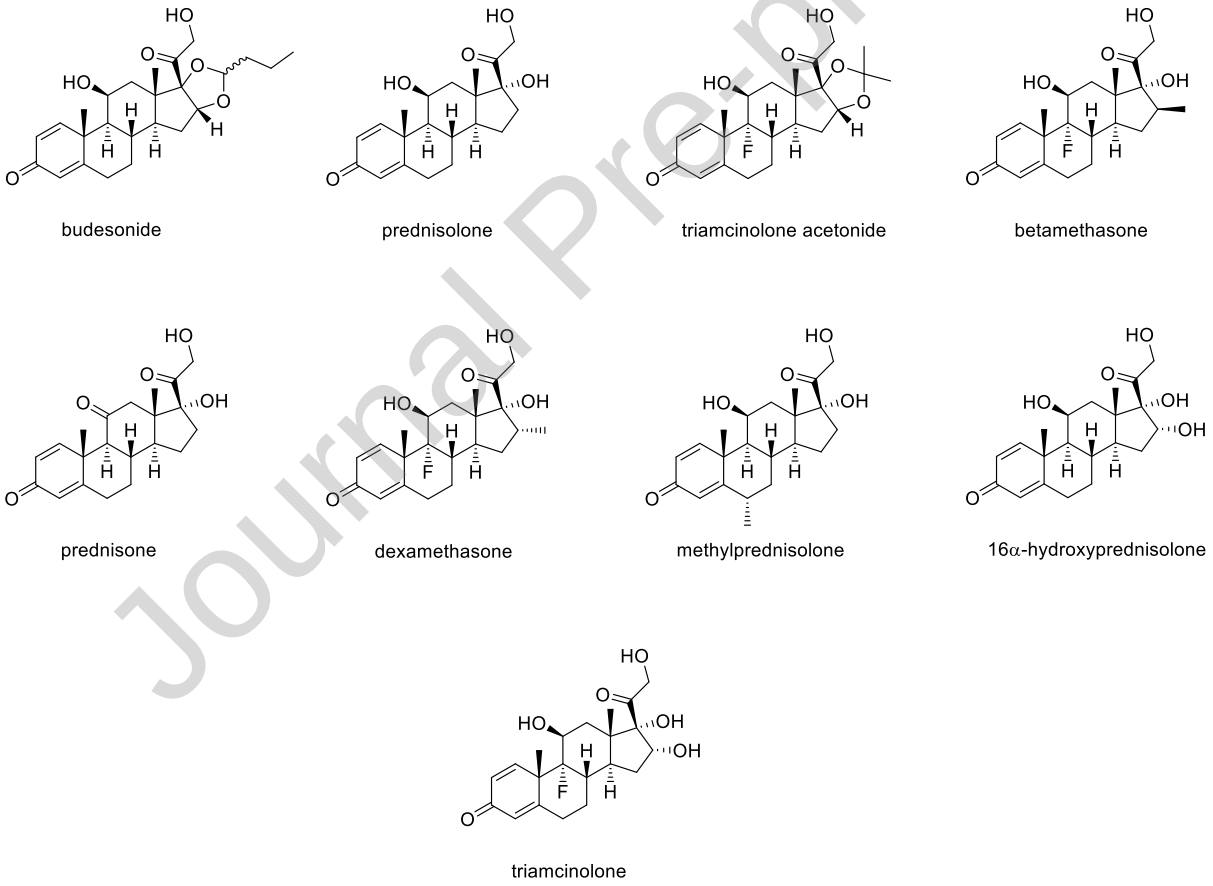


Figure 9. Structures of the most commonly detected glucocorticoids in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.11 β-Blockers (P1)

β-Blockers, also called as β-adrenergic receptor antagonists or β-antagonists, encompass a diverse group of compounds with diverse pharmacodynamic and pharmacokinetic properties

(Wiysonge et al. 2017). The name of this class, β-blockers, refers to the mode of their action which is mediated via blocking β-adrenergic receptors. β-Adrenergic receptors are transmembrane proteins belonging to the G protein-coupled receptors that involve three subtypes: β_1, β_2, and β_3. Activation of β_1-adrenergic receptors results in various physiological functions involving cardiac contraction and renin secretion. Activation of β_2-adrenergic receptor is associated with dilation of bronchi, relaxation of smooth muscle in uterus, increased insulin release, and increased liver and muscle glycogenolysis. Blocking of β-adrenergic receptors mediated by β-blockers causes an inhibition of these effects (Kwon and Rosendorff 2018). Based on the type of inhibition, there are three generations of β-blockers. First generation of β-blockers are non-selective inhibitors of both, β_1 and β_2-adrenergic receptors (propranolol, nadolol, timolol, celiprolol). Second generation of β-blockers are selective β_1-blockers that exhibit higher cardioselectivity (atenolol, bisoprolol, metoprolol, nebivolol). Third generation of β-blockers exhibit varied selectivity for β_1-adrenergic receptors, they may also block α_1-adrenergic receptors and activate β_3-adrenergic receptors which is associated with vasodilator effects (labetalol, carvedilol) (do Vale et al. 2019). In general, β-blockers are primarily used to treat cardiovascular diseases (Wiysonge et al. 2017). Adverse effects associated with the use of β-blockers are bradycardia and hypotension (Alali et al. 2017). As mentioned, the effects of β-blockers improve heart functions however, these compounds exhibit a powerful calming effect and stabilize motor performance. This action is beneficial in sports that require precision such as archery, billiards, darts, golf, shooting, and others (Hackney 2018c). By this reason, the use of β-blockers in particular sports is prohibited by WADA and β-blockers are listed as P1 section in the 2023 prohibited list (WADA 2022d). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the P1 section represents tenth commonly detected section of substances misused as a potential doping in all sports, representing 0.5% of overall findings (Table 1). The order of the most frequently detected ATFs and AAFs, reported by WADA during 2003–2021 in section of β-blockers (P1) is represented by: propranolol (29.0%), bisoprolol (21.1%), metoprolol (18.1%), atenolol (16.8%), carvedilol (5.2%), acebutolol (2.3%), sotalol (1.8%), nadolol (1.1%), carteolol and timolol (each represented by 0.9%), and others (2.7%). Findings of the commonly detected substances of the P1 section are summarized in Table 11, graphically expressed in Chart 11, and their structures are shown in Figure 10.

Table 11. β-Blockers in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021.

	propranolol	bisoprolol	metoprolol	atenolol	carvedilol	acebutolol	sotalol	nadolol	carteolol	timolol	others	Total (annually)	Percentage of total (annually)
2003	4	7	5	9		4			1			**30**	**6.8%**
2004	6	5	5	8			1					**25**	**5.7%**
2005	16	10	5	6	4	1						**42**	**9.5%**
2006	6	2	6	9	3					2		**28**	**6.3%**
2007	8	6	6	3	2				1	1		**27**	**6.1%**
2008	9	5	4	6	1	2					4	**31**	**7.0%**
2009	11	7	10	8	1				1			**38**	**8.6%**
2010	10	6	4	4	1	2	1	2				**30**	**6.8%**
2011	4	5	4	4	1			2			1	**21**	**4.8%**
2012	1	1	5	2		1	1				2	**13**	**2.9%**
2013	8	7	5	3					1		1	**25**	**5.7%**
2014	7	5	3	6	1		1			1	1	**25**	**5.7%**
2015	3	7	6		1		1	1				**19**	**4.3%**
2016	3	5	3	2							1	**14**	**3.2%**

Journal Pre-proof

2017	2	2	5		2		1					**12**	**2.7%**
2018	6	7	2		2		1					**18**	**4.1%**
2019	9	2	1	3	3		1				1	**20**	**4.5%**
2020	10	2	1	1								**14**	**3.2%**
2021	5	2			1						1	**9**	**2.0%**
Total (2003-2021)	**128**	**93**	**80**	**74**	**23**	**10**	**8**	**5**	**4**	**4**	**12**	**441**	
Percentage of total	**29.0%**	**21.1%**	**18.1%**	**16.8%**	**5.2%**	**2.3%**	**1.8%**	**1.1%**	**0.9%**	**0.9%**	**2.7%**		

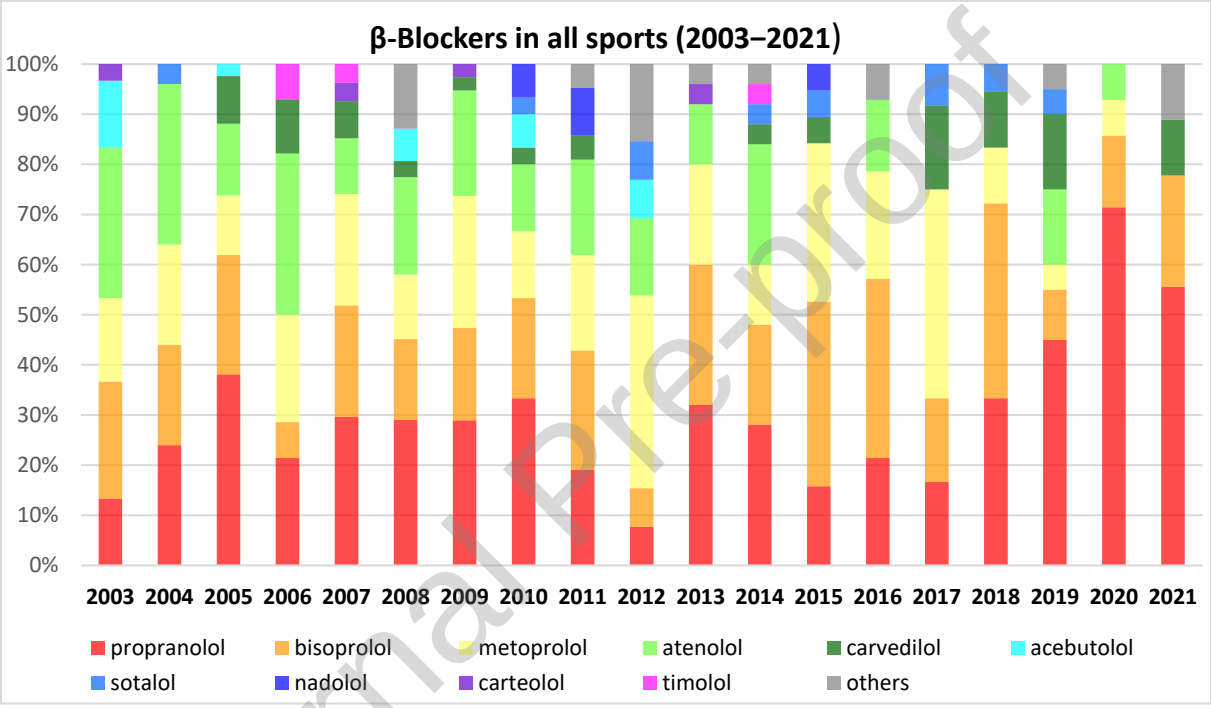


Chart 11. β-Blockers in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021.

Journal Pre-proof

Propranolol, the most commonly detected substance in the P1 section, contains 3-amino-1,2-propanediol scaffold which has incorporated isopropyl group on the atom of nitrogen and 1-naphtyl ring on the primary hydroxyl group. The structure has one chiral center and thus exists in a form of two enantiomers. Propranolol is therapeutically used as a racemic mixture however, (*S*)-propranolol acts as an antagonist of β-adrenergic receptors, while (*R*)-propranolol is considered to be inactive toward these receptors (do Vale et al. 2019). Propranolol, a member of the first generation of β-blockers, is a non-selective blocker of both, β_1 and β_2-adrenergic receptors, it has a lower affinity for β_3-adrenergic receptors. It is used to treat cardiovascular disorders such as hypertension, angina, arrhythmias, as well as to treat non-cardiovascular conditions such as essential tremor, migraine, anxiety, or hyperthyroidism. Propranolol administered orally is rapidly absorbed and has a relatively low bioavailability due to its first-pass metabolism (Bylund and Gruetter 2007). The major part of administered dose is removed via hepatic excretion and only about 25% of the drug reaches the systematic circulation. The plasmatic half-life of propranolol is 3–6 hours, while long-acting preparations of the drug have the half-life of 8–11 hours. The drug undergoes an extensive metabolism and the metabolites are excreted in the urine (Srinivasan 2019). Metabolism is mediated mainly via

CYP2D6 and cytochrome P450 1A2 (CYP1A2) and provides three main metabolites. Two metabolites, 4-hydroxypropranolol and 5-hydroxypropranolol, are products of aromatic hydroxylation, while one product, *N*-desisopropylpropranolol, is a product of *N*-dealkylation (Otey et al. 2006). Potential adverse effects associated with the use of propranolol include bronchospasm, bradycardia, dizziness, fatigue, lightheadedness, and erectile dysfunction (Longardner and Haubenberger 2022). In sports, propranolol is used by athletes to reduce anxiety and tremors associated with certain situations such as performance anxiety and acute anxiety (Patel et al. 2010).

Bisoprolol, second commonly detected substance in the P1 section, is a substituted 3-(isopropylamino)-1,2-propanediol in which primary hydroxyl group bears *para*-substituted benzene ring. Similar as propranolol, bisoprolol also has one chiral center. It is used as a racemic mixture in which the (*S*)-bisoprolol is responsible for most of the β-blocking activity (Malik and Bhushan 2017). Bisoprolol, a member of the second generation of β-blockers, is a cardioselective antagonist of β_1-adrenergic receptors. It has only a low affinity toward β_2-adrenergic receptors however, this effect is not observed at low doses of the drug. It is used to treat hypertension and heart failure. The drug administered orally is rapidly and almost completely absorbed (>90%), while only a small portion (<10%) undergoes hepatic metabolism. The peak plasma concentration is reached in about 3 hours after ingestion. Metabolism of bisoprolol is mediated mainly by CYP3A4 and appears to be non-selective. While CYP2D6 is only minimally involved in the metabolism and appears to be stereoselective (*R*/*S* = 1.50). Metabolites of bisoprolol are very polar and excreted in the urine. The drug has an elimination half-life of 10–12 hours (Bakheit et al. 2021). The major possible side effect for bisoprolol include bradycardia and asthma (Yamazaki et al. 2020). In sports, bisoprolol may be misused due to its effects on performance of athletes. A recent study, reported by Eregen et al., investigated and compared an effect of propranolol, a non-selective β-blocker, and bisoprolol, a selective β_1-blocker, on archery performance. The authors reported that despite the both β-blockers reduced hearth rates, 12.8% for bisoprolol and 8.6% for propranolol, a positive effect of both β-blockers on shooting score was not observed. In addition, the shooting behavior and body sway were also not affected by the use of the β-blockers (Ergen et al. 2021).

Metoprolol, third commonly detected substance in the P1 section, is structurally related to bisoprolol from which differs in type of substituent attached to *para*-position of benzene ring. Similar to previously described β-blockers, metoprolol has a single chiral center and it is also used as a racemic mixture in which (*S*)-metoprolol exhibits higher blocking effect of β-adrenergic receptors than its (*R*)-enantiomer. Metoprolol, a member of the second generation of β-blockers, is a selective blocker of β_1-adrenergic receptors (Antunes et al. 2013). Metoprolol is used to treat and moderate hypertension, serious conditions of myocardial infections, also in angina, tachycardia, extrasystole, and for secondary prophylaxis after a heart attack (Vardanyan and Hruby 2006b). Metoprolol is rapidly and almost completely absorbed. However, due to an extensive first-pass metabolism its bioavailability is only about 50%. The half-life of metoprolol is in a range of 3–4 hours for young adults and about 7–9 hours in elderly patients. Metoprolol is extensively metabolized. After oral intake, about 70% of the parent drug is metabolized via CYP2D6. The major routes of metabolism involve *O*-demethylation route, producing *O*-desmethylmetoprolol as a metabolite, which is further undergoes oxidation to produce metoprolol phenylacetate (65% of the oral dose recovered in the urine); *N*-dealkylation route, producing *N*-desisopropylmetoprolol (10%); and α-hydroxylation route resulting in α-hydroxymetoprolol (10%). Less than 5% of metoprolol is excreted by the kidney in unchanged form (Berger et al. 2018). The common adverse effects of metoprolol are fatigue, weakness, cold peripheries, headache, and gastrointestinal reactions (Che et al. 2018). Interestingly, a study reported in 1985 by Kruse et al. evaluated the effect of metoprolol in shooting performance. The authors reported an improvement in shooting performance after use of the drug, in comparison with placebo.

Journal Pre-proof

The authors concluded that the improvement is presumably mediated via effects of the drug on hand tremor (Kruse et al. 1986).

Atenolol, fourth commonly detected substance in the P1 section, is structurally related to metoprolol derivatized on the *para*-position of benzene ring attached to main scaffold. Atenolol has a single chiral center; it is used as a racemic mixture. The (*S*)-atenolol is a more potent β-blocker, whereas (*R*)-atenolol is largely devoid of this activity. Atenolol, a member of the second generation of β-blockers, is a selective antagonist of β_1-adrenergic receptors (Seebauer et al. 2022). Atenolol is used in the treatment of hypertension, chest pain, arrhythmia. It is also used to treat and prevent heart attacks, or to prevent migraines (Gad El-hak et al. 2019). Orally administered atenolol is incompletely absorbed from the intestine with bioavailability of about 50%. Peak plasma concentration is reached in about 3 hours after ingestion. The compound is predominantly eliminated via the kidneys, only about 5% of atenolol is metabolized in the liver. Its elimination half-life is in a range of 6–9 hours (Kirch and Görg 1982). Side effect of atenolol include bronchospasm, fatigue, depression, and others (Opie and Gersh 2012). In 1989, Bouckaert et al. reported that the use of atenolol results in decreased maximal work performance in young healthy volunteers (Bouckaert et al. 1989).

Carvedilol, fifth commonly detected substance in the P1 section, similarly to above-mentioned β-blockers has a structural motif of 3-amino-1,2-propanediol. However, its structure incorporates a carbazole core attached via oxygen atom as well as 2-(2-methoxyphenoxy)ethyl substituent attached via nitrogen atom. Similarly, carvedilol has a single chiral center and it is marketed as a racemic mixture (Diniz et al. 2019). (*S*)-Carvedilol acts as a blocker of both β and α_1-adrenergic receptors, whereas (*R*)-carvedilol is a weak blocker of α-adrenergic receptors. Antagonistic effect of carvedilol on β-adrenergic receptors is higher than those on α_1-adrenergic receptors (van Zwieten 1993). Carvedilol belongs to the third generation of β-blockers and it is used to treat heart failure, hypertension, and myocardial infarction (Diniz et al. 2019). Carvedilol is absorbed rapidly and undergoes an extensive first-pass metabolism in the liver. The peak concentration is reached within 1–2 hours and its elimination half-life is about 4–7 hours (Morgan 1994). Carvedilol undergoes an extensive metabolism mainly in the liver. The metabolism involves aromatic hydroxylation mediated by CYP2D6 which produces metabolites 4'-hydroxycarvedilol and 5'-hydroxycarvedilol; *O*-demethylation via CYP2C9 producing *O*-desmethylcarvedilol; aromatic hydroxylation of carbazole core producing 1-hydroxycarvedilol and 8-hydroxycarvedilol (Jung et al. 2018). Side effects associated with carvedilol include dizziness, fatigue, hypotension, bradycardia, heart block, and worsening heart failure (Wilson Tang and Fowler 1999). In comparison with above-mentioned β-blockers, the findings of carvedilol are considerably less frequent (Table 11).

Journal Pre-proof

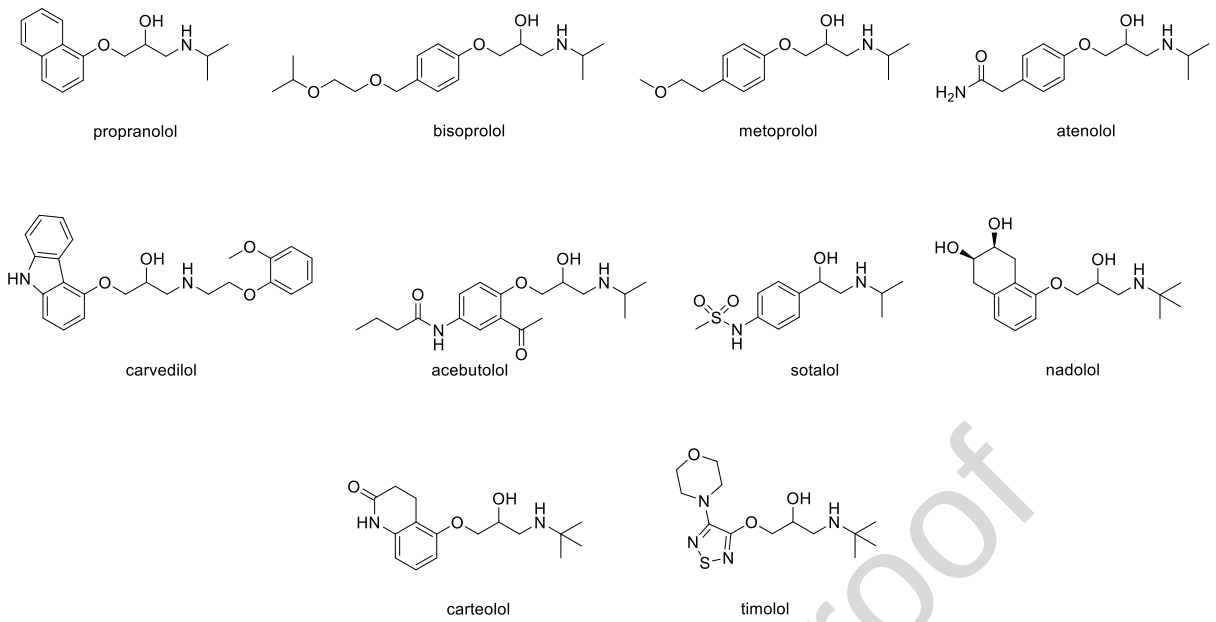


Figure 10. Structures of the most commonly detected β-blockers in all sports during 2003–2021. Based on the ATFs and AAFs reported by WADA-accredited laboratories in ADAMS.

4.12 Prohibited methods (M1-3)

In addition to a wide range of different substances prohibited by WADA and disclosed above, several methods are also prohibited in sports. These methods are prohibited at all times (in- and out-of-competition) and subdivided into three classes in the 2023 prohibited list (WADA 2022d). Based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, the prohibited methods section represents the least frequently detected (twelfth) section, ordered by a total number of findings, representing 0.1% of overall findings (Table 1). The order of the most frequently detected findings, reported by WADA during 2003–2021 in section of prohibited methods is represented by: chemical/physical manipulation (57.4%), blood transfusion (27.9%), urine substitution (7.4%), and efaproxiral (1.5%). All findings of prohibited methods are summarized in Table 12 and graphically expressed in Chart 12.

Table 12. Prohibited methods in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Notes to findings: blood transfusion - includes findings reported as "blood transfusion" and "homologous blood transfusion"; chem/phys manipulation - includes findings reported as "chemical and physical manipulation", "manipulation", and "physical manipulation"; includes two ATFs reported in 2020 in the summary table as "chemical and physical manipulation", however table summarizing identified substances for ATFs (specified findings) was not included in the report, despite the fact, the authors included these two findings because both ATFs were specified in the final summary table; tampering - includes findings reported as "tampering" and "attempted tampering". Abbreviations: chem/phys, chemical/physical.

	chem/phys manipulation (M2)	blood transfusion (M1)	urine substitution (M2)	tampering (M2)	efaproxiral (M1)	Total (annually)	Percentage of total (annually)
2003	2					**2**	**2.9%**
2004		2				**2**	**2.9%**
2005						**0**	**0.0%**
2006	4					**4**	**5.9%**
2007	3	3				**6**	**8.8%**

2008						**0**	**0.0%**
2009	5					**5**	**7.4%**
2010	6					**6**	**8.8%**
2011	3				1	**4**	**5.9%**
2012	1					**1**	**1.5%**
2013	1					**1**	**1.5%**
2014	3					**3**	**4.4%**
2015	1					**1**	**1.5%**
2016	4					**4**	**5.9%**
2017	1					**1**	**1.5%**
2018	1	8		3		**12**	**17.6%**
2019	2	2	5			**9**	**13.2%**
2020	2					**2**	**2.9%**
2021		4		1		**5**	**7.4%**
Total (2003-2021)	**39**	**19**	**5**	**4**	**1**	**68**	
Percentage of total	**57.4%**	**27.9%**	**7.4%**	**5.9%**	**1.5%**		

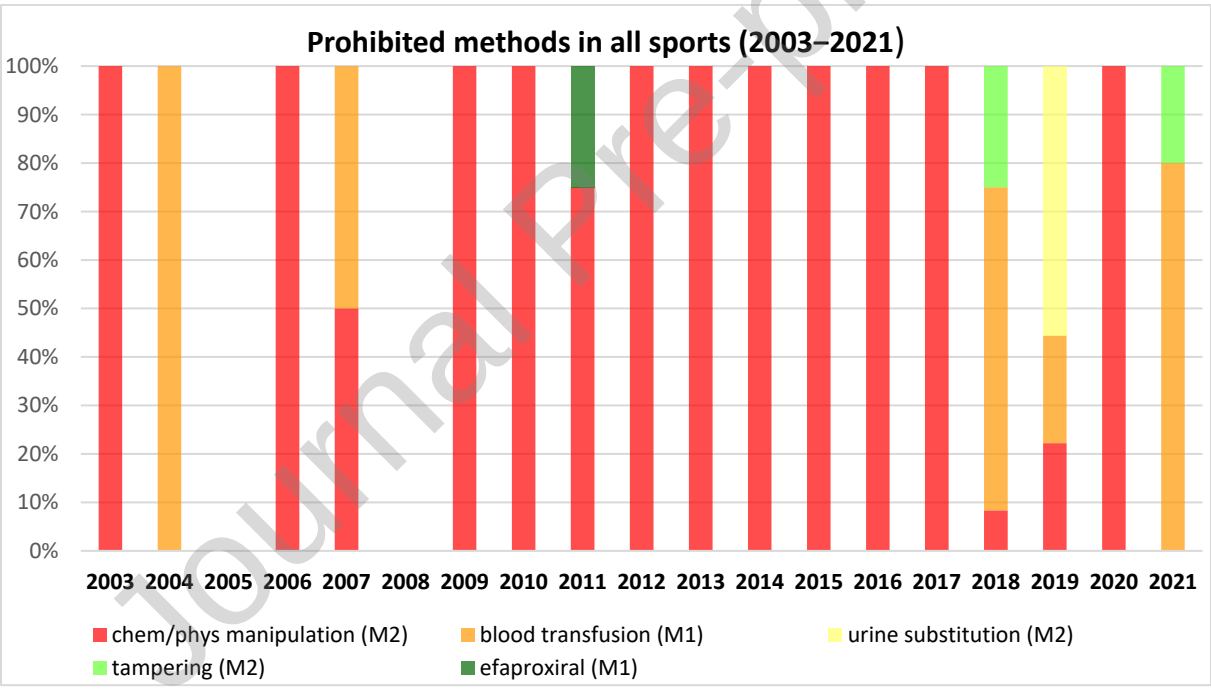


Chart 12. Prohibited methods in all sports identified as ATFs and AAFs in ADAMS and reported by WADA-accredited laboratories during 2003–2021. Abbreviations: chem/phys, chemical/physical.

4.12.1 Manipulation of blood and blood components (M1)

Manipulation of blood and blood components includes following methods: 1.) administration and reintroduction of any quality of autologous, allogenic (homologous) or heterologous, or red blood cell products of any origin into the circulatory system; 2.) artificially enhancing the uptake, transport, or delivery of oxygen, this includes but it is not limited to perfluorochemicals, efaproxiral, voxelotor and modified hemoglobin products, for example: hemoglobin-based blood substituents and microencapsulated hemoglobin products, excluding supplemental oxygen by inhalation; 3.) any form

Journal Pre-proof

of intravascular manipulation of the blood or blood components by physical or chemical means (WADA 2022d).

Blood transfusion, second commonly detected prohibited method, is a practice which can be misused by cheating athletes with the aim to increase capacity of the blood for oxygen transport to the tissues, particularly in endurance sport disciplines (Donati et al. 2022). In 1970s, blood transfusion became popular among elite endurance athletes. At the end of 1980s, the popularity of this method declined as the result of the introduction of recombinant erythropoietin. The introduction of the Hematological Module in the ABP, which considers a panel of biomarkers of blood doping that are measured in an athlete's blood sample, is a powerful tool to deter the blood transfusion doping in sports (Giraud et al. 2010).

Efaproxiral, representing only by one finding within the time period (2003–2021), is a synthetic allosteric modifier of hemoglobin with an ability to increase maximal muscle oxygen uptake (Jiménez et al. 2004). Thus, efaproxiral is incorporated in the 2023 prohibited list (WADA 2022d).

4.12.2 Chemical and physical manipulation (M2)

Chemical and physical manipulation includes following methods: 1.) tampering or attempting to tamper, to alter the integrity and validity of samples collected during doping control, this includes but it is not limited to sample substitution and/or adulteration; 2.) intravenous infusions and/or injections of more than a total of 100 mL per 12-hour period except for those legitimately received in the course of hospital treatments, surgical procedures, or clinical diagnostic investigations (WADA 2022d).

Chemical and physical manipulation, the most commonly reported prohibited method, can be used by athletes alone or in collaboration with other athletes. Using of a wide range of strategies and practices to avoid detection of prohibited substances or methods is classified by WADA as chemical and physical manipulation (Mottram 2018). In addition to general findings reported as chemical and physical manipulation, several findings included in this class have been reported as “urine substitution” (5 findings) or “tampering” (4 findings) (Table 12).

4.12.3 Gene and cell doping (M3)

Gene and cell doping includes following methods: 1.) the use of nucleic acids or their analogs that may alter genome sequences and/or gene expression by any mechanism, which includes but it is not limited to gene editing, gene silencing, and gene transfer technologies; 2.) the use of normal of genetically modified cells.

Genetic manipulations allow to add a genetic material to an organism or modify the expression of endogenous genes. The majority of genetic studies investigate the modifications of cells in culture or reproductive cells of animals, which results in an expression of the modification in all of the cells of a tissue-like muscle and the modification is heritable. However, the modification of non-reproductive (somatic) tissues in mature animals is possible as well. The second-mentioned technology is mostly intended for genetic therapies of inherited or acquired disease, which raises the possibility of genetic manipulation for an increased performance in sports, commonly known as gene doping (Wells 2018). To date, there have been no recorded of gene doping reported by WADA (Table 12).

Journal Pre-proof

4.13 Previously involved sections and others

The annual modification of the prohibited list by WADA is associated with an introduction of novel prohibited substances on the list, deletion of involved substances from the list, or their reclassification within the sections or classes. These modifications strengthens the world anti-doping system (Pokrywka et al. 2010). Several substances, such as caffeine, have been removed from the prohibited list. Moreover, several of previously reported findings were classified in sections that are not anymore incorporated in the 2023 prohibited list (WADA 2022d). The sections previously involved in prohibited lists with reported findings are represented by local anesthetics and alcohol. Local anesthetics were removed from the prohibited list in 2004 (Strano Rossi and Botrè 2011). Findings of local anesthetic were reported only in 2003 (82 findings) and are represented by lidocaine (57 findings, includes 2 findings reported as "lignocaine"), mepivacaine (20 findings, includes 1 finding reported as "carbocaine"), benzocaine (2 findings), bupivacaine (2 findings), and tetrazepam (1 finding) (Table 1). Alcohol (ethanol) has been removed from the prohibited list in 2018 (WADA 2017c). Finding reported in the section of alcohol (32 findings) were reported from 2009 to 2013 (5 findings in 2009, 9 findings in 2010, 5 findings in 2011, 5 findings in 2012, and 8 findings in 2013) (Table 1). Several findings were reported in the section termed as "other", these findings include "no indication" reported in 2014 (49 findings), "sample degradation" reported in 2015 (1 finding), "optical density of the sample (diluted 10 times) significantly above the threshold value for negative sample" also reported in 2015 (2 findings), "extensive microbial degradation" reported in 2018 (1 finding), and "unknown" reported in 2019 (7 findings) (Table 1). In summary, based on the ATFs and AAFs reported by accredited laboratories in ADAMS during 2003–2021, previously involved sections and others represent eleventh commonly detected section in all sports, representing 0.2% of overall findings (Table 1).

5 Discussion

In 1961, a creation of IOC Medical Commission can be considered as a modern anti-doping fight. In following decades, the anti-doping strategy was evolved and modified. In 1990s, the fight against doping gained increasing support inside as well as outside the sport community. In 1999, WADA has been established to develop, harmonize, and coordinate the anti-doping rules. To date, WADA is carrying on the anti-doping fight supported by the universally accepted WADA Code and an International Anti-Doping Convention under the United Nations Educational, Scientific and Cultural Organization (UNESCO) (Ljungqvist 2017). Nowadays, the constantly increasing demands in routine doping controls creating a pressure on the anti-doping laboratories, particularly to improve analytical sensitivity for detection of substances involved in the prohibited list. Advances in evaluation of the mechanism of prohibited compounds in combination with improvements in analytical instrumentation allowed to detect numerous urinary excreted substances in sub-ng/mL levels. These accomplishments are beneficial, for instance in comprehensive retesting programs such as reevaluation of long-term stored doping control samples collected at Olympic Games (Thevis et al. 2021a). The long-term detection of substances incorporated in the prohibited list represents an effective tool that enhances ability to detect doping practices. Commonly, the urine is utilized as a primary sample type due to non-invasive and rapid sample collection method. It is also important to mention that for a long-term detection of doping usually not the doping substance itself is detected but its metabolites. Although there is no unequivocal definition, as the long-term metabolites are considered metabolites excreted from the human body over an extended period of time. Therefore, the discovery and characterization of novel long-term metabolites represents a remarkable anti-doping strategy (Göschl et al. 2022). Indeed, analysis of phase I and II metabolites of doping agents have contributed to improved detectability in numerous cases (Piper et al. 2019). Due to an extensive biotransformation of AAS by

Journal Pre-proof

phase I and II metabolism that provide more polar substances with facilitated excretion via the urine, development of methods for monitoring long-term AAS metabolites becomes more popular in recent years (Thevis et al. 2021b). For example, two urinary long-term metabolites of methylstenbolone, a designer steroid found in a nutritional supplement, were detected until one week after administration of the steroid while unchanged parent structure, methylstenbolone, was detectable only in a period of 45 hours (Cavalcanti et al. 2013). More recently, Piper et al. detected metabolites of deuterated methylstenbolone even at the end of the study, 29 days after administration (Piper et al. 2019). The long-term metabolites of mesterolone and oxymesterone in human urine samples were detectable up to 15 and 46 days, respectively (Polet et al. 2017). These and additional studies (Loke et al. 2021; Göschl et al. 2022), demonstrate that long-term metabolites of AAS represent an efficient target of detection in anti-doping tests. The ability to detect metabolites of prohibited substances after a long period of time, weeks or even months, often after administration of a single dose in association with advancement of analytical techniques represent an effective anti-doping methodology. Moreover, methods able to detect prohibited substance or its metabolites over a long period of time as well as implementation of novel anti-doping tests have also a significant deterrent effect on athletes. At the 2020 Summer Olympics in Tokyo, the dried blood spots (DBSs) testing, a new anti-doping test to detect prohibited substances approved by WADA was partially implemented. The DBSs testing was fully introduced in the 2022 Winter Olympics in Beijing (Yan et al. 2023). Although, the potential of DBSs as a matrix for anti-doping testing was reported earlier (Peng et al. 2000), testing of DBSs have several benefits in comparison with other predominantly used samples for anti-doping evaluation (urine, blood, serum). In contrast to a venous blood sample, the DBSs samples are commonly represented by a small volume (10–50 µL per spot) of capillary blood collected by pricking of fingertip or upper arm. In contrast to a urine sample in cases when athletes are incapable of provide urine, for example due to dehydration or after a physical activity, the DBSs samples can be collected at any time. Further advantages of DBSs involve simplicity of the sampling procedure, DBSs can be collected by a trained doping control officer or athletes themselves; the size of DBSs cards allows convenient transport from the site of sampling to the analytical laboratory; the long-term storage of the samples; and lower cost. However, small volume of the DBSs samples could represent a certain challenge from an analytical point of view (Garzinsky et al. 2023). In addition to urine and blood, other matrices can be also used in anti-doping tests. Additional analysis, such as hair analysis can be performed on request of athletes to prove their innocence. Advantages of hair analysis involve for example the possibility to establish a consumption profile or to differentiate between single dose and repeated dose. Among limitations of the hair analysis belong poor incorporation of acidic and neutral molecules as well as the impossibility to distinguish route of exposure. The hair analysis should be considered as a complement, as stated in the regulations of WADA. However, future progress in analytical methods may lead to more frequent use of hair analysis in anti-doping tests (Kintz et al. 2023). Nail analysis can be also used for detection of doping in sport. Similar to hair, nail analysis can be useful to estimate the dates of administration. Analysis of nail is more complex since a proportion of the drug is also taken up at the nail bed; however, analysis of the nail sections could still be feasible to determine peak concentrations to estimate date of administration. In general, both hair and nail analysis seem to offer an advantage, as the detection windows in such samples have shown to be significantly prolonged (Brown and Perrett 2011). Although urine and blood are commonly utilized for the analysis, alternative matrices such as hair, nail, skin, saliva, sweat, and other can be also used for detection of doping in sport. Possibility to utilize different matrices in doping analysis increases the accuracy and credibility of the result. Moreover, future progress in analytical methods will result in more frequent incorporation and usage of other matrices for detection of doping in sport.

Journal Pre-proof

In addition to current progress in analytical methods, well documented cases of doping could also have a significant deterrent effect on the athlete's decision to abuse substances or methods in sport. Athletes may become more hesitant and cautious due to a fear of doping detection which can significantly damage athletes' career and reputation. Unveiling of misused substances or methods could also have financial and legal consequences. History of sport includes various doping scandals all around the globe, in specific cases, even worldwide-known athletes were revealed to use prohibited substances. In previous decades, several remarkable doping scandals that attracted a public attention were reported in different sports. In 1967, the British professional cyclist Tom Simpson died during the Tour de France under the influence of amphetamines (Vernec 2013). The information that Simpson's death is associated with amphetamines was reported in public media, which in turn raised public awareness on the health dimensions of doping (Hunt et al. 2012). From the 1960s through the 1980s, a systematic doping program has been established by the German Democratic Republic (GDR) for thousands of their athletes that includes use of drugs, notably AAS, in high-performance sports (Franke and Berendonk 1997; Barceloux and Palmer 2013). Doping scandal in the GDR damaged the health of numerous athletes and due to its complexity and durability is considered as fundamentally different to other historical and contemporary examples of doping (Dimeo et al. 2011). It is an example that, in contrast to individual cases, doping policy can be well-organized and widespread even on the government level. One of the most famous doping scandals in previous decade is a case of Lance Armstrong, an American former professional road cyclist. In 2012, Armstrong was sanctioned by USADA with a lifetime ban and disqualification of all competitive results achieved since August 1998, due to used illicit performance-enhancing drugs. In 2013, Armstrong in a television interview admitted doping, despite having made denials throughout his career. In the report published by USADA, detailing thousands of pages of evidence, the case was referred to as: “the most sophisticated, professionalized, and successful doping program … and more extensive than any previous revealed in professional sports histiory” (Zurloni et al. 2015; Bell et al. 2016; USADA 2021). These remarkable cases prove that prohibited substances and methods in sport can be misused by individual itself, individual with further collaborating support, or even on a nationwide level. However, public reports of doping can play a significant role in preventing the misuse of substances and methods in sport. These reports raising awareness, highlighting consequences, and supporting ethical values of fair play in sports.

Due to a complexity of the list of prohibited substances and methods as well as individuality among athletes, there are several exemptions regarding substances and methods incorporated in the prohibited list. An athlete suffering from an illness or medical condition may require a particular medication that contains a substance incorporated in the prohibited list. For this purpose, WADA allows to apply for therapeutic use exemption (TUE) to ensure requiring treatment and to protect right to medical care of athletes. A TUE is required by athletes during situations: to treat acute or chronic health condition, to restore normal health condition without offering any performance enhancement, when no other non-banned alternatives are available, when pervious therapies were ineffective, when it is not required as a consequence of prior use of a prohibited substance (Rogol et al. 2019). A TUE may be submitted in two different manners. Commonly submitted, prospective TUE, allows an athlete to use substance or method incorporated in the prohibited list in and out of competition for a specific duration from the particular day the prospective TUE is granted. In specific cases TUE may be submitted retrospectively. Retrospective TUE is applied after an athlete has already taken the prohibited substance or method. If granted, retrospective TUE backdates when an athlete first began using prohibited substance or method, thus the athlete cannot be liable for an anti-doping rule violation (Greene and Kaiser 2019). Examples of medications used by athletes that contain prohibited substances are anti-diabetic, anti-hypertensive, anti-arrhythmic drugs. For example, athletes treating

Journal Pre-proof

diabetes with insulin, a substance incorporated in the prohibited list, require a TUE (Fiorentini et al. 2022).

As described in this review, there is a plenty of substances and methods that can be misused by athletes as a potential doping. These act via different mechanisms and provide variable pharmacological effects. In 2019, Heuberger and Cohen reviewed performance-enhancing effects of substances from the 2018 prohibited list reported by WADA. The authors concluded that substances of only five sections from the list have an ability to enhance performance in sports. These are also incorporated in the 2023 prohibited list and are represented by following sections: anabolic agents, β_2 agonists, stimulants, glucocorticoids, and β-blockers (the section of prohibited methods was not discussed). In addition, growth hormone, a member of class of peptide hormones and their releasing factors, exhibits a performance enhancement but only in untrained subjects (Heuberger and Cohen 2019). According to the 2023 prohibited list reported by WADA, prohibited substances and methods are clustered in 12 sections, several of them are further subdivided into the classes. Non-approved substances (S0) are represented by pharmacological substances that have no current approval by any governmental regulatory health authority for human therapeutic use. This section includes various structures with variable pharmacological effects. However, as non-approved substances, these compounds may represent a potential health risk for athletes. In addition, their potential benefits as doping agents are also controversial. To date, none findings of non-approved substances have been reported (Table 1). Anabolic agents (S1) include a plenty of substances with potential anabolic and androgenic effects. Anabolic agents, as substances used with potential to increase protein synthesis and muscle growth, represent the most frequently detected adverse analytical/atypical findings in sports as reported by WADA (Table 1). The enhancement of muscle strength by anabolic agents can be observed at supratherapeutic doses (Heuberger and Cohen 2019). Findings for testosterone, the most frequently detected substance in this section, were notably affected by the implementation of Steroidal Module in 2014. The findings of anabolic agents are usually significantly higher in comparison to those of other sections. This fact also supports the role of anabolic agents as doping substances. Peptide hormones, growth factors, related substances, and mimetics (S2) involve substances that are subdivided into the three classes. Erythropoietins and agents affecting erythropoiesis that act via increasing of RBCs volume in order to increase oxygen transfer and thus improve performance. Peptide hormones control functions of gastrointestinal and cardiovascular systems as well as energy expenditure and reproduction, the class also involves releasing factors of peptide hormones. Growth factors that promote or inhibit mitosis or affect cellular differentiation as well as growth factor modulators. Growth hormone, a member of this section has an ability to improve sprint performance. β_2 Agonists (S3) acting via β_2-adrenergic receptors and at high concentrations these substances can increase muscle strength and peak sprint power (Heuberger and Cohen 2019). Hormone and metabolic modulators (S4) involve four classes of substances. This section consists of substances with different structural scaffolds. The findings of these substances increased within the time period from ones to few hundreds of occurrences (Table 1). Diuretics and masking agents (S5) are substances that are not primarily used to enhance performance in sport, but they might be potentially misused by athletes to dilute other doping agents or to reduce body weight. Interestingly, Heuberger and Cohen reported that evidenced of various studies indicate a negative effect of diuretics on athletic performance. Stimulants (S6) are commonly detected substances, represented by hundreds of findings reported annually (Table 1). These substances can increase muscle strength, peak sprint power, and improve time of run. Narcotics (S7) are used due to their analgesic effects that might enhance performance in athletes however, due to common side effects of these compounds any beneficial effects could be suppressed. Cannabinoids (S8) exhibit no effects on enhancement of performance. Moreover, cannabinoids presumably negatively affect sport performance which can be caused by an increased

Journal Pre-proof

reaction time and impaired information processing after their use. Glucocorticoids (S9) are involved in metabolic, inflammatory, cardiovascular, and behavioral processes via acting on glucocorticoid receptors. These functions presumably result in glucocorticoids-mediated muscle strength improvement. β-Blockers (P1) acting via inhibition of β-adrenergic receptors can improve performance in selected sports via improvement of accuracy (Heuberger and Cohen 2019). Prohibited methods (M1-3) involve three classes of prohibited methods. This section is the least commonly detected section within the time period (2003–2021). Each method is annually represented only by ones of the findings. Potential of these methods to enhance performance are limited.

As seen in this review, contemporary fight against doping requires a complex approach to address constantly changing trends in sports. The key elements essential for an effective anti-doping effort include comprehensive educational programs that focus on the understanding the risk and consequences of doping; continued development of novel testing methods and technologies to detect prohibited substances and methods; determined anti-doping policies and regulations to establish clear rules and consequences for doping violations; international collaboration among anti-doping organizations, sports federations, and governments; transparency and accountability to maintain public trust and confidence; and athlete support service to provide requiring medicinal and mental health resources. In sports, WADA represents an independent international organization to promote, coordinate, and monitor the fight against doping globally. The key objectives of WADA include harmonization of anti-doping policies, rules, and regulations; development and implementation of anti-doping programs; accreditation of anti-doping laboratories worldwide; supporting of scientific research and development to provide novel anti-doping technologies and techniques; promoting educational programs to raise awareness about the dangers and consequences of doping. To provide clean, fair, and drug-free sport, WADA is developing and updating Code; the core document; as the framework for anti-doping policies, regulations, and practices. This global harmonization of recognized anti-doping rules is applied across sports organizations in different countries. The Code protects rights and interests of fair-play athletes by deterring and sanctioning of doping violations. However, the Code has also several limitations and disadvantages. The Code is applied globally across different states, societies, and cultures that can have a diverse attitude towards doping. It is a comprehensive document that has to be carefully understood by athletes, coaches, and sports organizations. Moreover, the Code necessitates continuous updates and revisions to ensure its relevance and efficacy in combating doping. In contrast to the Code which is updated once in six years; the first Code was approved in 2003, second in 2009, third in 2015, fourth in 2021, the fifth will be updated in 2027; the prohibited list is updated annually (WADA 2023h). Updates and modifications are incorporated in the prohibited list to address novel emerging substances and methods; to ensure health and safety of athletes according to contemporary pharmacokinetic data; and to reflect an advancement in detection methods. However, the prohibited list also has several limitations and disadvantages, for example, the process of identifying and adding new substances and methods on the prohibited list can be delayed behind the emergence of novel doping techniques or substances. Although the prohibited list is updated annually, incorporation of all potential doping substances (such as designer drugs) or method cannot be completely encompassed on the list. Variability of incorporated substances and methods requires significant resources in terms of drug testing and monitoring. In summary, WADA, an international anti-doping organization, exerts a considerable effort to eliminate prohibited substances and methods, in order to provide fair-play and clean sport. Various methods and strategies are still developed with the aim to overcome detection of prohibited substances and methods; however, anti-doping rules and regulations are also continuously updated and modified. Therefore, the fight against doping is represented by continuous process of learning and implementation of newly acquired knowledge and strategies to combat misuse of prohibited substances and methods.

Journal Pre-proof

6 Conclusion

Anti-doping controls are still under development and novel methods and techniques for detection of prohibited substances and methods are continuously introduced. This process is necessary because novel doping substances and methods are under development as well. For a sake of completeness is also important to mention that an increase/decrease of the reported findings can be associated with modification and changes implemented by WADA. In addition, the number of reported findings can further be affected from external sources, such as during the global pandemic of coronavirus disease 2019 (COVID-19) which remarkably affected the findings reported in 2020 and 2021 (Table 1). In general, an extensive anti-doping analysis in sports results in thousands of adverse analytical/atypical findings reported annually (Table 1). These findings are represented by substances or methods that are clustered, in the 2023 prohibited list, into 12 sections. The most commonly detected findings, reported during last 20 years, belong to the section of anabolic agents with a potential to enhance muscle strength. Second commonly detected findings, members of stimulants, also have a potential to enhance performance of athletes. However, diuretics and masking agents that represent third commonly detected section are usually misused with the aim to dilute other doping agents or to reduce athlete's body mass. As summarized in this review, substances and methods potentially misused as doping agents exhibit a wide range of pharmacological functions and provide various potential cheating methods. It is also important to note that due to advances in analytical methods, the findings can be further detected after unintentional or unknown expositions. Nevertheless, anti-doping strategies and methods represent a valuable tool to protect health of athletes and to maintain fairness in sports.

Funding

This work was supported by the project UHK PrF Excelence project 2208/2023-2024 and by the the Deanship of Scientific Research, King Saud University for funding through Vice Deanship of Scientific Research Chairs; Research Chair of Doping.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Author Statement

Patrik Oleksak: Writing - original draft. **Eugenie Nepovimova**: Writing - review & editing. **Marian Valko**: Writing - review & editing. **Saleh Alwasel**: Writing - review & editing. **Suliman Alomar**: Resources, Writing - Reviewing and Editing. **Kamil Kuca**: Funding acquisition; Supervision; Correspondence

Journal Pre-proof

References

Abdalla MI, Herfarth H (2016) Budesonide for the treatment of ulcerative colitis. Expert Opinion on Pharmacotherapy 17:1549–1559. https://doi.org/10.1080/14656566.2016.1183648

Abramova N, Hubbard J, Schertz J, Richter E (2019) Safety of Follitropin Alfa/Lutropin Alfa for Stimulation of Follicular Development. Drug Saf 42:453–461. https://doi.org/10.1007/s40264-018-0742-3

Adami PE, Koutlianos N, Baggish A, et al (2022) Cardiovascular effects of doping substances, commonly prescribed medications and ergogenic aids in relation to sports: a position statement of the sport cardiology and exercise nucleus of the European Association of Preventive Cardiology. European Journal of Preventive Cardiology 29:559–575. https://doi.org/10.1093/eurjpc/zwab198

Adams BS, Nguyen H (2022) Salmeterol. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Agriesti F, Tataranni T, Pacelli C, et al (2020) Nandrolone induces a stem cell-like phenotype in human hepatocarcinoma-derived cell line inhibiting mitochondrial respiratory activity. Sci Rep 10:2287. https://doi.org/10.1038/s41598-020-58871-1

Akalestou E, Genser L, Rutter GA (2020) Glucocorticoid Metabolism in Obesity and Following Weight Loss. Frontiers in Endocrinology 11:

Alali AS, Mukherjee K, McCredie VA, et al (2017) Beta-Blockers and Traumatic Brain Injury: A Systematic Review and Meta-Analysis. Ann Surg 266:952–961. https://doi.org/10.1097/SLA.0000000000002286

Albertsdóttir AD, Van Gansbeke W, Van Eenoo P, Polet M (2021) Enabling the inclusion of non-hydrolysed sulfated long term anabolic steroid metabolites in a screening for doping substances by means of gas chromatography quadrupole time-of-flight mass spectrometry. Journal of Chromatography A 1642:462039. https://doi.org/10.1016/j.chroma.2021.462039

Alkaseem M, Baron M (2018) SERS and DFT study of 5F-PB-22. Journal of Raman Spectroscopy 49:1594–1606. https://doi.org/10.1002/jrs.5444

Altschule MD, Tillotson KJ (1948) The use of testosterone in the treatment of depressions. N Engl J Med 239:1036–1038. https://doi.org/10.1056/NEJM194812302392704

Alves P, Amaral C, Teixeira N, Correia-da-Silva G (2020) Cannabis sativa: Much more beyond Δ9-tetrahydrocannabinol. Pharmacological Research 157:104822. https://doi.org/10.1016/j.phrs.2020.104822

Ameline A, Gheddar L, Raul J-S, Kintz P (2021) Characterization of letrozole in human hair using LC-MS/MS and confirmation by LC-HRMS: Application to a doping case. Journal of Chromatography B 1162:122495. https://doi.org/10.1016/j.jchromb.2020.122495

Antunes NDJ, Cavalli RC, Marques MP, Lanchote VL (2013) Stereoselective determination of metoprolol and its metabolite α-hydroxymetoprolol in plasma by LC-MS/MS: application to pharmacokinetics during pregnancy. Chirality 25:1–7. https://doi.org/10.1002/chir.22102

Armanini D, Sabbadin C, Donà G, et al (2014) Aldosterone receptor blockers spironolactone and canrenone: two multivalent drugs. Expert Opin Pharmacother 15:909–912. https://doi.org/10.1517/14656566.2014.896901

Aronson JK (ed) (2016) Formoterol. In: Meyler's Side Effects of Drugs (Sixteenth Edition). Elsevier, Oxford, pp 444–449

Journal Pre-proof

Ashique S, Tyagi DS, Das R, et al (2022) An Essential textbook of Pharmaceutical Medicinal Chemistry: Fundamental book for Medicinal Chemistry. Shashwat Publication

Avois L, Robinson N, Saudan C, et al (2006) Central nervous system stimulants and sport practice. Br J Sports Med 40:i16–i20. https://doi.org/10.1136/bjsm.2006.027557

Ayotte C (2006) Significance of 19-norandrosterone in athletes' urine samples. Br J Sports Med 40:i25–i29. https://doi.org/10.1136/bjsm.2006.028027

Bakheit AH, Ali R, Alshahrani AD, El-Azab AS (2021) Chapter Two - Bisoprolol: A comprehensive profile. In: Al-Majed AA (ed) Profiles of Drug Substances, Excipients and Related Methodology. Academic Press, pp 51–89

Balcells G, Matabosch X, Ventura R (2017) Detection of stanozolol O- and N-sulfate metabolites and their evaluation as additional markers in doping control. Drug Test Anal 9:1001–1010. https://doi.org/10.1002/dta.2107

Banister SD, Stuart J, Kevin RC, et al (2015) Effects of Bioisosteric Fluorine in Synthetic Cannabinoid Designer Drugs JWH-018, AM-2201, UR-144, XLR-11, PB-22, 5F-PB-22, APICA, and STS-135. ACS Chem Neurosci 6:1445–1458. https://doi.org/10.1021/acschemneuro.5b00107

Barakat R, Park CJ, Perez PA, et al (2018) Female antiestrogens. In: Encyclopedia of Reproduction. Elsevier, pp 740–747

Barbieri M, Tirri M, Bilel S, et al (2022) Synthetic cannabinoid JWH-073 alters both acute behavior and in vivo/vitro electrophysiological responses in mice. Front Psychiatry 13:953909. https://doi.org/10.3389/fpsyt.2022.953909

Barceloux DG, Palmer RB (2013) Anabolic-androgenic steroids. Dis Mon 59:226–248. https://doi.org/10.1016/j.disamonth.2013.03.010

Barghi TS, Tavana MM, Amini E (2022) Medical Treatment in Men with Infertility Can Be Misinterpreted as Doping Practice: A Case of Unintentional World Anti-doping Agency (WADA) Code Violation. Asian J Sports Med 13:. https://doi.org/10.5812/asjsm-121004

Barros-Oliveira M da C, Costa-Silva DR, Andrade DB de, et al (2017) Use of anastrozole in the chemoprevention and treatment of breast cancer: A literature review. Rev Assoc Med Bras (1992) 63:371–378. https://doi.org/10.1590/1806-9282.63.04.371

Behairy A, Mohamed WAM, Ebraheim LLM, et al (2021) Boldenone Undecylenate-Mediated Hepatorenal Impairment by Oxidative Damage and Dysregulation of Heat Shock Protein 90 and Androgen Receptors Expressions: Vitamin C Preventive Role. Frontiers in Pharmacology 12:

Behonick G, Shanks KG, Firchau DJ, et al (2014) Four Postmortem Case Reports with Quantitative Detection of the Synthetic Cannabinoid, 5F-PB-22. J Anal Toxicol 38:559–562. https://doi.org/10.1093/jat/bku048

Bell P, Ten Have C, Lauchs M (2016) A case study analysis of a sophisticated sports doping network: Lance Armstrong and the USPS Team. International Journal of Law, Crime and Justice 46:57–68. https://doi.org/10.1016/j.ijlcj.2016.03.001

Benedetti F, Pollo A, Colloca L (2007) Opioid-Mediated Placebo Responses Boost Pain Endurance and Physical Performance: Is It Doping in Sport Competitions? J Neurosci 27:11934–11939. https://doi.org/10.1523/JNEUROSCI.3330-07.2007

Berezanskaya J, Cade W, Best TM, et al (2022) ADHD Prescription Medications and Their Effect on Athletic Performance: A Systematic Review and Meta-analysis. Sports Medicine - Open 8:5. https://doi.org/10.1186/s40798-021-00374-y

Journal Pre-proof

Berger B, Bachmann F, Duthaler U, et al (2018) Cytochrome P450 Enzymes Involved in Metoprolol Metabolism and Use of Metoprolol as a CYP2D6 Phenotyping Probe Drug. Front Pharmacol 9:774. https://doi.org/10.3389/fphar.2018.00774

Billington CK, Penn RB, Hall IP (2017) β2-agonists. Handb Exp Pharmacol 237:23–40. https://doi.org/10.1007/164_2016_64

Bonney CF, Hatten B, Wang GS (2019) Toxicity From Unintentional Pediatric Ingestion of a Performance-Enhancing Drug: A Case Report With Review of Clenbuterol Toxicity and Treatment. The Journal of Emergency Medicine 57:e105–e108. https://doi.org/10.1016/j.jemermed.2019.06.016

Boszkiewicz K, Piwowar A, Petryszyn P (2022) Aromatase Inhibitors and Risk of Metabolic and Cardiovascular Adverse Effects in Breast Cancer Patients—A Systematic Review and Meta-Analysis. Journal of Clinical Medicine 11:3133. https://doi.org/10.3390/jcm11113133

Botrè F, Georgakopoulos C, Elrayess MA (2020) Metabolomics and doping analysis: promises and pitfalls. Bioanalysis 12:719–722. https://doi.org/10.4155/bio-2020-0137

Boucher A, Payen C, Garayt C, et al (2011) Salbutamol misuse or abuse with fatal outcome: A case-report. Hum Exp Toxicol 30:1869–1871. https://doi.org/10.1177/0960327110388957

Bouckaert J, Lefebvre R, Pannier JL (1989) Effects of diltiazem and atenolol on exercise performance in man. J Sports Med Phys Fitness 29:240–244

Brinkman DJ, Tichelaar J, Van Agtmael MA (2021) [Can furosemide mask the use of performance-enhancing drugs?]. Ned Tijdschr Geneeskd 165:D5915

Brinkman JE, Tariq MA, Leavitt L, Sharma S (2022) Physiology, Growth Hormone. StatPearls Publishing

Brown HG, Perrett D (2011) Detection of Doping in Sport: Detecting Anabolic-androgenic Steroids in Human Fingernail Clippings. Med Leg J 79:67–69. https://doi.org/10.1258/mlj.2011.011013

Butenandt A, Hanisch G (1935) Über Testosteron. Umwandlung des Dehydro-androsterons in Androstendiol und Testosteron; ein Weg zur Darstellung des Testosterons aus Cholesterin. 237:89–97. https://doi.org/10.1515/bchm2.1935.237.1-3.89

Büttner A, Thieme D (2010) Side Effects of Anabolic Androgenic Steroids: Pathological Findings and Structure–Activity Relationships. In: Thieme D, Hemmersbach P (eds) Doping in Sports: Biochemical Principles, Effects and Analysis. Springer, Berlin, Heidelberg, pp 459–484

Bylund DB, Gruetter CA (2007) Propranolol. In: Enna SJ, Bylund DB (eds) xPharm: The Comprehensive Pharmacology Reference. Elsevier, New York, pp 1–9

Cadwallader AB, de la Torre X, Tieri A, Botrè F (2010) The abuse of diuretics as performance-enhancing drugs and masking agents in sport doping: pharmacology, toxicology and analysis. Br J Pharmacol 161:1–16. https://doi.org/10.1111/j.1476-5381.2010.00789.x

Campos DR, Yonamine M, de Moraes Moreau RL (2003) Marijuana as Doping in Sports. Sports Med 33:395–399. https://doi.org/10.2165/00007256-200333060-00001

Carone L, Oxberry SG, Twycross R, et al (2016) Furosemide. Journal of Pain and Symptom Management 52:144–150. https://doi.org/10.1016/j.jpainsymman.2016.05.004

Carson P, Hong CJ, Otero-Vinas M, et al (2015) Liver Enzymes and Lipid Levels in Patients With Lipodermatosclerosis and Venous Ulcers Treated With a Prototypic Anabolic Steroid (Stanozolol): A Prospective, Randomized, Double-Blinded, Placebo-Controlled Trial. The International Journal of Lower Extremity Wounds 14:11–18. https://doi.org/10.1177/1534734614562276

Journal Pre-proof

Cavalcanti G de A, Leal FD, Garrido BC, et al (2013) Detection of designer steroid methylstenbolone in “nutritional supplement” using gas chromatography and tandem mass spectrometry: elucidation of its urinary metabolites. Steroids 78:228–233. https://doi.org/10.1016/j.steroids.2012.11.009

CDC (2021) Commonly Used Terms | Opioids | CDC. In: Centers for Disease Control and Prevention. https://www.cdc.gov/opioids/basics/terms.html. Accessed 2 Apr 2023

Chang C-W, Huang T-Y, Tseng Y-C, et al (2014) Positive doping results caused by the single-dose local injection of triamcinolone acetonide. Forensic Science International 244:1–6. https://doi.org/10.1016/j.forsciint.2014.07.024

Chang M (2012) Tamoxifen Resistance in Breast Cancer. Biomol Ther (Seoul) 20:256–267. https://doi.org/10.4062/biomolther.2012.20.3.256

Che X, Li Y, Fang Y, et al (2018) Antiarrhythmic drug-induced smell and taste disturbances. Medicine (Baltimore) 97:e11112. https://doi.org/10.1097/MD.0000000000011112

Chen M-M, Zhao Y-P, Zhao Y, et al (2021) Regulation of Myostatin on the Growth and Development of Skeletal Muscle. Frontiers in Cell and Developmental Biology 9:

Chen W, Gao R, Xie X, et al (2015) A metabolomic study of the PPARδ agonist GW501516 for enhancing running endurance in Kunming mice. Sci Rep 5:9884. https://doi.org/10.1038/srep09884

Chester N (2022) Hormone and metabolic modulators. In: Drugs in Sport, 8th edn. Routledge

Chimalakonda KC, Bratton SM, Le V-H, et al (2011) Conjugation of Synthetic Cannabinoids JWH-018 and JWH-073, Metabolites by Human UDP-Glucuronosyltransferases. Drug Metab Dispos 39:1967–1976. https://doi.org/10.1124/dmd.111.040709

Choi J, Smitz J (2014) Luteinizing hormone and human chorionic gonadotropin: Origins of difference. Molecular and Cellular Endocrinology 383:203–213. https://doi.org/10.1016/j.mce.2013.12.009

Chourpiliadis C, Aeddula NR (2023) Physiology, Glucocorticoids. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Christiansen AR, Lipshultz LI, Hotaling JM, Pastuszak AW (2020) Selective androgen receptor modulators: the future of androgen therapy? Transl Androl Urol 9:S135–S148. https://doi.org/10.21037/tau.2019.11.02

Čižmáriková R, Valentová J, Horáková R (2020) Chirality of β2-agonists. An overview of pharmacological activity, stereoselective analysis, and synthesis. Open Chemistry 18:628–647. https://doi.org/10.1515/chem-2020-0056

Clarke R, Liu MC, Bouker KB, et al (2003) Antiestrogen resistance in breast cancer and the role of estrogen receptor signaling. Oncogene 22:7316–7339. https://doi.org/10.1038/sj.onc.1206937

Cohen PA, Wen A, Gerona R (2018) Prohibited Stimulants in Dietary Supplements After Enforcement Action by the US Food and Drug Administration. JAMA Internal Medicine 178:1721–1723. https://doi.org/10.1001/jamainternmed.2018.4846

Coll S, Monfort N, Alechaga É, et al (2021) Elimination profiles of betamethasone after different administration routes: Evaluation of the reporting level and washout periods to ensure safe therapeutic administrations. Drug Testing and Analysis 13:348–359. https://doi.org/10.1002/dta.2928

Cong Y, Dong H, Wei X, et al (2019) A novel murine antibody and an open sandwich immunoassay for the detection of clenbuterol. Ecotoxicology and Environmental Safety 182:109473. https://doi.org/10.1016/j.ecoenv.2019.109473

Journal Pre-proof

Connors NJ, Hoffman RS (2013) Experimental treatments for cocaine toxicity: a difficult transition to the bedside. J Pharmacol Exp Ther 347:251–257. https://doi.org/10.1124/jpet.113.206383

David K, Dingemanse E, Freud J, Laqueur E (1935) Über krystallinisches männliches Hormon aus Hoden (Testosteron), wirksamer als aus Harn oder aus Cholesterin bereitetes Androsteron. 233:281–283. https://doi.org/10.1515/bchm2.1935.233.5-6.281

Davis S (2007) Formoterol. In: Enna SJ, Bylund DB (eds) xPharm: The Comprehensive Pharmacology Reference. Elsevier, New York, pp 1–7

De Brabander HF, Poelmans S, Schilt R, et al (2004) Presence and metabolism of the anabolic steroid boldenone in various animal species: a review. Food Addit Contam 21:515–525. https://doi.org/10.1080/02652030410001687717

De Gregori S, De Gregori M, Ranzani GN, et al (2012) Morphine metabolism, transport and brain disposition. Metab Brain Dis 27:1–5. https://doi.org/10.1007/s11011-011-9274-6

de la Torre R, Farré M, Navarro M, et al (2004) Clinical pharmacokinetics of amfetamine and related substances: monitoring in conventional and non-conventional matrices. Clin Pharmacokinet 43:157–185. https://doi.org/10.2165/00003088-200443030-00002

de la Torre X, Iannone M, Botrè F (2021) Improving the detection of anabolic steroid esters in human serum by LC–MS. Journal of Pharmaceutical and Biomedical Analysis 194:113807. https://doi.org/10.1016/j.jpba.2020.113807

DeBerardinis RJ, Thompson CB (2008) Chapter 14 - Metabolism of Cell Growth and Proliferation. In: Mendelsohn J, Howley PM, Israel MA, et al. (eds) The Molecular Basis of Cancer (Third Edition). W.B. Saunders, Philadelphia, pp 189–203

Dellapasqua S, Colleoni M (2010) Letrozole. Expert Opinion on Drug Metabolism & Toxicology 6:251–259. https://doi.org/10.1517/17425250903540246

Derave W, Tipton KD (2014) Dietary Supplements for Aquatic Sports. International Journal of Sport Nutrition and Exercise Metabolism 24:437–449. https://doi.org/10.1123/ijsnem.2014-0017

Derosa G, Maffioli P, D'Avino M, et al (2017) Efficacy and safety of two dosages of canrenone as add-on therapy in hypertensive patients taking ace-inhibitors or angiotensin II receptor blockers and hydrochlorothiazide at maximum dosage in a randomized clinical trial: The ESCAPE-IT trial. Cardiovasc Ther 35:47–54. https://doi.org/10.1111/1755-5922.12235

Deventer K, Pozo OJ, Van Eenoo P, Delbeke FT (2009) Detection of urinary markers for thiazide diuretics after oral administration of hydrochlorothiazide and altizide-relevance to doping control analysis. Journal of Chromatography A 1216:2466–2473. https://doi.org/10.1016/j.chroma.2009.01.032

Dhingra K (1999) Antiestrogens – Tamoxifen, SERMs and Beyond. Invest New Drugs 17:285–311. https://doi.org/10.1023/A:1006348907994

Diel P (2020) Caffeine and Doping—What Have We Learned since 2004. Nutrients 12:2167. https://doi.org/10.3390/nu12082167

Dimeo P, Hunt TM, Horbury R (2011) The Individual and the State: A Social Historical Analysis of the East German 'Doping System.' Sport in History 31:218–237. https://doi.org/10.1080/17460263.2011.590026

Ding JB, Ng MZ, Huang SS, et al (2021) Anabolic-Androgenic Steroid Misuse: Mechanisms, Patterns of Misuse, User Typology, and Adverse Effects. J Sports Med (Hindawi Publ Corp) 2021:7497346. https://doi.org/10.1155/2021/7497346

Journal Pre-proof

Dinis-Oliveira RJ (2017) Metabolomics of Methylphenidate and Ethylphenidate: Implications in Pharmacological and Toxicological Effects. Eur J Drug Metab Pharmacokinet 42:11–16. https://doi.org/10.1007/s13318-016-0362-1

Diniz LF, Carvalho PSJr, da Nova Mussel W, et al (2019) Racemic Salts and Solid Solutions of Enantiomers of the Antihypertensive Drug Carvedilol. Crystal Growth & Design 19:4498–4509. https://doi.org/10.1021/acs.cgd.9b00263

do Vale GT, Ceron CS, Gonzaga NA, et al (2019) Three Generations of β-blockers: History, Class Differences and Clinical Applicability. Curr Hypertens Rev 15:22–31. https://doi.org/10.2174/1573402114666180918102735

Docter S, Khan M, Gohal C, et al (2020) Cannabis Use and Sport: A Systematic Review. Sports Health 12:189–199. https://doi.org/10.1177/1941738120901670

Dodd-Butera T, Broderick M (2005) Tamoxifen. In: Wexler P (ed) Encyclopedia of Toxicology (Second Edition). Elsevier, New York, pp 133–134

Donati F, de la Torre X, Pagliarosi S, et al (2022) Detection of Homologous Blood Transfusion in Sport Doping by Flow Cytofluorimetry: State of the Art and New Approaches to Reduce the Risk of False-Negative Results. Frontiers in Sports and Active Living 4:

Durdiakova J, Ostatnikova D, Celec P (2011) Testosterone and its metabolites--modulators of brain functions. Acta neurobiologiae experimentalis 71:

Durrani M, Bansal K (2023) Methadone. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Earl JA, Kim ED (2019) Enclomiphene citrate: A treatment that maintains fertility in men with secondary hypogonadism. Expert Review of Endocrinology & Metabolism 14:157–165. https://doi.org/10.1080/17446651.2019.1612239

Efimenko IV, Valancy D, Dubin JM, Ramasamy R (2022) Adverse effects and potential benefits among selective androgen receptor modulators users: a cross-sectional survey. Int J Impot Res 34:757–761. https://doi.org/10.1038/s41443-021-00465-0

Eggold JT, Rankin EB (2019) Erythropoiesis, EPO, Macrophages, and Bone. Bone 119:36–41. https://doi.org/10.1016/j.bone.2018.03.014

Eichner A, Lewis LA, Leonard B, et al (2021) Generic Pharmaceuticals as a Source of Diuretic Contamination in Athletes Subject to Sport Drug Testing. Front Sports Act Living 3:692244. https://doi.org/10.3389/fspor.2021.692244

Elkader A, Sproule B (2005) Buprenorphine. Clin Pharmacokinet 44:661–680. https://doi.org/10.2165/00003088-200544070-00001

Elliott S, Pham E, Macdougall IC (2008) Erythropoietins: a common mechanism of action. Exp Hematol 36:1573–1584. https://doi.org/10.1016/j.exphem.2008.08.003

Elmore JS, Baumann MH (2018) Repeated Exposure to the “Spice” Cannabinoid JWH-018 Induces Tolerance and Enhances Responsiveness to 5-HT1A Receptor Stimulation in Male Rats. Frontiers in Psychiatry 9:

ElSohly MA, Gul W, Tolbert C, et al (2015) Methylhexanamine is not detectable in Pelargonium or Geranium species and their essential oils: A multi-centre investigation. Drug Testing and Analysis 7:645–654. https://doi.org/10.1002/dta.1726

Ergen E, Hazir T, Celebi M, et al (2021) Effects of beta-blockers on archery performance, body sway and aiming behaviour. BMJ Open Sport Exerc Med 7:e001071. https://doi.org/10.1136/bmjsem-2021-001071

Journal Pre-proof

Euler L, Gillard N, Delahaut P, et al (2022) Assessing human urinary clomiphene metabolites after consumption of eggs from clomiphene-treated laying hens using chromatographic-mass spectrometric approaches. Analytica Chimica Acta 1202:339661. https://doi.org/10.1016/j.aca.2022.339661

Faraone SV (2018) The pharmacology of amphetamine and methylphenidate: Relevance to the neurobiology of attention-deficit/hyperactivity disorder and other psychiatric comorbidities. Neuroscience & Biobehavioral Reviews 87:255–270. https://doi.org/10.1016/j.neubiorev.2018.02.001

Farzam K, Faizy RM, Saadabadi A (2023) Stimulants. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Favretto D, Stocchero G, Pertile R, et al (2022) Post-mortem investigation into a death involving doping agents: The case of a body builder. Drug Testing and Analysis 14:1795–1799. https://doi.org/10.1002/dta.3350

Favretto D, Visentin S, Scrivano S, et al (2019) Multiple incidence of the prescription diuretic hydrochlorothiazide in compounded nutritional supplements. Drug Test Anal 11:512–522. https://doi.org/10.1002/dta.2499

Fink J, Schoenfeld BJ, Hackney AC, et al (2021) Human chorionic gonadotropin treatment: a viable option for management of secondary hypogonadism and male infertility. Expert Review of Endocrinology & Metabolism 16:1–8. https://doi.org/10.1080/17446651.2021.1863783

Fiorentini C, Esefeld K, Halle M (2022) Medication for patient-athletes—the therapeutic use exemption procedure. European Journal of Preventive Cardiology 29:556–558. https://doi.org/10.1093/eurjpc/zwac005

Franke WW, Berendonk B (1997) Hormonal doping and androgenization of athletes: a secret program of the German Democratic Republic government. Clinical Chemistry 43:1262–1279. https://doi.org/10.1093/clinchem/43.7.1262

Franz F, Angerer V, Hermanns-Clausen M, et al (2016) Metabolites of synthetic cannabinoids in hair—proof of consumption or false friends for interpretation? Anal Bioanal Chem 408:3445–3452. https://doi.org/10.1007/s00216-016-9422-2

Fraser AD (2004) Doping control from a global and national perspective. Ther Drug Monit 26:171–174. https://doi.org/10.1097/00007691-200404000-00015

Furman BL (2017) Nandrolone☆. In: Reference Module in Biomedical Sciences. Elsevier

Furman BL (2019) Prednisolone☆. In: Reference Module in Biomedical Sciences. Elsevier

Gad El-hak ASM, Mohammed AAK, Abdel Hakiem AF, Mahfouz RM (2019) Molecular conformation, vibrational spectroscopic and NBO analysis of atenolol and atenolol-hydrochlorothiazide cocrystals. Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy 222:117200. https://doi.org/10.1016/j.saa.2019.117200

Ganchev B, Heinkele G, Kerb R, et al (2011) Quantification of clomiphene metabolite isomers in human plasma by rapid-resolution liquid chromatography–electrospray ionization–tandem mass spectrometry. Anal Bioanal Chem 400:3429–3441. https://doi.org/10.1007/s00216-011-5045-9

Ganesan K, Rahman S, Zito PM (2022) Anabolic Steroids. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Garzinsky A-M, Thomas A, Guddat S, et al (2023) Dried blood spots for doping controls—Development of a comprehensive initial testing procedure with fully automated sample preparation. Biomedical Chromatography 37:e5633. https://doi.org/10.1002/bmc.5633

Gawchik SM, Saccar CL (2000) A Risk-Benefit Assessment of Intranasal Triamcinolone Acetonide in Allergic Rhinitis. Drug-Safety 23:309–322. https://doi.org/10.2165/00002018-200023040-00004

Journal Pre-proof

Gerace E, Salomone A, Abbadessa G, et al (2012) Rapid determination of anti-estrogens by gas chromatography/mass spectrometry in urine: Method validation and application to real samples. Journal of Pharmaceutical Analysis 2:1–11. https://doi.org/10.1016/j.jpha.2011.09.011

Ghobadi C, Gregory A, Crewe HK, et al (2008) CYP2D6 is primarily responsible for the metabolism of clomiphene. Drug Metab Pharmacokinet 23:101–105. https://doi.org/10.2133/dmpk.23.101

Giovanelli L, Quinton R (2022) Therapeutic effects of androgens for cachexia. Best Practice & Research Clinical Endocrinology & Metabolism 36:101598. https://doi.org/10.1016/j.beem.2021.101598

Giraud S, Sottas P-E, Robinson N, Saugy M (2010) Blood Transfusion in Sports. In: Thieme D, Hemmersbach P (eds) Doping in Sports: Biochemical Principles, Effects and Analysis. Springer, Berlin, Heidelberg, pp 295–304

Gómez-Guerrero NA, González-López NM, Zapata-Velásquez JD, et al (2022) Synthetic Peptides in Doping Control: A Powerful Tool for an Analytical Challenge. ACS Omega 7:38193–38206. https://doi.org/10.1021/acsomega.2c05296

Göschl L, Gmeiner G, Gärtner P, et al (2022) Detection of DHCMT long-term metabolite glucuronides with LC-MSMS as an alternative approach to conventional GC-MSMS analysis. Steroids 180:108979. https://doi.org/10.1016/j.steroids.2022.108979

Gou P, Cheng X, Leng J, Su N (2022) A Real-World Study of Recombinant Human Growth Hormone in the Treatment of Idiopathic Short Stature and Growth Hormone Deficiency. Ther Clin Risk Manag 18:113–124. https://doi.org/10.2147/TCRM.S363564

Gould D, Carson S (2008) Life skills development through sport: current status and future directions. International Review of Sport and Exercise Psychology 1:58–78. https://doi.org/10.1080/17509840701834573

Grant KS, Petroff R, Isoherranen N, et al (2018) Cannabis use during pregnancy: Pharmacokinetics and effects on child development. Pharmacology & Therapeutics 182:133–151. https://doi.org/10.1016/j.pharmthera.2017.08.014

Greene P, Kaiser M (2019) How athletes' rights can be better protected when it comes to retroactive Therapeutic Use Exemptions. Int Sports Law J 19:29–37. https://doi.org/10.1007/s40318-019-00154-7

Grewal N, Huecker MR (2023) Opioid Prescribing. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Gruetter CA (2007) Terbutaline. In: Enna SJ, Bylund DB (eds) xPharm: The Comprehensive Pharmacology Reference. Elsevier, New York, pp 1–4

Guevara N, Lorier M, Vázquez M, et al (2018) Revisiting Pharmacokinetics and Pharmacogenetics of Methadone in Healthy Volunteers. IntechOpen

Guo Q, Peng Y, Zhao X, Chen Y (2022) Rapid Detection of Clenbuterol Residues in Pork Using Enhanced Raman Spectroscopy. Biosensors 12:859. https://doi.org/10.3390/bios12100859

Hackney AC (2018a) Chapter 6 - Beta-2 Agonists. In: Hackney AC (ed) Doping, Performance Enhancing Drugs, and Hormones in Sport. Elsevier, pp 65–76

Hackney AC (2018b) Chapter 7 - Hormone and Metabolic Modulators. In: Hackney AC (ed) Doping, Performance Enhancing Drugs, and Hormones in Sport. Elsevier, pp 77–89

Hackney AC (2018c) Chapter 9 - Beta Blockers. In: Hackney AC (ed) Doping, Performance Enhancing Drugs, and Hormones in Sport. Elsevier, pp 103–111

Journal Pre-proof

Hall RCW, Hall RCW (2005) Abuse of supraphysiologic doses of anabolic steroids. South Med J 98:550–555. https://doi.org/10.1097/01.SMJ.0000157531.04472.B2

Han S (2011) Chapter Two - Crystal Structure of Activin Receptor Type IIB Kinase Domain. In: Litwack G (ed) Vitamins & Hormones. Academic Press, pp 29–38

Han Y, Cao L, Yuan K, et al (2022) Unique Pharmacology, Brain Dysfunction, and Therapeutic Advancements for Fentanyl Misuse and Abuse. Neurosci Bull 38:1365–1382. https://doi.org/10.1007/s12264-022-00872-3

Hawksworth DJ, Burnett AL (2019) Chapter 4.2 - Other Hormonal Therapies and Men's Health. In: Yafi FA, Yafi NR (eds) Effects of Lifestyle on Men's Health. Academic Press, pp 253–268

Heidet M, Abdel Wahab A, Ebadi V, et al (2019) Severe Hypoglycemia Due to Cryptic Insulin Use in a Bodybuilder. The Journal of Emergency Medicine 56:279–281. https://doi.org/10.1016/j.jemermed.2018.10.030

Heiland CE, Ericsson M, Pohanka A, et al (2022) Optimizing detection of erythropoietin receptor agonists from dried blood spots for anti-doping application. Drug Testing and Analysis 14:1377–1386. https://doi.org/10.1002/dta.3260

Helmlin H-J, Mürner A, Steiner S, et al (2016) Detection of the diuretic hydrochlorothiazide in a doping control urine sample as the result of a non-steroidal anti-inflammatory drug (NSAID) tablet contamination. Forensic Science International 267:166–172. https://doi.org/10.1016/j.forsciint.2016.08.029

Herman LL, Bashir K (2023) Hydrochlorothiazide. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Heuberger JAAC, Cohen AF (2019) Review of WADA Prohibited Substances: Limited Evidence for Performance-Enhancing Effects. Sports Med 49:525–539. https://doi.org/10.1007/s40279-018-1014-1

Holt RIG, Ho KKY (2019) The Use and Abuse of Growth Hormone in Sports. Endocrine Reviews 40:1163–1185. https://doi.org/10.1210/er.2018-00265

Horstman AM, Dillon EL, Urban RJ, Sheffield-Moore M (2012) The Role of Androgens and Estrogens on Healthy Aging and Longevity. J Gerontol A Biol Sci Med Sci 67:1140–1152. https://doi.org/10.1093/gerona/gls068

Höybye C, Christiansen JS (2015) Growth hormone replacement in adults – current standards and new perspectives. Best Practice & Research Clinical Endocrinology & Metabolism 29:115–123. https://doi.org/10.1016/j.beem.2014.09.006

Hsu E, Bajaj T (2022) Beta 2 Agonists. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Hudzik TJ, Patel M, Brown A (2021) β2-Adrenoceptor agonist activity of higenamine. Drug Testing and Analysis 13:261–267. https://doi.org/10.1002/dta.2992

Humans IWG on the E of CR to, Cancer IA for R on (2001) Some Thyrotropic Agents. World Health Organization

Hunt TM, Dimeo P, Jedlicka SR (2012) The historical roots of today's problems: A critical appraisal of the international anti-doping movement. Performance Enhancement & Health 1:55–60. https://doi.org/10.1016/j.peh.2012.05.001

Irvine MA, Oller D, Boggis J, et al (2022) Estimating naloxone need in the USA across fentanyl, heroin, and prescription opioid epidemics: a modelling study. The Lancet Public Health 7:e210–e218. https://doi.org/10.1016/S2468-2667(21)00304-2

Ivanova S, Ivanov K, Petkova E, et al (2017) Methods for detection of the misuse of "anti-oestrogens and aromatase inhibitors" in sport. Biomedical Research 28:7157–7166

Journal Pre-proof

Jacobson GA, Hostrup M, Narkowicz CK, et al (2017) Enantioselective disposition of (R)-salmeterol and (S)-salmeterol in urine following inhaled dosing and application to doping control. Drug Test Anal 9:1262–1266. https://doi.org/10.1002/dta.2131

Jelkmann W (2016) Erythropoietin. Sports Endocrinology 47:115–127. https://doi.org/10.1159/000445174

Jermak CM, Dellacroce JT, Heffez J, Peyman GA (2007) Triamcinolone Acetonide in Ocular Therapeutics. Survey of Ophthalmology 52:503–522. https://doi.org/10.1016/j.survophthal.2007.06.004

Jessen S, Solheim SA, Jacobson GA, et al (2020) Beta2 -adrenergic agonist clenbuterol increases energy expenditure and fat oxidation, and induces mTOR phosphorylation in skeletal muscle of young healthy men. Drug Test Anal 12:610–618. https://doi.org/10.1002/dta.2755

Jiménez C, Ventura R, Segura J (2004) Detection in urine of efaproxiral (RSR13), a potential doping agent, by a routine screening procedure based on methylation followed by gas chromatography/mass spectrometry. Analytica Chimica Acta 505:227–229. https://doi.org/10.1016/j.aca.2003.10.069

Jobe AH, Milad MA, Peppard T, Jusko WJ (2020) Pharmacokinetics and Pharmacodynamics of Intramuscular and Oral Betamethasone and Dexamethasone in Reproductive Age Women in India. Clinical and Translational Science 13:391–399. https://doi.org/10.1111/cts.12724

Jones AW, Karlsson L (2005) Relation between blood- and urine-amphetamine concentrations in impaired drivers as influenced by urinary pH and creatinine. Hum Exp Toxicol 24:615–622. https://doi.org/10.1191/0960327105ht586oa

Jordan MR, Morrisonponce D (2023) Naloxone. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Jordan VC (2003) Antiestrogens and Selective Estrogen Receptor Modulators as Multifunctional Medicines. 1. Receptor Interactions. J Med Chem 46:883–908. https://doi.org/10.1021/jm020449y

Jung E, Ryu S, Park Z, et al (2018) Influence of CYP2D6 Polymorphism on the Pharmacokinetic/Pharmacodynamic Characteristics of Carvedilol in Healthy Korean Volunteers. J Korean Med Sci 33:e182. https://doi.org/10.3346/jkms.2018.33.e182

Jung Y-H, Kim H, Kim H, et al (2022) The anti-nociceptive effect of BPC-157 on the incisional pain model in rats. J Dent Anesth Pain Med 22:97–105. https://doi.org/10.17245/jdapm.2022.22.2.97

Kahyaoğlu S, Yılmaz B, Işık AZ (2017) Pharmacokinetic, pharmacodynamic, and clinical aspects of ovulation induction agents: A review of the literature. J Turk Ger Gynecol Assoc 18:48–55. https://doi.org/10.4274/jtgga.2016.0107

Kalsen A, Hostrup M, Backer V, Bangsbo J (2016a) Effect of formoterol, a long-acting β2-adrenergic agonist, on muscle strength and power output, metabolism, and fatigue during maximal sprinting in men. American Journal of Physiology-Regulatory, Integrative and Comparative Physiology 310:R1312–R1321. https://doi.org/10.1152/ajpregu.00364.2015

Kalsen A, Hostrup M, Söderlund K, et al (2016b) Inhaled Beta2-Agonist Increases Power Output and Glycolysis during Sprinting in Men. Medicine & Science in Sports & Exercise 48:39. https://doi.org/10.1249/MSS.0000000000000732

Kanavos PG, Saka ÖR (2005) How Should Functionally Equivalent Drugs be Reimbursed? Dis-Manage-Health-Outcomes 13:359–370. https://doi.org/10.2165/00115677-200513060-00002

Kelly CM, Buzdar AU (2010) Anastrozole. Expert Opinion on Drug Safety 9:995–1003. https://doi.org/10.1517/14740338.2010.515977

Journal Pre-proof

Kerpel dos Santos M, Walber GB, Kreutz T, et al (2019) Evaluation of the Presence of 1,3-Dimethylamylamine in Pelargonium Leaves and Essential Oils by Mass Spectrometric and Chromatographic Methods. Chromatographia

Kerrigan S, Goldberger BA (2020) Opioids. In: Levine BS, KERRIGAN S (eds) Principles of Forensic Toxicology. Springer International Publishing, Cham, pp 347–369

Khan TM, Patel R, Siddiqui AH (2023) Furosemide. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Kim M-J, Byeon J-Y, Kim Y-H, et al (2018) Effect of the CYP2D6*10 allele on the pharmacokinetics of clomiphene and its active metabolites. Arch Pharm Res 41:347–353. https://doi.org/10.1007/s12272-018-1005-7

Kinahan A, Budgett R, Mazzoni I (2017) Structure and Development of the List of Prohibited Substances and Methods. Med Sport Sci 62:39–54. https://doi.org/10.1159/000460699

Kinnunen M, Piirainen P, Kokki H, et al (2019) Updated Clinical Pharmacokinetics and Pharmacodynamics of Oxycodone. Clin Pharmacokinet 58:705–725. https://doi.org/10.1007/s40262-018-00731-3

Kintz P (2022) The forensic response after an adverse analytical finding (doping) involving a selective androgen receptor modulator (SARM) in human athlete. Journal of Pharmaceutical and Biomedical Analysis 207:114433. https://doi.org/10.1016/j.jpba.2021.114433

Kintz P, Salomone A, Vincenti M (2023) Perspectives and Challenges of Hair Analysis. Royal Society of Chemistry

Kirch W, Görg KG (1982) Clinical pharmacokinetics of atenolol — A review. European Journal of Drug Metabolism and Pharmacokinetics 7:81–91. https://doi.org/10.1007/BF03188723

Klei TRL, Meinderts SM, van den Berg TK, van Bruggen R (2017) From the Cradle to the Grave: The Role of Macrophages in Erythropoiesis and Erythrophagocytosis. Frontiers in Immunology 8:

Kołodziejski PA, Pruszyńska-Oszmałek E, Wojciechowicz T, et al (2021) The Role of Peptide Hormones Discovered in the 21st Century in the Regulation of Adipose Tissue Functions. Genes (Basel) 12:756. https://doi.org/10.3390/genes12050756

Kozhuharov VR, Ivanov K, Ivanova S (2022) Dietary Supplements as Source of Unintentional Doping. BioMed Research International 2022:e8387271. https://doi.org/10.1155/2022/8387271

Kröner P, Heinkele G, Kerb R, et al (2021) Stereoselective quantification of phase 1 and 2 metabolites of clomiphene in human plasma and urine. Talanta 221:121658. https://doi.org/10.1016/j.talanta.2020.121658

Kruse P, Ladefoged J, Nielsen U, et al (1986) beta-Blockade used in precision sports: effect on pistol shooting performance. J Appl Physiol (1985) 61:417–420. https://doi.org/10.1152/jappl.1986.61.2.417

Ksenofontov AA, Stupikova SA, Bocharov PS, et al (2019) Novel fluorescent sensors based on zinc(II) bis(dipyrromethenate)s for furosemide detection in organic media. Journal of Photochemistry and Photobiology A: Chemistry 382:111899. https://doi.org/10.1016/j.jphotochem.2019.111899

Kumar R, Viswanath O, Saadabadi A (2023) Buprenorphine. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Kwasnik A, Abreu A, Chediak A (2020) Cannabinoids and Sleep: Helpful or Harmful? Curr Pulmonol Rep 9:96–101. https://doi.org/10.1007/s13665-020-00254-y

Kwon L, Rosendorff C (2018) 20 - The Medical Treatment of Stable Angina. In: de Lemos JA, Omland T (eds) Chronic Coronary Artery Disease. Elsevier, pp 280–302

Journal Pre-proof

Lappin KM, Mills KI, Lappin TR (2021) Erythropoietin in bone homeostasis—Implications for efficacious anemia therapy. Stem Cells Translational Medicine 10:836–843. https://doi.org/10.1002/sctm.20-0387

Lee P, Birzniece V, Umpleby AM, et al (2015) Formoterol, a Highly β2-Selective Agonist, Induces Gender-Dimorphic Whole Body Leucine Metabolism in Humans. Metabolism 64:506–512. https://doi.org/10.1016/j.metabol.2014.12.005

Limberger RP, Jacques ALB, Schmitt GC, Arbo MD (2013) Pharmacological Effects of Ephedrine. In: Ramawat KG, Mérillon J-M (eds) Natural Products: Phytochemistry, Botany and Metabolism of Alkaloids, Phenolics and Terpenes. Springer, Berlin, Heidelberg, pp 1217–1237

Lin Y, Liu L, Ou G, et al (2023) Development of Nafion/single-walled carbon nanotube integrated arrays for the rapid detection of salbutamol doping. Analytica Chimica Acta 1249:340907. https://doi.org/10.1016/j.aca.2023.340907

Lippi G, Mattiuzzi C (2017) Misuse of the metabolic modulator meldonium in sports. J Sport Health Sci 6:49–51. https://doi.org/10.1016/j.jshs.2016.06.008

Listos J, Łupina M, Talarek S, et al (2019) The Mechanisms Involved in Morphine Addiction: An Overview. International Journal of Molecular Sciences 20:4302. https://doi.org/10.3390/ijms20174302

Ljungqvist A (2017) Brief History of Anti-Doping. Med Sport Sci 62:1–10. https://doi.org/10.1159/000460680

Loke S, de la Torre X, Iannone M, et al (2021) Controlled administration of dehydrochloromethyltestosterone in humans: Urinary excretion and long-term detection of metabolites for anti-doping purpose. J Steroid Biochem Mol Biol 214:105978. https://doi.org/10.1016/j.jsbmb.2021.105978

Longardner K, Haubenberger D (2022) 3.03 - Essential Tremor. In: Kenakin T (ed) Comprehensive Pharmacology. Elsevier, Oxford, pp 18–33

Longer MA, Ch'Ng HS, Robinson JR (1985) Bioadhesive Polymers as Platforms for Oral Controlled Drug Delivery III: Oral Delivery of Chlorothiazide Using a Bioadhesive Polymer. Journal of Pharmaceutical Sciences 74:406–411. https://doi.org/10.1002/jps.2600740408

Losacker M, Kraemer M, Philipsen A, et al (2021) Enantioselective Quantification of Amphetamine and Metabolites in Serum Samples: Forensic Evaluation and Estimation of Consumption Time. Metabolites 11:521. https://doi.org/10.3390/metabo11080521

Lu J, He G, Wang X, et al (2013a) Mass spectrometric analyses of urinary clomiphene and toremifene metabolites in doping control by liquid chromatography quadrupole time-of-flight mass spectrometry (LC-QTOF). Anal Methods 5:6677–6681. https://doi.org/10.1039/C3AY41334A

Lu Y, Sun Z, Zhang Y, et al (2013b) Simultaneous quantification of 22R and 22S epimers of budesonide in human plasma by ultra-high-performance liquid chromatography-tandem mass spectrometry: application in a stereoselective pharmacokinetic study. J Chromatogr B Analyt Technol Biomed Life Sci 921–922:27–34. https://doi.org/10.1016/j.jchromb.2013.01.015

Lugo RA, Kern SE (2002) Clinical Pharmacokinetics of Morphine. Journal of Pain & Palliative Care Pharmacotherapy 16:5–18. https://doi.org/10.1080/J354v16n04_02

Mahendru D, Kumaravel J, Mahalmani VM, Medhi B (2020) Athlete Biological Passport: Need and Challenges. Indian J Orthop 54:264–270. https://doi.org/10.1007/s43465-020-00040-7

Maini AAN, Maxwell-Scott H, Marks DJB (2014) Severe alkalosis and hypokalemia with stanozolol misuse. The American Journal of Emergency Medicine 32:196.e3-196.e4. https://doi.org/10.1016/j.ajem.2013.09.027

Journal Pre-proof

Malik P, Bhushan R (2017) Development of liquid chromatographic methods for enantioseparation and sensitive detection of β-adrenolytics/β2-agonists in human plasma using a single enantiomer reagent. Journal of Chromatography B 1061–1062:117–122. https://doi.org/10.1016/j.jchromb.2017.06.041

Malve HO (2018) Sports Pharmacology: A Medical Pharmacologist's Perspective. J Pharm Bioallied Sci 10:126–136. https://doi.org/10.4103/jpbs.JPBS_229_17

Margo K, Winn R (2006) Testosterone Treatments: Why, When, and How? afp 73:1591–1598

Markowitz JS, Patrick KS (2017) The Clinical Pharmacokinetics of Amphetamines Utilized in the Treatment of Attention-Deficit/Hyperactivity Disorder. J Child Adolesc Psychopharmacol 27:678–689. https://doi.org/10.1089/cap.2017.0071

Marques L, Vale N (2022) Salbutamol in the Management of Asthma: A Review. International Journal of Molecular Sciences 23:14207. https://doi.org/10.3390/ijms232214207

Martin D, Le JK (2023) Amphetamine. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Matabosch X, Pozo OJ, Monfort N, et al (2015a) Detection and characterization of betamethasone metabolites in human urine by LC-MS/MS. Drug Test Anal 7:663–672. https://doi.org/10.1002/dta.1770

Matabosch X, Pozo OJ, Pérez-Mañá C, et al (2012) Identification of budesonide metabolites in human urine after oral administration. Anal Bioanal Chem 404:325–340. https://doi.org/10.1007/s00216-012-6037-0

Matabosch X, Pozo OJ, Pérez-Mañá C, et al (2015b) Detection and characterization of prednisolone metabolites in human urine by LC-MS/MS. Journal of Mass Spectrometry 50:633–642. https://doi.org/10.1002/jms.3571

Mazzarino M, Biava M, de la Torre X, et al (2013) Characterization of the biotransformation pathways of clomiphene, tamoxifen and toremifene as assessed by LC-MS/(MS) following in vitro and excretion studies. Anal Bioanal Chem 405:5467–5487. https://doi.org/10.1007/s00216-013-6961-7

Mazzarino M, Khevenhüller-Metsch FL, Fiacco I, et al (2018) Drug–drug interaction and doping: Effect of non-prohibited drugs on the urinary excretion profile of methandienone. Drug Testing and Analysis 10:1554–1565. https://doi.org/10.1002/dta.2406

Mbi Feh MK, Wadhwa R (2023) Clomiphene. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Mesonzhnik NV, Postnikov PV, Appolonova SA, Krotov GI (2018) Characterization and Detection of Erythropoietin Fc Fusion Proteins Using Liquid Chromatography–Mass Spectrometry. J Proteome Res 17:689–697. https://doi.org/10.1021/acs.jproteome.7b00739

Miller GD, Moore C, Nair V, et al (2019) Hypothalamic-Pituitary-Testicular Axis Effects and Urinary Detection Following Clomiphene Administration in Males. J Clin Endocrinol Metab 104:906–914. https://doi.org/10.1210/jc.2018-01159

Miyake T, Kung CK, Goldwasser E (1977) Purification of human erythropoietin. Journal of Biological Chemistry 252:5558–5564. https://doi.org/10.1016/S0021-9258(19)63387-9

Molphy J, Dickinson JW, Chester NJ, et al (2019) The Effects of Inhaled Terbutaline on 3-km Running Time-Trial Performance. International Journal of Sports Physiology and Performance 14:822–828. https://doi.org/10.1123/ijspp.2018-0633

Morgan T (1994) Clinical Pharmacokinetics and Pharmacodynamics of Carvedilol. Clin Pharmacokinet 26:335–346. https://doi.org/10.2165/00003088-199426050-00002

Mottram DR (2018) Chemical and physical manipulation. In: Drugs in Sport, 7th edn. Routledge

Journal Pre-proof

Mottram DR, Chester N (2018) Drugs in Sport, Seventh edition. Routledge

Mottram DR, George AJ (2000) Anabolic steroids. Best Practice & Research Clinical Endocrinology & Metabolism 14:55–69. https://doi.org/10.1053/beem.2000.0053

Mukherjee AG, Wanjari UR, Nagarajan D, et al (2022) Letrozole: Pharmacology, toxicity and potential therapeutic effects. Life Sciences 310:121074. https://doi.org/10.1016/j.lfs.2022.121074

Muniz-Santos R, Avezum J, Abidão-Neto B, Cameron LC (2023) Dietary higenamine from Annonaceae family fruits as a possible source of unintentional doping. Forensic Science International 342:111539. https://doi.org/10.1016/j.forsciint.2022.111539

Mürdter TE, Kerb R, Turpeinen M, et al (2012) Genetic polymorphism of cytochrome P450 2D6 determines oestrogen receptor activity of the major infertility drug clomiphene via its active metabolites. Hum Mol Genet 21:1145–1154. https://doi.org/10.1093/hmg/ddr543

Murphy PB, Bechmann S, Barrett MJ (2023) Morphine. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Narayanan R, Kuppermann BD (2010) CHAPTER 30 - Corticosteroids: triamcinolone, dexamethasone, fluocinolone, and others. In: Nguyen QD, Rodrigues EB, Farah ME, Mieler WF (eds) Retinal Pharmacotherapy. W.B. Saunders, Edinburgh, pp 201–207

Nardulli AM (2003) Estrogen Receptor-α Structure and Function. In: Henry HL, Norman AW (eds) Encyclopedia of Hormones. Academic Press, New York, pp 594–599

Neamat-Allah ANF (2014) Effect of boldenone undecylenate on haematological and biochemical parameters in veal calves. Global Veterinaria 13:1092–1096

Nedresky D, Singh G (2022) Physiology, Luteinizing Hormone. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Ng T, Gupta V (2023) Tetrahydrocannabinol (THC). In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Niederberger E, King TS, Russe OQ, Geisslinger G (2015) Activation of AMPK and its Impact on Exercise Capacity. Sports Med 45:1497–1509. https://doi.org/10.1007/s40279-015-0366-z

Ocaña-Rios I, Araujo-González F, Olmos-Espejel JJ, Peña-Alvarez A (2022) Miniaturized Analysis of Methylhexanamine in Urine by Gas Chromatography Applying In Situ Derivatization. Chromatographia 85:95–104. https://doi.org/10.1007/s10337-021-04116-w

O'Donnell S, O'Morain CA (2010) Therapeutic benefits of budesonide in gastroenterology. Therapeutic Advances in Chronic Disease 1:177–186. https://doi.org/10.1177/2040622310379293

Ogbonna AO, Arinze CO, Anusiem CA, Nwokike MO (2021) The Abuse of Central Nervous System Stimulants and its Impact on the Youth of Eastern Nigeria. International Neuropsychiatric Disease Journal 15:20–33. https://doi.org/10.9734/indj/2021/v15i330156

Oh SW, Han SY (2015) Loop Diuretics in Clinical Practice. Electrolyte Blood Press 13:17–21. https://doi.org/10.5049/EBP.2015.13.1.17

Opie LH, Gersh BJ (2012) Drugs for the Heart. Elsevier Health Sciences

Otey CR, Bandara G, Lalonde J, et al (2006) Preparation of human metabolites of propranolol using laboratory-evolved bacterial cytochromes P450. Biotechnol Bioeng 93:494–499. https://doi.org/10.1002/bit.20744

Patel DR, Omar H, Terry M (2010) Sport-related Performance Anxiety in Young Female Athletes. Journal of Pediatric and Adolescent Gynecology 23:325–335. https://doi.org/10.1016/j.jpag.2010.04.004

Journal Pre-proof

Patel MMR, Patel DOB, Patel DHU (2022) TEXTBOOK OF MEDICINAL CHEMISTRY - II. Darshan Publishers

Pathan H, Williams J (2012) Basic opioid pharmacology: an update. Br J Pain 6:11–16. https://doi.org/10.1177/2049463712438493

Peng SH, Segura J, Farré M, de la Torre X (2000) Oral testosterone administration detected by testosterone glucuronidation measured in blood spots dried on filter paper. Clin Chem 46:515–522

Pereira HMG, Sardela VF, Padilha MC, et al (2017) Doping control analysis at the Rio 2016 Olympic and Paralympic Games. Drug Testing and Analysis 9:1658–1672. https://doi.org/10.1002/dta.2329

Pickup ME (1979) Clinical Pharmacokinetics of Prednisone and Prednisolone. Clin-Pharmacokinet 4:111–128. https://doi.org/10.2165/00003088-197904020-00004

Piotrowska I, Piotrowska M (2019) Anastrozole as aromatase inhibitor – new approaches to breast cancer treatment in postmenopausal women. Nowotwory Journal of Oncology 69:26–35. https://doi.org/10.5603/NJO.2019.0005

Piper T, Fusshöller G, Schänzer W, et al (2019) Studies on the in vivo metabolism of methylstenbolone and detection of novel long term metabolites for doping control analysis. Drug Testing and Analysis 11:1644–1655. https://doi.org/10.1002/dta.2736

Pistilli EE, Bogdanovich S, Goncalves MD, et al (2011) Targeting the Activin Type IIB Receptor to Improve Muscle Mass and Function in the mdx Mouse Model of Duchenne Muscular Dystrophy. Am J Pathol 178:1287–1297. https://doi.org/10.1016/j.ajpath.2010.11.071

Pohl O, Homery M-C, Lemaux F, et al (2015) Pharmacokinetic interactions of OBE001 and betamethasone in healthy female volunteers. Journal of Clinical Pharmacy and Therapeutics 40:328–332. https://doi.org/10.1111/jcpt.12258

Pokrywka A, Kwiatkowska D, Kaliszewski P, Grucza R (2010) Some aspects concerning modifications of the List of Prohibited Substances and Methods in sport. Biol Sport 27:307–314. https://doi.org/10.5604/20831862.927498

Polet M, Van Gansbeke W, Geldof L, et al (2017) Identification and characterization of novel long-term metabolites of oxymesterone and mesterolone in human urine by application of selected reaction monitoring GC-CI-MS/MS. Drug Test Anal 9:1673–1684. https://doi.org/10.1002/dta.2183

Pomara C, Cassano T, D'Errico S, et al (2012) Data available on the extent of cocaine use and dependence: biochemistry, pharmacologic effects and global burden of disease of cocaine abusers. Curr Med Chem 19:5647–5657. https://doi.org/10.2174/092986712803988811

Postnikov P, Krotov G, Mesonzhnik N, et al (2015) Fc-fragment removal allows the EPO-Fc fusion protein to be detected in blood samples by IEF-PAGE. Drug Testing and Analysis 7:999–1008. https://doi.org/10.1002/dta.1916

Powell J, Gurk-Turner C (2002) Darbepoetin alfa (Aranesp). Proc (Bayl Univ Med Cent) 15:332–335

Puckett Y, Gabbar A, Bokhari AA (2022) Prednisone. StatPearls Publishing

Rahimi-Ardabili B, Pourandarjani R, Habibollahi P, Mualeki A (2006) Finasteride induced depression: a prospective study. BMC Clinical Pharmacology 6:7. https://doi.org/10.1186/1472-6904-6-7

Ramos-Matos CF, Bistas KG, Lopez-Ojeda W (2023) Fentanyl. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Journal Pre-proof

Rangelov Kozhuharov V, Ivanov K, Ivanova S (2022) Higenamine in Plants as a Source of Unintentional Doping. Plants 11:354. https://doi.org/10.3390/plants11030354

Rankin GO (2008) Canrenone. In: Enna SJ, Bylund DB (eds) xPharm: The Comprehensive Pharmacology Reference. Elsevier, New York, pp 1–6

Rankins EM, Salem K, Manso Filho HC, et al (2022) Effect of Clenbuterol on Muscle Activity During Exercise in Standardbred Horses. Journal of Equine Veterinary Science 118:104126. https://doi.org/10.1016/j.jevs.2022.104126

Rao PS (2014) Androgens. In: Wexler P (ed) Encyclopedia of Toxicology (Third Edition). Academic Press, Oxford, pp 230–233

Rice J, Proctor K, Lopardo L, et al (2018) Stereochemistry of ephedrine and its environmental significance: Exposure and effects directed approach. Journal of Hazardous Materials 348:39–46. https://doi.org/10.1016/j.jhazmat.2018.01.020

Richards JR, Laurin EG (2023) Cocaine. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Rogol AD, Miller GD, Eichner D (2019) Growth hormone, growth hormone secretagogues, and insulin-like growth factor-1 in sports: prohibited status, therapeutic use exemptions, and analytical detectability. Current Opinion in Endocrine and Metabolic Research 9:8–13. https://doi.org/10.1016/j.coemr.2019.05.003

Rosenfeld GA, Chang A, Poulin M, et al (2011) Cholestatic jaundice, acute kidney injury and acute pancreatitis secondary to the recreational use of methandrostenolone: a case report. J Med Case Reports 5:138. https://doi.org/10.1186/1752-1947-5-138

Ruzicka L, Wettstein A (1935) Sexualhormone VII. Über die künstliche Herstellung des Testikelhormons Testosteron (Androsten-3-on-17-ol). Helvetica Chimica Acta 18:1264–1275. https://doi.org/10.1002/hlca.193501801176

Sadiq NM, Dice TJ, Mead T (2023) Oxycodone. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Saiz-Rodríguez M, Ochoa D, Herrador C, et al (2019) Polymorphisms associated with fentanyl pharmacokinetics, pharmacodynamics and adverse effects. Basic & Clinical Pharmacology & Toxicology 124:321–329. https://doi.org/10.1111/bcpt.13141

Sakellariou P, Petrou M, Lyris E, et al (2021) Determination of salmeterol, α-hydroxysalmeterol and fluticasone propionate in human urine and plasma for doping control using UHPLC–QTOF–MS. Biomedical Chromatography 35:e5114. https://doi.org/10.1002/bmc.5114

Saudan C, Baume N, Robinson N, et al (2006) Testosterone and doping control. Br J Sports Med 40:i21–i24. https://doi.org/10.1136/bjsm.2006.027482

Schilling BK, Hammond KG, Bloomer RJ, et al (2013) Physiological and pharmacokinetic effects of oral 1,3-dimethylamylamine administration in men. BMC Pharmacol Toxicol 14:52. https://doi.org/10.1186/2050-6511-14-52

Schlegel PN (2012) Aromatase inhibitors for male infertility. Fertil Steril 98:1359–1362. https://doi.org/10.1016/j.fertnstert.2012.10.023

Schobersberger W, Dünnwald T, Gmeiner G, Blank C (2017) Story behind meldonium–from pharmacology to performance enhancement: a narrative review. Br J Sports Med 51:22–25. https://doi.org/10.1136/bjsports-2016-096357

Schoener B, Borger J (2022) Erythropoietin Stimulating Agents. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Journal Pre-proof

Seebauer CT, Graus MS, Huang L, et al (2022) Non–beta blocker enantiomers of propranolol and atenolol inhibit vasculogenesis in infantile hemangioma. J Clin Invest 132:. https://doi.org/10.1172/JCI151109

Seif-Barghi T, Moghadam N, Kobarfard F (2015) Morphine/Codeine Ratio, a Key in Investigating a Case of Doping. Asian J Sports Med 6:e28798. https://doi.org/10.5812/asjsm.28798

Sessa F, Salerno M, Di Mizio G, et al (2018) Anabolic Androgenic Steroids: Searching New Molecular Biomarkers. Frontiers in Pharmacology 9:

Seyerlein L, Gillard N, Delahaut P, et al (2021) Depletion of clomiphene residues in eggs and muscle after oral administration to laying hens. Food Addit Contam Part A Chem Anal Control Expo Risk Assess 38:1875–1882. https://doi.org/10.1080/19440049.2021.1949497

Shagufta, Ahmad I (2018) Tamoxifen a pioneering drug: An update on the therapeutic potential of tamoxifen derivatives. European Journal of Medicinal Chemistry 143:515–531. https://doi.org/10.1016/j.ejmech.2017.11.056

Shaker K, Nillas A, Ellison R, et al (2023) Delta-8-Tetrahydrocannabinol Exposure and Confirmation in Four Pediatric Patients. J Med Toxicol 19:190–195. https://doi.org/10.1007/s13181-022-00927-x

Sharma RK, Gupta M, Rani R (2022) Delineating Injectable Triamcinolone-Induced Cutaneous Atrophy and Therapeutic Options in 24 Patients—A Retrospective Study. Indian Dermatol Online J 13:199–206. https://doi.org/10.4103/idoj.idoj_483_21

Sheikh NK, Dua A (2023) Cannabinoids. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Siddiqui MAA, Keating GM (2006) Darbepoetin alfa: a review of its use in the treatment of anaemia in patients with cancer receiving chemotherapy. Drugs 66:997–1012. https://doi.org/10.2165/00003495-200666070-00018

Sim YE, Kim JW, Ko BJ, et al (2023) Determination of urinary metabolites of cannabidiol, Δ8-tetrahydrocannabinol, and Δ9-tetrahydrocannabinol by automated online µSPE–LC–MS/MS method. Journal of Chromatography B 1214:123568. https://doi.org/10.1016/j.jchromb.2022.123568

Sismondo S, Greene JA (2015) The Pharmaceutical Studies Reader. John Wiley & Sons

Skauby RH, Bergan S, Andersen AM, et al (2021) In vitro assessments predict that CYP3A4 contributes to a greater extent than CYP3A5 to prednisolone clearance. Basic & Clinical Pharmacology & Toxicology 129:427–436. https://doi.org/10.1111/bcpt.13645

Skoupá K, Šťastný K, Sládek Z (2022) Anabolic Steroids in Fattening Food-Producing Animals—A Review. Animals 12:2115. https://doi.org/10.3390/ani12162115

Slíž K, Olešová D, Piešťanský J, Mikuš P (2022) Simple and Sensitive Analysis of Clenbuterol in Urine Matrices by UHPLC-MS/MS Method with Online-SPE Sample Preparation. Separations 9:440. https://doi.org/10.3390/separations9120440

Solagna F, Tezze C, Lindenmeyer MT, et al (2021) Pro-cachectic factors link experimental and human chronic kidney disease to skeletal muscle wasting programs. J Clin Invest 131:. https://doi.org/10.1172/JCI135821

Solomon ZJ, Mirabal JR, Mazur DJ, et al (2019) Selective Androgen Receptor Modulators: Current Knowledge and Clinical Applications. Sex Med Rev 7:84–94. https://doi.org/10.1016/j.sxmr.2018.09.006

Spies CM, Strehl C, van der Goes MC, et al (2011) Glucocorticoids. Best Practice & Research Clinical Rheumatology 25:891–900. https://doi.org/10.1016/j.berh.2011.11.002

Journal Pre-proof

Srinivasan AV (2019) Propranolol: A 50-Year Historical Perspective. Ann Indian Acad Neurol 22:21–26. https://doi.org/10.4103/aian.AIAN_201_18

Statler AK, Maani CV, Kohli A (2023) Ephedrine. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Stenman U-H, Hotakainen K, Alfthan H (2008) Gonadotropins in doping: pharmacological basis and detection of illicit use. Br J Pharmacol 154:569–583. https://doi.org/10.1038/bjp.2008.102

Stone WL, Leavitt L, Varacallo M (2022) Physiology, Growth Factor. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Strano Rossi S, Botrè F (2011) Prevalence of illicit drug use among the Italian athlete population with special attention on drugs of abuse: a 10-year review. J Sports Sci 29:471–476. https://doi.org/10.1080/02640414.2010.543915

Stuart M, Schneider C, Steinbach K (2016) Meldonium use by athletes at the Baku 2015 European Games. Br J Sports Med 50:694–698. https://doi.org/10.1136/bjsports-2015-095906

Sultan K, Zamir A, Ashraf W, et al (2023) Clinical pharmacokinetics of terbutaline in humans: a systematic review. Naunyn-Schmiedeberg's Arch Pharmacol 396:213–227. https://doi.org/10.1007/s00210-022-02304-5

Suresh S, Rajvanshi PK, Noguchi CT (2020) The Many Facets of Erythropoietin Physiologic and Metabolic Response. Frontiers in Physiology 10:

Tacey A, Parker L, Yeap BB, et al (2019) Single-dose prednisolone alters endocrine and haematologic responses and exercise performance in men. Endocrine Connections 8:111–119. https://doi.org/10.1530/EC-18-0473

Tagen M, Klumpers LE (2022) Review of delta-8-tetrahydrocannabinol (Δ8-THC): Comparative pharmacology with Δ9-THC. British Journal of Pharmacology 179:3915–3933. https://doi.org/10.1111/bph.15865

Tang K, Zhang P (2011) Enantioselective extraction of terbutaline enantiomers with β-cyclodextrin derivatives as hydrophilic selectors. Chem Pap 65:273–279. https://doi.org/10.2478/s11696-011-0011-8

Tauchen J, Jurášek M, Huml L, Rimpelová S (2021) Medicinal Use of Testosterone and Related Steroids Revisited. Molecules 26:1032. https://doi.org/10.3390/molecules26041032

Teng R, Wang L, Chen M, et al (2020) Amino acid based pharmaceutical cocrystals and hydrate cocrystals of the chlorothiazide: Structural studies and physicochemical properties. Journal of Molecular Structure 1217:128432. https://doi.org/10.1016/j.molstruc.2020.128432

Thau L, Gandhi J, Sharma S (2022) Physiology, Cortisol. StatPearls Publishing

Thevis M, Geyer H, Mareck U, et al (2007) Doping-control analysis of the 5alpha-reductase inhibitor finasteride: determination of its influence on urinary steroid profiles and detection of its major urinary metabolite. Ther Drug Monit 29:236–247. https://doi.org/10.1097/FTD.0b013e31803bb85d

Thevis M, Kuuranne T, Geyer H, Schänzer W (2010a) Annual banned-substance review: analytical approaches in human sports drug testing. Drug Testing and Analysis 2:149–161. https://doi.org/10.1002/dta.128

Thevis M, Kuuranne T, Thomas A, Geyer H (2021a) Do dried blood spots have the potential to support result management processes in routine sports drug testing?—Part 2: Proactive sampling for follow-up investigations concerning atypical or adverse analytical findings. Drug Testing and Analysis 13:505–509. https://doi.org/10.1002/dta.3011

Journal Pre-proof

Thevis M, Piper T, Thomas A (2021b) Recent advances in identifying and utilizing metabolites of selected doping agents in human sports drug testing. Journal of Pharmaceutical and Biomedical Analysis 205:114312. https://doi.org/10.1016/j.jpba.2021.114312

Thevis M, Sigmund G, Geyer H, Schänzer W (2010b) Stimulants and doping in sport. Endocrinol Metab Clin North Am 39:89–105, ix. https://doi.org/10.1016/j.ecl.2009.10.011

Thevis M, Thomas A, Schänzer W (2010c) Insulin. In: Thieme D, Hemmersbach P (eds) Doping in Sports: Biochemical Principles, Effects and Analysis. Springer, Berlin, Heidelberg, pp 209–226

Thevis M, Walpurgis K, Thomas A, Geyer H (2019) Peptidic drugs and drug candidates in sports drug testing: agents affecting mitochondrial biogenesis or preventing activin receptor II activation. Current Opinion in Endocrine and Metabolic Research 9:22–27. https://doi.org/10.1016/j.coemr.2019.05.004

Thuyne WV, Delbeke FT (2008) Declared Use of Medication in Sports. Clinical Journal of Sport Medicine 18:143. https://doi.org/10.1097/JSM.0b013e318163f220

Torrisi M, Pennisi G, Russo I, et al (2020) Sudden Cardiac Death in Anabolic-Androgenic Steroid Users: A Literature Review. Medicina (Kaunas) 56:587. https://doi.org/10.3390/medicina56110587

Trifilieff P, Martinez D (2014) Chapter Five - Cocaine: Mechanism and Effects in the Human Brain. In: Madras B, Kuhar M (eds) The Effects of Drug Abuse on the Human Nervous System. Academic Press, Boston, pp 103–133

Trinh KV, Diep D, Chen KJQ, et al (2020) Effect of erythropoietin on athletic performance: a systematic review and meta-analysis. BMJ Open Sport Exerc Med 6:e000716. https://doi.org/10.1136/bmjsem-2019-000716

Tse KY, Zuraw BL, Chen Q, Christiansen SC (2017) Anabolic androgen use in the management of hereditary angioedema: Not so cheap after all. Annals of Allergy, Asthma & Immunology 118:456-460.e1. https://doi.org/10.1016/j.anai.2017.01.015

USADA (2019) World Anti-Doping Agency (WADA) Prohibited List | USADA. In: U.S. Anti-Doping Agency. https://www.usada.org/athletes/substances/prohibited-list/. Accessed 13 Apr 2023

USADA (2021) U.S. Postal Service Pro Cycling Team Investigation | USADA. In: United States Anti-Doping Agency. https://www.usada.org/athletes/results/u-s-postal-service-pro-cycling-team-investigation/. Accessed 19 Mar 2024

van Amsterdam J, Opperhuizen A, Hartgens F (2010) Adverse health effects of anabolic–androgenic steroids. Regulatory Toxicology and Pharmacology 57:117–123. https://doi.org/10.1016/j.yrtph.2010.02.001

van Zwieten PA (1993) Pharmacodynamic profile of carvedilol. Cardiology 82 Suppl 3:19–23. https://doi.org/10.1159/000175939

Vanol PG, Singhal P, Shah PA, et al (2016) SPE–UPLC–MS/MS assay for determination of letrozole in human plasma and its application to bioequivalence study in healthy postmenopausal Indian women. Journal of Pharmaceutical Analysis 6:276–281. https://doi.org/10.1016/j.jpha.2016.05.004

Vardanyan R, Hruby V (2016) Chapter 23 - Drugs for Treating Respiratory System Diseases. In: Vardanyan R, Hruby V (eds) Synthesis of Best-Seller Drugs. Academic Press, Boston, pp 357–381

Vardanyan RS, Hruby VJ (2006a) 27 - Corticosteroids. In: Vardanyan RS, Hruby VJ (eds) Synthesis of Essential Drugs. Elsevier, Amsterdam, pp 349–363

Vardanyan RS, Hruby VJ (2006b) 12 - Adrenoblocking Drugs. In: Vardanyan RS, Hruby VJ (eds) Synthesis of Essential Drugs. Elsevier, Amsterdam, pp 161–177

Journal Pre-proof

Vari C-E, Bianca O, Miklos A, et al (2016) Aromatase inhibitors in men – off-label use, misuse, abuse and doping. Farmacia 64:813–818

Varmus M, Kubina M, Adámik R (2021) From Management to Sport Management. In: Varmus M, Kubina M, Adámik R (eds) Strategic Sport Management: Sustainability of Sports Clubs. Springer International Publishing, Cham, pp 3–34

Ventura R, Segura J (2010) Masking and Manipulation. In: Thieme D, Hemmersbach P (eds) Doping in Sports: Biochemical Principles, Effects and Analysis. Springer, Berlin, Heidelberg, pp 327–354

Verghese C, Abdijadid S (2023) Methylphenidate. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Vernec A, Slack A, Harcourt PR, et al (2020) Glucocorticoids in elite sport: current status, controversies and innovative management strategies-a narrative review. Br J Sports Med 54:8–12. https://doi.org/10.1136/bjsports-2018-100196

Vernec AR (2013) Doping, Ethics, and the Sport Physician. Current Sports Medicine Reports 12:283. https://doi.org/10.1249/JSR.0b013e3182a4b877

Vuckovic S, Prostran M, Ivanovic M, et al (2009) Fentanyl Analogs: Structure-Activity-Relationship Study. Current Medicinal Chemistry 16:2468–2474. https://doi.org/10.2174/092986709788682074

WADA (2006) 2006 Adverse Analytical Findings Reported by Accredited Laboratories. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/WADA_LaboStatistics_2006.pdf. Accessed 3 Feb 2023

WADA (2010) The 2011 Prohibited List International Standard. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/WADA_Prohibited_List_2011_EN.pdf. Accessed 21 Jan 2023

WADA (2014) TD2014MRPL. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/wada-td2014mrpl-summary-modifications-en.pdf. Accessed 8 Apr 2023

WADA (2015) Reporting & Management of Urinary Human Chorionic Gonadothropin (hCG) and Luteinizing Hormone (LH) Findings in Male Athletes. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/wada-guidelines-hcg-lh-findings-v2.0-2015-en.pdf. Accessed 1 Feb 2023

WADA (2017a) TD2017CG/LH. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/resources/files/wada-td2017cglh-en_0.pdf. Accessed 10 Jan 2023

WADA (2017b) Novel Biomarkers in Recombinant Human Growth Hormone Detection. LC-MRM-MS Method for Fibronectin Quantification in AntiDoping Routin. In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/scientific-research/novel-biomarkers-recombinant-human-growth-hormone-detection-lc-mrm-ms. Accessed 17 Mar 2024

WADA (2017c) Summary of Major Modifications and Explanatory Notes 2018 Prohibited List. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/prohibited_list_2018_summary_of_modifications_en.pdf. Accessed 12 Apr 2023

WADA (2019a) Improving detection of the confounding factor dutasteride. In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/scientific-research/improving-detection-confounding-factor-dutasteride. Accessed 18 Mar 2024

Journal Pre-proof

WADA (2019b) Is tramadol a Performance Enhancing Drug? In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/scientific-research/tramadol-performance-enhancing-drug. Accessed 18 Mar 2024

WADA (2021a) 2021 Anti-Doping Testing Figures Questions and Answers. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/2023-01/2021_testing_figures_report_qa_en_0.pdf. Accessed 12 Apr 2023

WADA (2021b) TD2021NA. In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/lab-documents/td2021na. Accessed 10 Jan 2023

WADA (2022a) Anti-Doping Testing Figures Report. In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/anti-doping-stats/anti-doping-testing-figures-report. Accessed 8 Dec 2022

WADA (2022b) WADA publishes 2023 Prohibited List. In: World Anti Doping Agency. https://www.wada-ama.org/en/news/wada-publishes-2023-prohibited-list. Accessed 13 Apr 2023

WADA (2022c) Summary of Major Modifications and Explanatory Notes 2022 Prohibited List. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/2022-01/2022list_draft_explanatory_list_en_0.pdf. Accessed 21 Jan 2023

WADA (2022d) 2023 Prohibited List. In: World Anti Doping Agency. https://www.wada-ama.org/en/resources/world-anti-doping-program/2023-prohibited-list. Accessed 4 Dec 2022

WADA (2022e) TD2022MRPL. In: World Anti Doping Agency. https://www.wada-ama.org/sites/default/files/2022-01/td2022mrpl_v1.1_eng_0.pdf. Accessed 8 Apr 2023

WADA (2023a) Who We Are. In: World Anti Doping Agency. https://www.wada-ama.org/en/who-we-are. Accessed 13 Apr 2023

WADA (2023b) The Anti-Doping Administration & Management System (ADAMS). In: World Anti Doping Agency. https://www.wada-ama.org/en/what-we-do/adams. Accessed 13 Apr 2023

WADA (2023c) The World Anti-Doping Code. In: World Anti Doping Agency. https://www.wada-ama.org/en/what-we-do/world-anti-doping-code. Accessed 13 Apr 2023

WADA (2023d) Athlete Biological Passport. In: World Anti Doping Agency. https://www.wada-ama.org/en/athlete-biological-passport. Accessed 16 Jan 2023

WADA (2023e) WADA's Athlete Biological Passport: an important tool for protecting clean sport. In: World Anti Doping Agency. https://www.wada-ama.org/en/news/wadas-athlete-biological-passport-important-tool-protecting-clean-sport. Accessed 16 Jan 2023

WADA (2023f) Anti-Doping Statistics. In: World Anti Doping Agency. https://www.wada-ama.org/en/data-research/anti-doping-statistics. Accessed 13 Apr 2023

WADA (2023g) WADA publishes updates to documents relating to the narcotic tramadol that is on the 2024 Prohibited List. In: World Anti Doping Agency. https://www.wada-ama.org/en/news/wada-publishes-updates-documents-relating-narcotic-tramadol-2024-prohibited-list

WADA (2023h) WADA launches first phase of 2027 World Anti-Doping Code and International Standards Update Process. In: World Anti Doping Agency. https://www.wada-ama.org/en/news/wada-launches-first-phase-2027-world-anti-doping-code-and-international-standards-update. Accessed 20 Mar 2024

Weber C, Krug O, Kamber M, Thevis M (2017) Qualitative and Semiquantitative Analysis of Doping Products Seized at the Swiss Border. Substance Use & Misuse 52:742–753. https://doi.org/10.1080/10826084.2016.1263665

Journal Pre-proof

Wells DJ (2018) Gene doping. In: Drugs in Sport, 7th edn. Routledge

Wen J, Li M, Zhang W, et al (2022) Role of Higenamine in Heart Diseases: A Mini-Review. Frontiers in Pharmacology 12:

Wilson Tang WH, Fowler MB (1999) Clinical Trials of Carvedilol in Heart Failure. Heart Fail Rev 4:79–88. https://doi.org/10.1023/A:1009859902685

Wiysonge CS, Bradley HA, Volmink J, et al (2017) Beta-blockers for hypertension. Cochrane Database of Systematic Reviews. https://doi.org/10.1002/14651858.CD002003.pub5

Wu C, Kovac JR (2016) Novel Uses for the Anabolic Androgenic Steroids Nandrolone and Oxandrolone in the Management of Male Health. Curr Urol Rep 17:72. https://doi.org/10.1007/s11934-016-0629-8

Wu X, Gao F, Zhang W, Ni J (2015) Metabolism study of boldenone in human urine by gas chromatography-tandem mass spectrometry. J Pharm Biomed Anal 115:570–575. https://doi.org/10.1016/j.jpba.2015.08.014

Yamazaki D, Shimura Y, Enjoji Y, et al (2020) Pilot study of bisoprolol 8mg transdermal patch at bedtime before coronary computed tomography angiography. Journal of Cardiology 75:394–399. https://doi.org/10.1016/j.jjcc.2019.09.013

Yan X, Yan S, Chang W, et al (2023) Ultraviolet spectra determination and computational analysis of 44 E/Z steroid isomers in dried blood spot. Drug Test Anal. https://doi.org/10.1002/dta.3598

Yu NH, Ho ENM, Wan TSM, Wong ASY (2010) Doping control analysis of recombinant human erythropoietin, darbepoetin alfa and methoxy polyethylene glycol-epoetin beta in equine plasma by nano-liquid chromatography-tandem mass spectrometry. Anal Bioanal Chem 396:2513–2521. https://doi.org/10.1007/s00216-010-3455-8

Zadik Y, Elad S, Shapira A, Shapira MY (2017) Treatment of oral mucosal manifestations of chronic graft-versus-host disease: dexamethasone vs. budesonide. Expert Opinion on Pharmacotherapy 18:235–242. https://doi.org/10.1080/14656566.2017.1282464

Zito PM, Bistas KG, Syed K (2023) Finasteride. In: StatPearls. StatPearls Publishing, Treasure Island (FL)

Zurloni V, Diana B, Cavalera C, et al (2015) Deceptive behavior in doping related interviews: The case of Lance Armstrong. Psychology of Sport and Exercise 16:191–200. https://doi.org/10.1016/j.psychsport.2014.02.008

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Journal Pre-proof